INFINITY

MULTIDIMENSIONAL KNOWLEDGE-BASED ANNOTATION FOR ETHICAL CONTEXT-AWARE HERITAGE DATA LIFE CYCLES

#@APP-FORM-HERIAIA@#

List of participants

Participant No.	Participant organisation	Acronym	Country
1 (Coordinator)	ALMA MATER STUDIORUM - UNIVERSITA DI BOLOGNA	UNIBO	IT
2	Consiglio Nazionale delle Ricerche	CNR	IT
3	STORYPACT GMBH	SPA	AT
4	FRIEDRICH-SCHILLER-UNIVERSITÄT JENA	UNI JENA	DE
5	STICHTING EUROPEANA	EF	NL
6	STICHTING NEDERLANDS INSTITUUT VOORBEELD EN GELUID	NISV	NL
7	TIME MACHINE ORGANISATION (TMO)- ORGANISATION FUR INTERNATIONALE ZUSAMMENARBEIT IN TECHNOLOGIE UND WISSENSCHAFT UND KULTURELLEM ERBE	TMO	AT
8	INSTYTUT CHEMII BIOORGANICZNEJ POLSKIEJ AKADEMII NAUK	PSNC	PL
9	DIGITAL PATHS SRL	DP	IT
10	FACTS & FILES DIGITAL SERVICES GMBH	F&F DS	DE
11	FACTS & FILES - HISTORISCHES FORSCHUNGSINSTITUT BERLIN DRAUSCHKE - SCHREIBER PARTNERSCHAFTSGESELLSCHAFT -ARCHIV-, GESCHICHTS- U. LITERATURWISSENSCHAFTLER	F&F PartG	DE
12	MINISTERO DELLA CULTURA	MIC	IT
13	THE UNIVERSITY OF EDINBURGH	UEDIN	UK
14	UNIWERSYTET WROCŁAWSKI	UWr	PL
15	EIT CULTURE & CREATIVITY SOUTH WEST SL	EIT-CC-SW	ES
16	ECOLE POLYTECHNIQUE FEDERALE DE LAUSANNE	EPFL	CH
17	AIT AUSTRIAN INSTITUTE OF TECHNOLOGY GMBH	AIT	AT

1. Excellence #@REL-EVA-RE@#

1.1 Objectives and ambition #@PRJ-OBJ-PO@#

"INFINITY" symbolises the boundless potential and continuous lifecycle of cultural heritage digital objects (CHDOs) within the European Collaborative Cloud for Cultural Heritage (ECCCH). The name of the project metaphorically represents its core concept and objectives: empowering cultural organisations to unlock new perspectives and connections across cultural heritage collections, enabling the responsible and creative reuse of CHDOs. This fosters new opportunities for collaboration across cultural organisations, researchers and creative industries, ensuring the sustainability and relevance of CHDOs for present and future generations.

1.1.1 Overcoming the limits of digitisation in the cultural heritage sector

Digitisation is transforming the way the cultural heritage (CH) sector manages collections, enabling unprecedented global access and reuse of distributed collections. While this digital shift has enriched audience engagement and democratized access, it also presents challenges. Many CHDOs lack nuanced, multi-contextual metadata, limiting their accessibility, understanding, and ethical reuse. Non-interoperable platforms with disparate, often unintuitive interfaces, hinder the seamless discovery, exploration and reuse of these resources. Current systems frequently **fail to capture the multidimensional nature of CHDOs** (Tugce et al, 2020), resulting in **fragmented access and overly simplistic or biased representations**. Even when different representations of a CHDO exist, they are typically stored in disconnected datasets, shaped by the specific perspectives of individual projects, areas of expertise, administrative priorities, political objectives or research goals. This disconnection obscures the rich and diverse perspectives CHDOs embody **- cultural, historical, social, legal and ethical -** that vary by context and audience. These perspectives often include heritage knowledge passed down through family ties or community networks. Equally critical is the **provenance** of data: the history of **key steps, the individuals and their expertise, the scientific and professional values, and the workflows** that shaped the **creation and evolution** of CHDOs. This includes **intellectual property rights and associated permissions**. These diverse perspectives and provenance details are essential for **researchers, students and professionals in the cultural heritage and creative industries**, and for presenting a fairer rendition of European digital heritage collections. They also help **citizens** to better understand and appreciate the complexity of cultural heritage. For example, an Indigenous artefact may hold spiritual significance, historical value and artistic importance, requiring metadata that captures all these dimensions to ensure responsible use. Legal considerations such as **intellectual property rights** and **ethical factors**, including the **prevention of cultural appropriation or unintended AI reuse** (Tiribelli et al, 2024) (Tóth-Czifra et al, 2023), are vital for informed and respectful repurposing.

Addressing these concerns not only safeguards cultural integrity but also opens pathways for sustainable and **innovative business models** that exploit CHDOs responsibly. Fragmented collections and inadequate contextualisation hinder the development of coherent narratives, increasing the risk of misrepresentation, ethical breaches and harm to the community to whom the heritage belongs. This ultimately erodes trust in CH institutions. For example, the collections of the Benin Bronzes in UK institutions are often described as having been 'collected' in 1897, without acknowledging that they were looted by the British army as a punitive measure to consolidate control in Southern Nigeria. Much museum documentation obscures this contentious history, celebrating the artistic achievements of the bronzes while perpetuating stereotypes of Nigerian culture as backward (Hicks 2020).

Cultural Heritage Institutions (CHIs) require systems that link CHDOs to multiple interconnected perspectives while upholding ethical principles. This ensures that users can access data from **reliable information providers**, reflecting **diverse interpretative perspectives** (Freire et al, 2018) and fostering trust. A notable example is the Sloane Lab, which aims to facilitate access to the collections amassed in the eighteenth century by Hans Sloane (1660-1753) and the establishment of the British Museum. The Sloane Lab[1] ensures that voices of knowledge communities outside universities and heritage institutions can shape future digital collections. This is achieved through insights drawn from participatory design processes and contemporary understandings of cultural heritage collections, which emphasise the needs and requirements of decolonised and multivocal heritage (Humbel et al 2024). This aligns with the European Commission's 2021 Recommendation on a common European data space for cultural heritage (DS4CH)[2] and the ambitions for the European Collaborative Cloud for Cultural Heritage (ECCCH). For the ECCCH to succeed, it must prioritise solutions that support **multi-context representation**, foster trust and enable the **ethical and appropriate reuse** of CHDOs.

[1] https://sloanelab.org

[2] https://www.dataspace-culturalheritage.eu/en

INFINITY's approach

The goal of INFINITY is to create **a digital ecosystem as part of the ECCCH**, encompassing materials, models, methods and software components. To manage CHDOs subject to multi-perspective interpretations, the project introduces a technological shift by enhancing CHDO management systems (CMS). This includes supporting diverse perspectives and ensuring these viewpoints are interconnected on a large scale. This will be achieved through a **well-documented, open library of software components leveraging Semantic Web (SW) and Artificial Intelligence (AI)** technologies that support large-scale collaborative (meta)data enrichment of CHDOs, ensuring compliance with legal regulations and ethical principles:

1. Creating and deploying a **cross-sector ontology network** to support the CH domain. This involves reusing and extending established ontologies to capture, alongside data provenance information, multiple descriptions of CHDOs. These descriptions include their spatial-temporal and cultural-socio-political interpretations, reuse conditions, IPR and other associated rights, as well as the key actions, agents, scientific and professional value, methods, and significant changes that have affected their lifecycle and preservation status. This technology ensures the creation of high-quality digital twins of cultural heritage objects and can be leveraged to enhance the robustness of AI tools through explainability.
2. Developing and managing **multidimensional knowledge graphs** (mKG) that interconnect, represent, display, store and explore CHDOs from multiple perspectives. These mKGs will adhere to the FAIR principles and align with the cross-sector ontology network. This ensures that the ECCCH can fully achieve its potential by supporting diverse, context-sensitive interpretations of CHDOs, tracing key interactions in their lifecycle (e.g. curation and evolution workflows), and embedding legal and ethical practices into its framework by design
3. Incorporating **advanced AI-driven metadata enrichment and explainable tools** into existing CMS, including functionalities such as multilingual comprehension, geolocation alignment, manuscript transcription, and entity and concept extraction and linking. These tools will operate within rigorous legal and ethical frameworks, ensuring **robustness** through IPR encoding and verification, and ethical validation. This empowers a wide range of stakeholders - developers, researchers, educators, cultural institutions and policymakers - to engage responsibly with CHDOs, driving innovation while preserving the integrity and authenticity of cultural heritage.
4. Providing a **scalable infrastructure** to support large-scale crowdsourcing and citizen science campaigns. These campaigns will integrate AI functionalities to enable collaborative multi-context annotation enrichment of CHDOs, encoding the resulting descriptions as mKGs. The enriched data will be released as multidimensional, interconnected heritage data on the European Data Space for Cultural Heritage DS4CH via the ECCCH. This approach captures knowledge about CHDOs, including traditional cultural heritage knowledge at risk of loss, and connecting it with existing CHDO collections and contributing to the concrete realisation of digital commons.
5. Designing, deploying and experimenting with **intuitive and engaging interfaces** integrated into existing CMS to support users with varying levels of technical expertise. These interfaces will enable interaction with the multiple interpretations and descriptive contexts of CHDOs, such as annotation, browsing, visualisation and querying, alongside AI-based functionalities and associated explanation, verification, legal and ethical validation tools. By broadening and simplifying access to CHDO collections, this technology will foster significant scientific, societal and economic impacts.

The development of these **models, methods, tools and intuitive interfaces** follows a **user-centric approach**. It is grounded in, guided by and continuously intertwined with experiments and validations performed in **five real-world use cases** identified by cultural heritage stakeholders represented in the consortium, such as cultural institutions, cultural associations and collection owners. These use cases encompass large-scale, collaborative efforts to enrich metadata across various multimedia assets (including text, images, audio, video and 3D models) and represent CHDO **reuse scenarios in both commercial and non-commercial settings**, covering a wide range of legal and ethical considerations. The use cases will perform three *validation goals*, aligned with the three core issues identified in the CL2-2024-HERITAGE-ECCCH-01-03 call text:

1. **(Meta)data enrichment**
2. **Embedding IPR and professional value**
3. **Collaboration and business models**

All computational components (ontologies, knowledge graphs, and software) and accompanying documentation, guidelines and training materials will be shared through the **INFINITY digital ecosystem,** ensuring continuous and full alignment with the ECCCH data models and infrastructure. The software components will be released both as reusable plugin components (libraries) and REST interfaces for seamless integration with existing CMS. These outcomes will help establish the ECCCH as a pivotal platform for cultural heritage preservation and access, strengthening society's connection to history and expanding opportunities for research and creative exploration. Moreover, INFINITY is committed to **developing new business models** around the ECCCH by fostering partnerships with the cultural and creative industries. By utilising enriched metadata and the versatile representations provided by the cross-sector ontology network and mKGs, IPR rights of CHDOs produced and stored in the ECCCH will be recorded and traceable, enabling the project to open novel economic and sustainability opportunities.

Through these innovations, INFINITY addresses key needs in the field of cultural heritage:

- How can cultural institutions and other providers effectively and ethically represent and share CHDOs on collaborative platforms such as the ECCCH, balancing accessibility with cultural sensitivities?
- What strategies and supports can improve cultural institutions' workflows for creating and managing CHDOs to ensure responsible, transparent and appropriate reuse?
- How can collaborative digital environments enhance stakeholder collaboration in CHDO creation, management and reuse, fostering transparency and responsibility?
- How can CH heterogeneous collections be enriched and linked on a large scale using efficient AI and human-in-the-loop methods for high-quality research and exploitation?
- How can CH and creative professionals and researchers leverage distributed CHDO collections for social and cultural research, as well as for sustainable, ethical business opportunities?
- What methodologies and tools can assist researchers, professionals and the public in accessing and understanding the complexity of CHDOs while ensuring adherence to ethical standards?

The outcomes of INFINITY will significantly improve the workflows of various target groups accessing CH collections via the ECCCH for reuse in both commercial and non-commercial settings. The three stories provide a glimpse into the diversity of applications and impact areas enabled by the innovations INFINITY brings to the ECCCH.

Story 1. Pawel, a 24-year-old from Poznań, is struggling to read his grandfather's diary, handwritten in Polish and German. He brings it to a collection day for post-WWII memorabilia organized by heritage organisations contributing data to the ECCCH. Using the Transcribathon citizen science platform, enhanced with INFINITY's crowdsourcing, transcription, translation and geolocation functionalities, Pawel uncovers his grandfather's story as a forced labourer in Germany. He learns of a secret romance with a French woman and the child they had together. The diary's data, including photos of locations and individuals, is automatically released on the ECCCH and aligned with its schemas. This allows Pawel to discover related materials preserved by European heritage organisations, such as archival records documenting the fate of his newly found relatives. Armed with this information, Pawel is able to locate his relatives in France and visit the exact locations where his grandfather lived and worked. In parallel, a historian in Portugal discovers these new materials as a link to them is suggested by the content management system he is using to analyse government archival data. This helps him fill critical gaps in government archives for his thesis.

Story 2. Paolo, a digital content strategist working at Digital Paths - a consultancy supporting cultural and creative organisations - wants to use heritage collections in marketing campaigns for his clients. However, he struggles to understand reuse possibilities and is therefore forced to rely on "safe" datasets from stock imagery platforms like Unsplash or Shutterstock. While these sources offer certainty in terms of copyright compliance, they limit the diversity and originality of materials available. The ECCCH offers an alternative. Intuitive user interfaces connected to INFINITY's mKGs provide Paolo with clear information on ethical reuse principles and IRP compliance, including steps for reusing materials that are still under copyright. This enables Digital Paths to expand their portfolio of marketing and communication techniques with ethically sourced cultural collections, helping them attract new clients.

Story 3. Kristina, an architectural historian, is working on a research project to support the reconstruction of the Schleiz Castle in Thuringia. To do this, she needs access to historical imagery as photos, drawings and plans, documenting architectural changes in the castle since its construction. While such exist, they are scattered across heritage organisations, making them difficult to discover. Furthermore, limited metadata complicates spatio-temporal comparisons and alignments. AI-based segmentation, classification and spatio-temporal enrichment techniques introduced by INFINITY to the ECCCH allow Kristina to quickly locate all the relevant content. The enriched metadata enables 4D digital twins, helping her to visually assess parts of the castle and their transition over time and gaining historical evidence for constructions. This significantly simplifies the research process and enables Kristina's team to better support the restoration work.

INFINITY Six core objectives

- **O1 ALIGN: Connect and align CH metadata.** To transform disconnected heritage collections, their metadata and the scattered schemas and ontologies they are based on into a cohesive, scalable network of cultural heritage knowledge. This includes ensuring the consistent application of data and metadata models across various heritage collections, fostering continuous interoperability and accuracy. It also aims to support cross-domain representation and multi-context understanding of cultural heritage digital objects (CHDOs) sourced from distributed collections, while preserving their inherent meanings and respecting the multiplicity of viewpoints and audiences.
- **O2 CODIFY: Ensure the ethical use of CHDOs.** To establish principles and guidelines for the ethical use of CHDOs, integrating these principles into computational methods and tools that support the full lifecycle of CHDOs (selection, creation, description, management, discovery and reuse). This will facilitate appropriate reuse in all curation settings, from small to large collaborative scales. This objective focuses on the robust recording, tracking and embedding of scientific insights, diverse cultural and historical perspectives, end-user contributions, and intellectual property rights within the digital content management chain. It also aims to enhance the quality of content and metadata through acquisition, annotation and categorisation of CHDOs according to diverse contexts and from multiple viewpoints, both individually and in relation to other CHDOs. Importantly, it addresses complex ethical frameworks, acknowledging situations where reuse may not be appropriate and determining how such scenarios can be managed within a digital infrastructure.
- **O3 ENHANCE: Deploy Intelligent and semantic data enrichment tools.** To develop and deploy reusable software components, tailored and made available to the CH community through the ECCCH. These components will: 1) Semantically capture multi-contextual descriptions of CHDOs; 2) Implement mechanisms for continuous data updates to ensure consistency across collections; 3) Enhance interoperability among diverse metadata models, knowledge graphs, multilingual vocabularies, and ontologies; 4) Provide reliable, robust AI-assisted annotations. This objective also supports the broader integration with the DS4CH and ECCCH, ensuring seamless technical compatibility with various collection management systems used by European CHIs.
- **O4 INTEGRATE: Ensure consistency and integration with the ECCCH.** To enable CHIs to transition from local to global data providers, addressing issues around provenance, access control, security and rights-respecting reuse. This involves aligning heritage data structures from local data environments with a wider, shared data ecosystem, including the ECCCH and the DS4CH. Meeting this objective ensures enriched data is easily consumable and reusable by source systems while maintaining high quality. It also supports small and medium-sized cultural heritage institutions in aligning their systems and data with the ECCCH.
- **O5 INTERACT: Improve the interaction experience with CHDOs.** To move beyond tedious, database-driven interfaces and create fulfilling user experiences. This involves developing and experimenting with interfaces (through the use-case platforms) that improve user experience in contributing to and consuming CHDOs with a multi-context description approach. The aim is to facilitate the study of CHDOs and reuse of results by enabling the visual, interactive exploration of interlinked, multidimensional collections. It seeks to transform how CHDOs are perceived, experienced and accessed.
- **O6 ENGAGE: Drive adoption, sustainable business models, and technological shifts.** To facilitate the adoption and integration of INFINITY innovations into CHI systems and workflows while fostering sustainable business models and cross-sector collaboration. By ensuring compatibility with existing infrastructures, publishing open FSTP calls, providing training resources and hosting engagement activities like hackathons, this objective unlocks the potential of enriched cultural heritage data within the ECCCH and beyond.

Objectives, results and success indicators. Table 1.1 summarises the objectives and core results of INFINITY. The right-hand column lists measurable indicators that will be used, alongside formal evaluations within the associated WP tasks, to assess the successful achievement of each objective.

Table 1.1 Objectives and results ("#" stands for "number of")

O1: ALIGN - Connect and align Cultural Heritage collections (knowledge graphs)	
INFINITY results	**Success indicators**
R1: An ontology network resulting from the systematic review, extension and alignment of ontologies supporting CHDO multi-context representations, covering the CH, legal, ethical, procedural, spatio-temporal, socio-political, and trust/provenance domains	▪ Dimensions covered by the ontology network: CH, Perspectives, IPR and other rights, provenance, CHDO curation and management workflows, ethical principles, Space, Time, (historical, socio-political) narratives ▪ # of collections harmonized across at least 3 dimensions ≥ 10

R2: Technologies for harmonizing metadata and for multilingual, multimodal entity linking across collection knowledge graphs **R3:** Knowledge graphs of cross-collections alignments	▪ # links between collection knowledge graphs: in the order of thousands
O2: CODIFY - Ensuring the ethical use of Cultural Heritage Digital Objects	
INFINITY results **R4:** Guidelines for the ethical annotation and use of CHDO **R5:** Tools for supporting ethical annotation and use of CHDO (multi-context annotation, explanation, IPR management, content provenance tracking, curation and usage workflow tracking, bias detection, ethical validation, user consent management for AI-generated datasets) **R6:** Tool for user consent management for AI-generated datasets	**Success indicators** ▪ # of CHI consulted on adoption of guidelines: ≥ 10 ▪ # CHDO enriched with multi-context metadata (including IPR and provenance) ≥ 70% ▪ # ethical and legal validation checkpoints integrated into annotation/reuse workflows: ≥ 10 ▪ # datasets validated using the bias detection toolset: ≥ 10 ▪ Tracked data points with human-AI differentiation ≥ 10k ▪ # CMS platforms integrating workflow tracking tools: ≥ 10 ▪ # of consents managed through the tool: ≥ 500
O3: ENHANCE - Intelligent and semantic data enrichment tools plugged into CMS	
INFINITY results **R7:** Benchmark for validating CHDO collection management processes and tools **R8:** Data ingestion pipelines from text, images, audiovisuals, maps, archives **R9:** mKG infrastructure: deployment, query, evolution management; pipelines for mKG creation from structured data **R10:** Fine-tuned AI models for multilingual textual and concept understanding in multiple contexts **R11:** AI tools for multilingual entity and concept recognition (from text and images) **R12:** AI tools for geospatial data alignment and enrichment **R13:** AI tools for multilingual manuscript transcription **R14:** AI-assisted multi-context annotation tools **R15**: Extended Transcribathon toolset and crowdsourcing tools **R16:** Plug-in and web service architecture for integrating INFINITY components into existing platforms (e.g., dLibra, CLARIAH Media Suite)	**Success indicators** ▪ Benchmark validation reports from the CMSs supporting the five use cases + ≥ 2 additional CMSs ▪ # data formats successfully ingested: ≥ 5 (text, images, audiovisuals, maps, archives), accuracy ≥ 80% ▪ # mKG instances deployed in production: ≥ 5 ▪ # languages supported with understanding accuracy ≥ 85%: ≥ 5 ▪ # entities and concepts recognized with accuracy ≥ 85%: ≥ 1000 per dataset ▪ # geospatial entities correctly aligned across datasets (e.g., historical maps and modern geographic data): ≥ 50000 ▪ # manuscripts automatically transcribed with accuracy ≥ 85%: ≥ 10 ▪ # automated annotations suggested with ≥ 85% acceptance rate by human validators: ≥ 1,000. ▪ % CHDO enriched with semantic annotations in each use case collection(s) ≥ 90% ▪ # large-scale crowdsourcing campaigns completed: 3 (PL, NL, GER); # institutions involved in each campaign: ≥ 5, active participants ≥ 1000 ▪ # additional CMS enhanced through plug-ins or web services ≥ 2
O4: INTEGRATE - Ensure consistency and integration with the ECCCH	
INFINITY results **R17:** The INFINITY Rulebook (including guidelines and constraints for alignment with ECCCH and DS4CH) **R18:** Training material: INFINITY software ecosystem for CH research infrastructure **R19:** Integration with ECCCH and DS4CH metadata models (schema mapping and extension tools)	**Success indicators** ▪ % ecosystem components reviewed for compliance with the Rulebook by the Technical Board: 100% ▪ # representatives from INFINITY actively contributing to the ECHOES Integration TF ≥ 2 ▪ # formal feedback sessions with the ECHOES Integration TF confirming alignment ≥ 2 ▪ % alignment with the core ECCCH framework and DS4CH metadata model: 100% ▪ # schema mapping and extension tools developed and submitted for ECCCH and DS4CH ≥ 3
O5: INTERACT - Improving user experience in interacting with cultural heritage digital objects (CHDO)	

INFINITY results	**Success indicators**
Interfaces developed for the CMS supporting the use cases: **R19:** GUI for browsing/querying/visualizing mKG with context-specific visualizations: IPR, workflow, socio-political, geospatial, temporal, etc **R20:** GUIs to interact with AI tools offering manual and automatic options, providing explanations, and validation support **R21:** GUI for bias detection and ethical validation **R22:** GUI for interacting with workflow tracking tools (annotating, querying, visualizing) **R23:** Transcription interface: interactive tool for transcribing handwritten materials with AI-assisted suggestions **R24:** GUI for collaborative multi-perspective annotation enrichment and verification **R25:** Permission and rights management interface **R26:** Extended multilingual 4D viewer from mKG	▪ Percentage of target users actively using the GUIs after 6 months from deployment: ≥ 70% ▪ Percentage of users rating the GUIs as "intuitive" or "easy to use" in surveys: ≥ 85% ▪ Average satisfaction rating across all GUIs (e.g., through surveys): ≥ 85% ▪ Percentage of users encountering interface-related errors or failures: ≤ 5% ▪ Average system response time for querying or interactions: ≤ 2 seconds ▪ % of users satisfied with the clarity of bias detection results and validation workflows: ≥ 80%.
O6: ENGAGE – Adoption, sustainable business models and technological shift	
INFINITY results	**Success indicators**
R27: GitHub INFINITY project(s) collecting open-source licensed software, resources, documentation and guidelines by the project **R28:** Open call (Financial Support for Third Parties - FSTP): targeting creative industry (business models) and cultural institutions with their technology providers (early adopters) **R29:** Capacity building workshops/hackathons engaging stakeholders (INFINITY Stakeholder Network) **R30:** Five use cases addressing CHDO metadata enrichment, large scale crowdsourcing campaigns, AI-assisted annotation, data harmonization and alignment, multiple visualizations, IPR and curation workflow management, and ethical use **R31:** Open days targeting citizens	▪ # forks, downloads, and visits to GitHub: in the order of hundreds ▪ # proposals received per FSTP call: ≥ 50 ▪ # FSTP selected projects: 16 ▪ # attendees to each workshop/hackathon ≥ 40 ▪ % coverage of member states and associated countries in the Stakeholder Network ≥ 90% ▪ # users participating in use case testing: [CROWD]≥1000, [4DGRID]≥1000, [MANUSCRIPT]≥ 25, [AUDIOVISUAL]≥ 50, [ARCHIVE]≥100 ▪ # citizens outreached through Open Days ≥ 1000

1.1.2 Ambition (State of the Art)

Advancement in cross-collection interlinking at large scale (O1: ALIGN)

State of the art: Currently it is **very difficult to connect and align heterogeneous CHDO metadata** representations such that different collections may be browsed, queried or merged seamlessly to reflect the cultural policies of the EU, e.g. in the ECCCH (Capurro et al, 2024). The challenge of aligning metadata models, especially addressing the limitations of 1:1 mapping between classes and properties, is a key focus in KG research. Simple mappings often fail to capture the nuances of semantic relationships, as exemplified by SKOS relations like "broader" and "narrower," which do not distinguish degrees of broadness. The Ontology Alignment Evaluation Initiative[3] (OAEI) provides benchmarks and cutting-edge results ontology and schema mapping addressing also complex ontology alignments, beyond the 1-to-1 cases, however implementing these methods to support large scale applications is a challenge. A promising approach involves utilizing graph embeddings to represent and compare associations between classes and properties across diverse models. This technique goes beyond traditional ontology matching by leveraging the capabilities of graph embeddings, such as TransE, DistMult, and QuatE, for semantic similarity assessment (Hogan et al, 2021). This approach is exemplified in research on refining the SemOpenAlex concept ontology (Erten et al, 2024), where knowledge graph embeddings are used to predict missing SKOS relations, enhancing the semantic quality of the knowledge graph. Explainable graph neural networks (Agarwal et al, 2023) are typically employed to capture complex spatial dependencies and provide insights into the factors influencing predictions, helping solve

[3] https://oaei.ontologymatching.org/

classic ML problems like semantic drift.

INFINITY advancement: We will model the aggregate set of metadata properties and their values through embeddings generated by a fine-tuned language model, once pre-trained for general language understanding, it may be fine-tuned by public CHDO descriptions e.g. Europeana and Data Space for CH data. Open source LLMs such as Gemma[4], Llama[16] or Mistral[5] will be evaluated for this task. We will use **embeddings to capture both explicit and implicit semantic relationships** between different terms (Hou et al, 2024) used in the original CHDO metadata files and represent those relationships in a fine-grained model (not just 1-to-1) which captures multiple dimensions (different types of relationships) and the approximation of one term to another along any dimension. Our approach will leverage the recent results from OAEI[Fout! Bladwijzer niet gedefinieerd.] and will be evaluated against their benchmarks. Established ontologies developed in other projects will be reused as reference and possibly extended to address cross-domains representations (to create the INFINITY ontology network): provenance (e.g. PROV-O[6]), CH (e.g. EDM[7], CIDOC CRM[8], ArCo[9] (Carriero et al, 2021), PON[10] (de Berardinis et al, 2023), ECHOES[11]), IPR (e.g. RightStatements[12]), workflows and processes (DUL[13]), perspectives and narratives (SON[14]). This will enable a fine-grained shared understanding across the different underlying metadata models and vocabularies, such that contextualisation and the precise meaning of CHDO metadata and entities is not lost (Jiang et al, 2024). The INFINITY ontology network will enrich the DS4CH with an unprecedented set of interconnected cross-domains ontologies tailored to the needs of the CH domain.

Advancement in ethical and multi-perspective use of CHDO (O2: CODIFY)

State of the art: Explicit knowledge about relationships among general concepts can be externalized into a knowledge model. Public knowledge graphs like DBpedia and Wikidata provide freely accessible, machine-processable knowledge but prioritize a minimal subset of widely accepted, global knowledge. Current metadata extraction approaches, including semantic annotation (linking property values to knowledge graph concepts via URIs), overlook the need for **ethical and multi-perspective use of CHDOs**. To address this, a knowledge layer for CHDO annotation must account for multidimensionality, reflecting how concept associations change across spatial, temporal, socio-cultural, and political dimensions. Previous efforts, like (Fafalios, 2018), add single dimensions to knowledge graphs, modelling political leanings of politicians in TweetsKB, while recent advancement leverage the multidimensional nature of neural network embeddings for automated knowledge graph completion (AKGC) tasks (Lin et al., 2015).. However, **biased collections** and dominant cultural narratives remain a major challenge in CH (Zhitomirsky-Geffet, Kizhner et al., 2023), exacerbated by AI's tendency to replicate dominant features and produce limited explainable results (Osoba and Welser IV, 2017; Tribelli et al., 2024). To move forward, it is critical to ensure that **AI systems respect cultural sensitivities and avoid perpetuating biases** (European Commission). To support the ethical curation and reuse of CHDOs, CMS must integrate bias detection and IPR management tools. Bias detection frameworks such as Fairlearn (Bird et al., 2020), AIF360 (Bellamy et al., 2019), and TextAttack (Morris et al., 2020) provide robust fairness metrics and can uncover hidden biases in textual models. Advanced models like Hugging Face's BERT (Devlin et al., 2018; Liu et al., 2019) and LLMs like GPT-7[15] and Llama[16] enhance pattern recognition but require careful fine-tuning. For IPR management, tools like RightsStatements.org and ccREL enable ownership tracking, supported by standards such as PROV-O and platforms like DSpace and IIIF. However, effective integration demands a thorough understanding of these tools' workflows to prevent introducing additional biases.

INFINITY advancement: CHDOs often embody concepts interpreted differently across dimensions (e.g., social, cultural, political), suggesting that a multi-perspective representation can support their ethical and context-sensitive use and reuse. A systematic review of data ethics in relation to CHDOs will be conducted to analyse existing

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4 https://ai.google.dev/gemma

5 https://mistral.ai/

6 https://www.w3.org/TR/prov-o/

7 https://pro.europeana.eu/page/edm-documentation

8 http://www.cidoc-crm.org/

9 https://github.com/ICCD-MiBACT/ArCo

10 https://github.com/polifonia-project/ontology-network

11 https://www.echoes-eccch.eu/

12 https://rightsstatements.org/en/

13 http://www.ontologydesignpatterns.org/ont/dul/DUL.owl

14 https://github.com/spice-h2020/SON

15 https://openai.com/index/gpt-4/

16 https://www.llama.com/

approaches related to linked data and metadata annotation, given the limited research on this topic. The findings will inform guidelines for integrating advanced software components into metadata annotation workflows. We will test the hypothesis that **multidimensional knowledge graphs enable CHDOs to be interpreted from multiple perspectives**. Building on the state of the art (Schuetz, 2020), which currently lacks multidimensionality in KG creation and evolution workflows, we plan to use graph embeddings (e.g., via Graph Neural Networks or GNNs) for representation learning (Gesese et al., 2021). Our approach will extend this by incorporating multidimensionality through architectures like MGCN (Ma et al., 2019) for multi-layer graph understanding. However, user testing of these methods remains limited, as does an understanding of how they intersect with recent AI, NLP advancements, and the needs of CHI professionals. As we explore the use of multidimensional knowledge graphs, it is important to recognise that bias in cultural heritage cannot and should not be erased but must be identified and mitigated to ensure it is not perpetuated (Havens et al., 2023). To address this, we design and implement modular software components for bias detection and IPR management that integrate into CMS via REST APIs, e.g. combining tools like Hugging Face models and LLMs with preconfigured workflows tailored to cultural heritage applications, for advanced analysis of textual and visual datasets (bias detection), and encapsulating standards like PROV-O and RightsStatements.org within scalable microservices, ensuring efficient management of rights and provenance across diverse digital ecosystems (IPR). We are confident that this modular design will improve adaptability, reduce integration barriers, and enable tailored solutions across CMS.

Advancement in semantic enrichment of CHDO metadata (O3: ENHANCE)

State of the art: Current data enrichment tools in the CH domain are capable and validated only for specific datasets at small scale. Most AI approaches are currently trained with contemporary datasets while historical training datasets are rather small (Münster et al. 2024). The challenge is being able to deal with “in the wild” data at large scale, as well as understanding and integrating the socio-cultural aspects related to them (Kumar et al, 2024). A related challenge is that, unlike in fields such as medical AI (Salahuddin et al. 2022) (Salahuddin et al. 2022) an optimised heuristic interpretation is insufficient for historical sources and their singularity (Fiorucci et al. 2020). Current approaches to employ AI in heritage validate their results only for some examples (Farella et al. 2021). The challenge is to include mechanisms to validate results at large scale and support de-biasing. Another challenge is integration, as any metadata model, vocabulary or knowledge graph needs to be included in a metadata workflow (e.g. (Patel et al, 2024) or (Bu et al, 2025). Against the background of the heterogeneity of data infrastructures for European CH the challenge is to integrate tools and data with a plurality of extant data ecosystems (Münster 2023, Storeide, George et al. 2023). This requires a system architecture capable to connect services to various data infrastructures in a flexible and versatile way.

INFINITY advancement: INFINITY introduces an interoperability layer by developing a solution that integrates metadata interoperability and multidimensional knowledge graphs. This enables intelligent, contextualised CHDO metadata enrichment, with a focus on extending existing tools in use at the CHI. INFINITY will develop flexible and simple APIs to enable wrapping existing AI tools for knowledge extraction (such as entity recognition and linking, geospatial alignment, manuscript transcription, etc.) and assisted annotation enrichment, into plugins for existing CMS. Although relying on state-of-the-art tools, the innovation lies in extending these tools to maximise their customisation to the need of the CH domain, both in terms of knowledge specialisation on the domain and in easiness of integration in existing CMS. The inclusion of full-scale cross-validation for AI-based analysis of CHDO and associated contents (4DGRID, CROWD) - using mixed-methods (Rao et al., 2008) and human-in-the-loop approaches – will help account for the uniqueness of the domain and addresses the challenges posed by the scarcity of historical sources, which may complicate training language models. INFINITY approach also addresses KG evolution, often neglected, which is crucial for maintaining KG relevance and accuracy over time. KG evolution is essential for addressing semantic drifting, the changes in the meaning of concepts over time. This can be achieved by updating embedding models through various methods (e.g., incremental, temporal, etc) and allowing them to adapt to the evolving nature of knowledge (Cheng et al, 2024). Incorporating data from dynamic sources like public KG (e.g., Wikidata) and public web data allows to capture these shifts in meaning and ensures the KG stays current. This continuous evolution allows KGs to remain valuable resources for various applications, including semantic search, question answering, and recommendation systems.

Advancement in the implementation of the ECCCH (O4: INTEGRATE)

State of the art: For CHI’s, guaranteeing data integrity and synchronicity still remains a big challenge due to the heterogeneity of systems and protocols. OAI-PMH[17] (Open Archives Initiative Protocol for Metadata Harvesting),

[17] https://www.openarchives.org/pmh/

a widely used standard for metadata harvesting, supports one-directional data exchange. Emerging or alternative frameworks and technologies, such as the IIIF Change Discovery API[18] (International Image Interoperability Framework), Solid[19] or COAR Notify[20] (Confederation of Open Access Repositories), offer advanced capabilities for discovery, interlinking or interoperability. At this stage of the ECCCH implementation project, ECHOES, exact specifications of mechanisms for integration with the cloud platform have not yet been determined. However, a key principle of the ECCCH is to ensure the interoperability of diverse systems contributing to the platform. It is recommended to use common data formats (e.g., RDF), open community data and metadata models, open-access REST APIs, and adaptable data models. This approach enables data consistency and seamless integration with the ECCCH, which includes synchronisation of the enriched CHDO metadata between the cloud and CHIs' local metadata management systems. Implementation may involve well-designed notification workflows, ensuring that CHIs are informed about the accurate and consistent representation of their CHDOs throughout the entire metadata lifecycle. However, this is not enough — enriched metadata needs to be accompanied by paradata (information about how the metadata was collected), provenance data (where the data came from and who is responsible for it, including computer-generated enrichments), and usage data (the rights of other users or systems to reuse metadata and/or enrichments and for what purposes) — all of which should also be maintained consistently across cloud and local systems. All this information must be maintained consistently across cloud and local systems. Rapid integration with ECCCH will therefore be crucial for CHIs. However, current market tools, such as CMS software, lack direct compatibility with ECCCH because it is still in development.

INFINITY advancement: We will develop a strategic framework for data governance within INFINITY, a vital part of our solution which ensures data is managed responsibly, ethically and in line with CHI, national and European legal and ethical regulations. It aims for accountability, transparency, regulatory compliance (e.g. GDPR) and data quality. For example, our data management will not just be FAIR (findable, accessible, interoperable and reusable) (Wilkinson et al. 2016) but will also CARE (Carroll et al. 2020) about power differentials and historical contexts of the creation of CHDOs and what they represent to different social and ethnic groups. The CARE principles for indigenous data governance ensure that data from indigenous groups are used in ways grounded in indigenous worldviews and respect indigenous innovation and self-determination. The multi-contextual aspect of our metadata enrichment approach makes indigenous ethnic groups just one of many possible dimensions of interpretation of CHDOs, meaning we can generalise such data governance principles further to preserve and respect all possible other historical, social, cultural etc. perspectives on the CHDOs. Our tools will adhere to ECHOES and DS4CH guidelines for data and service integration and synchronization. Developing a plugin solution for the dLibra CMS platform that will enable efficient integration with the aforementioned platforms addresses the need to facilitate seamless data exchange and will make it easier to populate them with relevant data. It will also serve as a foundation for creating similar solutions for other CMS systems, promoting broader integration possibilities in the cultural heritage domain across Europe. The development process will emphasize the use of open standards and interoperable formats, ensuring that software development aligns with FAIR (Barker et al., 2022) principles.

Advancement in interacting with CHDO (O5: INTERACT)

State of the art: Building upon these challenges, INFINITY addresses the limitations of current SOTA by developing a strategic framework for data governance that not only adheres to the requirements of ECHOES but also advances the field with innovative principles and tools. Our framework ensures data is managed responsibly, ethically, and in compliance with CHI, national, and European legal and ethical regulations, such as GDPR. While existing methodologies emphasise **FAIR** (findable, accessible, interoperable, reusable) principles (Wilkinson et al., 2016), we extend this by embedding **CARE** principles (Carroll et al., 2020) into our approach. This ensures data governance respects power differentials, acknowledges historical contexts, and supports self-determination, particularly for indigenous and marginalised groups. Considering the use of AI systems for enrichment, adhering to both FAIR and CARE principles is essential to ensure trustworthiness and accountability. Furthermore, unlike the limited capabilities of current frameworks to **synchronise enriched metadata back into CHIs' systems**, the multi-contextual metadata enrichment approach in INFINITY integrates paradata, provenance data, and usage data, ensuring these are preserved across both cloud and local systems. Our tools will adhere to ECHOES and DS4CH guidelines for data and service integration and synchronisation. Developing a plugin solution for the widely used open source dLibra CMS platform that will enable efficient integration with the aforementioned platforms addresses the need to facilitate seamless data exchange and will make it easier to populate them with relevant data. It will also serve as a baseline for creating similar solutions for other CMS systems, promoting broader integration possibilities in the cultural heritage domain

[18] https://iiif.io/api/discovery/1.0/
[19] https://solidproject.org
[20] https://coar-notify.net/

across Europe.

INFINITY advancement: To anticipate changes in the future, a modular adaptability of technological frameworks is envisioned, i.e. providing APIs to integrate specific functionalities also in other applications or the capability to integrate different viewer or HCI stacks in the applications. Thus ensures the widespread and sustainable use of INFINITY results. Both *findability and accessibility* of collections of CHDOs will be extended by making use of the multi-contextual representations to include exploration of objects and their metadata along multiple dimensions and therefore enabling multimodal access to data. Beside spatial and temporal dimensions (i.e. geomaps for spatial, timelines for temporal), INFINITY will explore means to intuitively give users access to CHDO exploration along arbitrary dimensions that may or may not explicitly identifiable, i.e. both labelled and unlabelled historical, social, cultural, ethnic etc. Analyses of CHDO metadata will support additional visual analytics based on metadata values and with a focus on generic use for multiple user communities, e.g. geomap of all locations present in a selected collection or keyword graphs which represent co-occurrences of keywords in the collection metadata. Regarding *stability and ease of use* the interfaces used in INFINITY are already at high TRL and with an extensive track record of development and maintenance as well as sustained plannings to ensure their sustainability also after INFINITY.

Advancement in the repurposing and valorisation of CHDO through INFINITY technology (O6: ENGAGE)

State of the art: CHI face significant challenges in adopting advanced digital technologies, despite their potential to transform the management and reuse of cultural heritage data. Many CHIs operate within fragmented infrastructures, relying on disparate systems that lack interoperability, making it difficult to integrate tools like those envisioned by the European Collaborative Cloud for Cultural Heritage (ECCCH). Limited resources, particularly in smaller institutions, exacerbate the difficulty of implementing scalable and sustainable solutions. Existing training programs often fail to equip users with the skills needed to fully leverage advanced technologies, while opportunities for collaborative learning and innovation remain scarce. Furthermore, the absence of structured business models hinders the long-term sustainability of these advancements, restricting their potential to foster creativity and economic growth (Martinelli 2022). Ethical concerns, including intellectual property rights and cultural sensitivity, are not consistently addressed in current systems, leaving gaps in responsible data management.

INFINITY advancement: The **adoption and integration** of INFINITY technologies into CHI's existing IT infrastructure and work processes will be facilitated, as well as integration in European data spaces. This will be ensured through the use cases which will work closely with the technology adopters to design technology and tools which not only meets their technical requirements but also follow an **iterative human-centric design methodology** which ensures that they also address user needs and meet user expectations through continuous testing, refinement and collaboration. To unlock the full potential of cultural data integration through the ECCCH and our multi-contextual CHDOs, CHIs and other relevant stakeholders will receive training and tutorials, learn how to understand and use the multi-contextual metadata representations and participate in on-site or online activities where they discover the value of their and others' CHDOs via the ECCCH and our enriched metadata.

1.1.3 Technological and societal relevance

Table 1.1.3a lists the Key Exploitable Results of the INFINITY project and the Technology Readiness Level at the beginning of the project ("Start") and target by the end of the project (Planned), for each result. The second column indicates the WP(s) that mostly contribute to the development of the technological artefact. The third column indicate which project's results (as listed in Table 1.1) is part of each KER. Table 2.4 (Section 2) provides additional information about the expected impact of INFINITY KERs.

Table 1.1.3a Technical Innovation in INFINITY - Key Exploitable Results (KER)

KER	WPs	Result	Technology / Component / Methodology	Innovation in INFINITY	Start	Planned
1	WP8 WP9 WP10	R1 R3 R19	Ontology network and mapping schemas. Knowledge graphs of cross-collections alignments.	Cross-sector alignments specialized for the CH domain: legal, ethical, procedural, provenance/trust, time/space, etc. Ontology extensions. Multi-context representation Compliance with ECCCH and DS4CH schemas	TRL6	TRL8

2	WP9 WP10	R2 R11	Multilingual, multimodal (text, images) cross-collection entity and concept recognition/linking and metadata alignment	Multilinguality, specialization in the CH domain. Easy integration with CMS (plugins/web services)	TRL4	TRL7
3	WP6-7 WP9-10	R13	Multilingual AI-aided manuscript transcription	Easy integration with CMS (plugins/web services)	TRL6	TRL7
4	WP6-7 WP9-10	R12	AI tools for geolocalisation of images, data synchronization with authority data providers, toponym extraction	Easy integration with CMS (plugins/web services). Validation for large scale datasets, integration with mKGs	TRL6	TRL7
5	WP6 WP8-9-10	R5 R14	AI-assisted annotation tools supporting multi-context representation of CHDO	Easy integration with CMS (plugins/web services). Multi-context representation	TRL3	TRL7
6	WP8-9-10	R10	(Large) Language Models tuned to learn multilingual text and concept embeddings and fine-tuning pipeline	Coverage of historical and contemporary document corpora Multi-context layer to create mKG	TRL6	TRL7
7	WP8-9-10	R5	Ethical validation and bias detection tools	Customization for CHDO curation and reuse workflow. Easy integration with CMS (plugins/web services).	TRL7	TRL8
8	WP8-9-10	R5 R6	Content provenance tracking tools User consent management for AI-generated datasets.	Customization for CHDO curation and reuse workflow. Easy integration with CMS (plugins/web services).	TRL6	TRL8
9	WP8-9-10	R5	IPR tools to encode and manage ownership, usage rights and reusability	Customization for CHDO curation and reuse workflow. Easy integration with CMS (plugins/web services).	TRL5	TRL7
10	WP8-9-10	R9	mKG data model and software for creation, deployment, querying, evolution management	Multidimensionality in KG Large scale support for deployment, querying, update	TRL5	TRL7
11	WP6-7	R19	Web interfaces to edit, browse, query, and visualise multidimensional KGs	Customization for CH CMS	TRL5	TRL8
12	WP6-7	R5 R20	GUIs to interact with AI tools offering manual and automatic options, providing explanations, and validation support	Customization for CH CMS	TRL5	TRL8
13	WP8-9-10	R5	Tools for embedding and tracking the lifecycle of CHDOs as mKG: skills, methods, preservation status, history of changes	Customization for CHDO curation and reuse workflow. Easy integration with CMS (plugins/web services).	TRL6	TRL8
14	WP6 WP7	R23	Transcription interface: interactive tool for transcribing handwritten materials with AI-assisted suggestions	Customization for CH CMS	TRL6	TRL8
15	WP6-7 WP9-10	R26	4D multilingual visualization tools (geospatial and temporal layering of data)	Extension of mobile and desktop viewers for visualization from mKG	TRL5	TRL8
16	WP6 WP7	R25	Permission and rights management interface	Customization for CH CMS	TRL7	TRL8

17	WP5-6-7 WP9	R15	Extended Transcribathon toolset and crowdsourcing tools	support for large-scale citizen science and crowdsourcing campaigns	TRL6	TRL8
18	WP6-7	R24	GUI for collaborative multi-perspective annotation enrichment and verification	Customization for CH CMS	TRL6	TRL8
19	WP9 WP10	R16	Plug-in and web service architecture for integrating INFINITY components into existing platforms (e.g., dLibra, CLARIAH Media Suite)	APIs for smooth integration of AI and service components into existing CMS	TRL7	TRL8
20	WP8 WP9	R4	Guidelines for the ethical annotation and use of CHDO	Significant reform in the integration of ethics within CHDO management	TRL2	TRL7
21	WP6-7	R22	GUI for encoding, querying and visualizing lifecycle history of CHDO	Customization for CH CMS	TRL6	TRL8
22	WP6 WP7	R21	GUI for bias detection and ethical validation	Customization for CH CMS	TRL7	TRL8
23	WP4 WP11 WP12 WP13	R7 R17 R18	Benchmark for validating the capability and maturity level of CHDO collection management processes and tools. Guidelines and training material for reusing INFINITY ecosystem software components	Multi-context representation Sustainable and ethical reuse Alignment with ECCCH and DS4CH	TRL2	TRL7

Table 1.1.3b Societal Readiness Levels in Infinity[21]

Solution	Innovation in INFINITY	SRL start	SRL end
24. Multi-perspective data governance principles	Inspired by individual efforts to reflect and represent specific contexts in data governance (e.g. CARE for indigenous groups), we will leverage the multi-contextual aspect of our enriched metadata to generalise data governance principles to preserve and respect all relevant perspectives on the contextual interpretation of CHDO.	4	7
25. Significant reform in the integration of ethics within CHDO management.	In response to the evolving landscape of digital ethics, our approach to digital ethics is not merely an adjunct but a core component of the creation, maintenance, and deployment of digital infrastructures for CHDO. Our frameworks will ensure assets are handled in a way that respects cultural significance, promotes inclusivity, and protects against the misuse of digital heritage, emphasising a holistic and ethical stewardship throughout the lifecycle of digital artifacts, addressing critical concerns such as consent, transparency, and cultural sensitivity.	3	6

#§PRJ-OBJ-PO§#

1.2 Methodology #@CON-MET-CM@# #@COM-PLE-CP@#

This section outlines the overall concept of the INFINITY project (Section 1.2.1), followed by six methodological principles underpinning the project design (section 1.2.2) , and the position of the five use cases (1.2.3) in the overall project's execution. This is followed by sections highlighting related projects, Interdisciplinarity, SSH Integration, Gender dimensions, Open science and finally Data Management.

[21] SRL is a way of assessing the level of societal adaptation https://innovationsfonden.dk/sites/default/files/2019-03/societal_readiness_levels_-_srl.pdf

1.2.1 Concept: The INFINITY ecosystem as part of the ECCCH

The core concept of the INFINITY methodology involves creating a decentralized, open-source digital ecosystem of digital tools, methods and intuitive interfaces. This ecosystem is crafted by an interdisciplinary team from computer science, social sciences, and humanities. The success of the ecosystem relies heavily on public and societal engagement, achieved through **use cases and the FSTP program**. Alongside these elements, the project will deploy effective strategies for dissemination, exploitation, and communication, detailed further in Sections 2 and 3.

INFINITY will set a new standard for integrating multimodal and multi-contextual metadata through multidimensional knowledge graphs (mKG), enhancing the Digital Commons with a rich, semantically enriched digital representation of cultural heritage. The project will utilize widely adopted tools in the Cultural Heritage and Creative Industries such as the CMS that allows CHI to manage, organize and document their collections, validating and integrating plugins to build up a toolkit for different types of CHDO as part of a **modular, extensible and evolutionary data lifecycle** supporting digital continuum by **tracing all interactions** with CHDO and related data. The INFINITY use cases (See 1.2.3) are based on well-established concepts and proven demands and therefore will reach a large number of users and equip them to interact with, manipulate and enrich digital twins, leading to new co-created scientific knowledge. Web interfaces will support data exploration and visualization: 1) visualization of the generated metadata enrichments of CHDO with version control, discussion/explanation, metadata quality rating, review/correction/approval/reject of metadata and AI-generated content, handling user consent, and bias detection, and displaying confidence score for AI-generated data, 2) a mKG browser to view and update mKG, 3) a 4D viewer for spatio-temporal exploration of annotated collections, 4) a permissions and rights management interface for CHDO, including highlights of AI-generated content, flags for content in need of legal/ethical review/action, possibility to request permission to reuse, etc. Other Web based GUIs will allow users to explore the outputs of the various technical components outlined above, e.g. in the interests of explainable AI, including the element of responsible and ethical digitality as an important building block towards a **sustainable and ethical approach of cultural heritage**. INFINITY involves key players from ECHOES, DS4CH, EIT Culture and Creativity, EOSC, and Research and Academic institutions, maximizing synergies and outreach, and leveraging resources such as data, computing power and domain knowledge available in the respective communities and connecting the **digital, sustainable, and innovative** aspects of cultural heritage.

Figure 1.2.1 The INFINITY digital ecosystem as part of the ECCCH

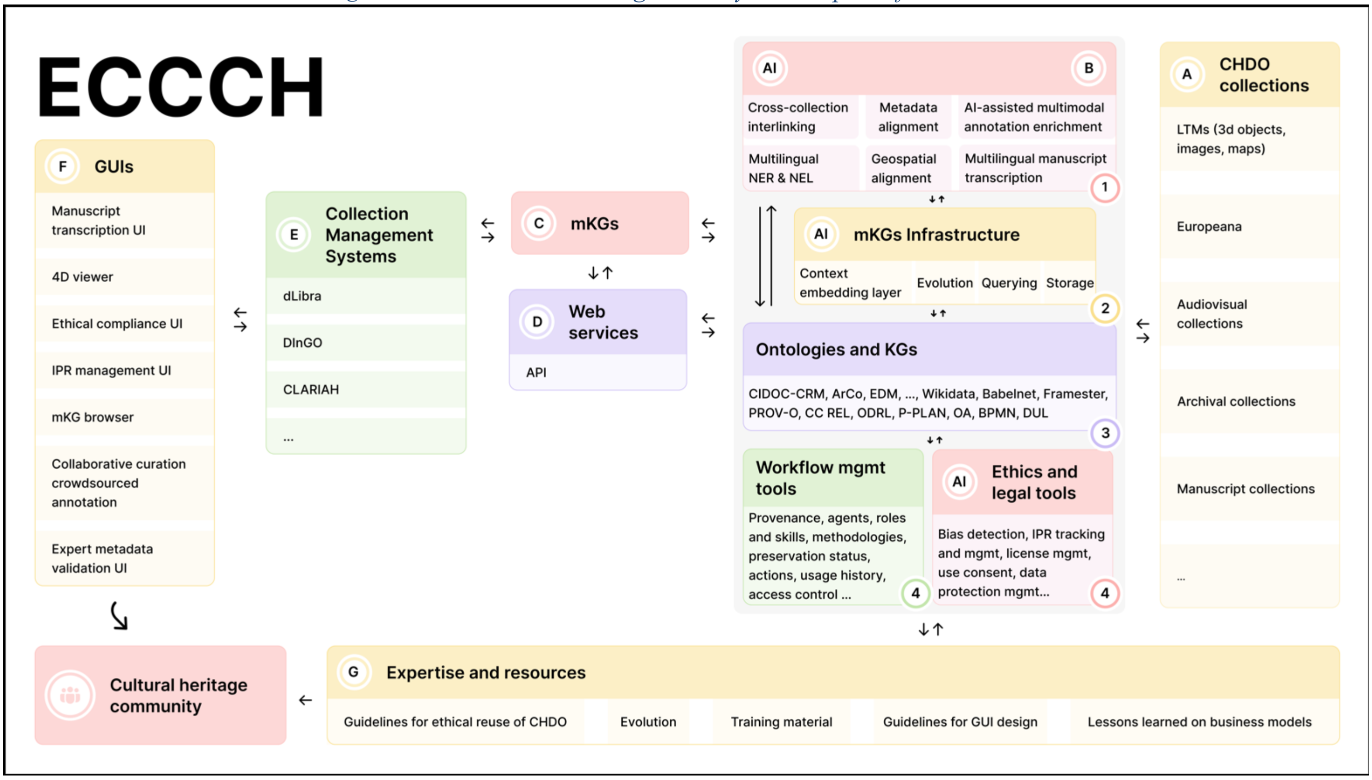


Figure 1.2.1 depicts the **INFINITY digital ecosystem, as part of the European Collaborative Cloud for Cultural Heritage** highlighting its main components. The software components are made available to the CH community with a “plug and play” approach to foster innovation by enabling easy integration with existing collection management systems (CMS). The picture can be read starting from the right-hand side, where heterogeneous CH digital collections are represented within a grey box (A). These sources are the protagonists of INFINITY and represent the state of play at the beginning of the project. Many of them, apart from notable examples, are still unavailable through the

DS4CH making it difficult to discover, access, explore, and reuse them. A key ambition of INFINITY is to maximize their accessibility through the ECCCH and systematize its smooth integration and synchronization with existing CH infrastructure. From (A) we move to the centre of the picture, where a set of services are visualized (B). These services concern AI methods for processing CH sources and annotating them with multi-dimensional knowledge graphs (mKG)(C). They include multimodal metadata enrichment of CHDO, multilingual entity recognition and linking (to mKG), cross-collection interlinking, metadata alignment, geo-spatial alignment (map historical locations to present day coordinates), and multilingual manuscript transcription (B-1). State of the art libraries and tools will be identified, evaluated and selected for reuse and possibly extended to address the specific needs emerging from the analysis of use case requirements. A mKG (C) is a KG augmented with multi-context interpretation of entities and concepts supporting the annotation and fruition of CHDO from diverse perspectives (cultural, historical, political, temporal, spatial, legal, etc.). These contexts are extracted by means of fine-tuned large language models that produce multilingual text and concept embeddings, creating a context embedding layer capturing changes in meaning and association of lexical terms and concepts (e.g. named entities) along multiple dimensions such as time, space, socio-cultural group, political bias etc. These models will rely on ingestion pipelines for training (from public Web data, social media, CH metadata such as Europeana, archives, etc.) and be extended with RAG (retrieval augmented generation) to include online reference corpora and contemporary documents (e.g. news articles, cultural heritage publications) (B-2). The mKG (C) can be fed back into the service layer (B) to perform additional processing such as classification, enrichment and interlinking. The results can further evolve CH knowledge graphs (C). A core component of the INFINITY ecosystem is a reusable infrastructure for storing, querying, and managing the evolution of mKG, at large scale (B-2). This will be developed as extensions of existing implementations for (unidimensional) knowledge graphs. Scalability and robustness of this infrastructure will be a crucial requirement of our use cases, especially [CROWD] that will realize an extended Transcribathon toolset and crowdsourcing tools to support large scale citizen science campaigns. In addition to knowledge extraction from CH sources and mKG management, a suite of services is developed to support the lifecycle of CHDO, ensuring their appropriate reuse from legal, historical, and ethical perspectives (B-4). IPR information (licenses, copyright conditions, etc.) will be recorded as part of the CHDO mKG descriptions and used as reference knowledge for tools that help end-users verifying compliance of their planned exploitation of CHDO (in commercial as well as non-commercial settings) with legal status and conditions (IPR tracking and management).

INFINITY places emphasis on making ethical principles natively embedded into processes supporting CHDO lifecycle. This is substantiated by releasing libraries and tools for bias detection (biased phrasing or culturally inappropriate terms, detection of imbalance of metadata attributes, potential data privacy infringement, etc.). Additionally, key workflow actions and produced content reflecting the biography of CHDO (agents and their roles, preservation status, main changes, usage history) will be recorded in their associated mKG descriptions along with providing access control mechanisms, to support appropriate reuse of CHDO. All software components will comply with a plug-in and web service architecture (D) to ensure the smooth integration of ecosystem components into existing platforms (E) (e.g. dLibra/DInGO, CLARIAH Media Suite). Customized GUIs will be designed and implemented as part of the specific CMS platforms (F) to support easy interaction with CHDO, their metadata and the processes enabled by INFINITY services. The resulting GUIs will be validated in the context of the INFINITY use cases and the results and experience gained will contribute to provide the CH community with reusable guidelines and designs for an effective and engaging user experience through intuitive and user-friendly interfaces for seamless interaction with CHDO throughout their lifecycle. In addition to its software components, INFINITY will provide a comprehensive suite of materials and resources as part of its digital ecosystem (G). These resources aim to foster expertise within the CH community and facilitate the adoption and exploitation of the project's innovative solutions. They include: a benchmark for evaluating CMS platforms against core capability goals: i) annotation enrichment, embedding of IPR and professional value, and fostering collaboration and sustainable business models; ii) Ethical guidelines for the responsible reuse of CHDO; iii) Best practices for GUI design, tailored to improve user interaction with CHDO; iv) Training materials to support the technical integration of INFINITY components within existing systems; v) Documentation of lessons learned from real-world experiments involving collaboration and business models that leverage CHDO in both commercial and non-commercial contexts. This multifaceted approach ensures that INFINITY not only provides technical tools but also empowers the CH community with the knowledge and resources needed to maximize their impact and sustainability.

1.2.2 Methodological principles

Six principles will underpin the methodological approach of INFINITY: decentralisation, robustness of AI tools, open collaboration, public and societal engagement, and interdisciplinarity and iterative design.

Decentralized approach. The INFINITY digital ecosystem adopts a decentralized approach, consistent with the principles of the ECCCH. This approach ensures scalability and sustainability, critical for success in the highly diverse Cultural Heritage (CH) sector, which encompasses various disciplines, methodologies, and data formats. While this diversity offers valuable opportunities for research, management, and the reuse of CHDO, it remains underutilized due to fragmented tools, formats, and practices, as well as the challenges of handling large-scale distributed knowledge. A centralized solution would be impractical, achieving limited adoption while requiring significant resources and being prone to obsolescence. In contrast, decentralization means to develop common interfaces that allow cultural heritage institutions to overcome local limitations, contributing to and benefiting from shared knowledge and tools. Concretely, this concept substantiates in pursuing a strong interdisciplinary approach, ensuring the system's adaptability and broad applicability.

Robustness of AI tools. INFINITY will reuse and implement AI tools (T8.4,T9.4, WP6). The respect of ethical principles and legal regulations is a driver of the project, therefore ensuring robustness of the technology that the project delivers are paramount and a key driver of the project's methodology. State-of-the-art advancements in explainable AI (XAI) to ensure that all AI-based methodologies integrated into our components and tools meet rigorous standards of robustness, scalability, and trustworthiness will be leveraged. This will involve ensuring transparency in model decision-making processes, promoting accountability, and mitigating biases, thus enhancing the reliability of AI systems. Our strategies will adhere to relevant regulatory frameworks, including the EU AI Act and the EU Data Services Act, which emphasize fairness, transparency, and the prevention of discriminatory practices. By embedding these ethical principles and robust interpretability mechanisms, we aim to facilitate the responsible and compliant deployment of AI technologies within a regulatory context. The INFINITY methodology is designed to maximise technical robustness, accuracy and reproducibility of the AI tools that it delivers. All tools will be designed and deployed to be able to provide a suitable explanation of their decision-making processes, by integrating state-of-the-art tools for AI explainability. The workplan (Section 3) includes WP8, dedicated to elaborating the formal requirements and design of the software components of the INFINITY digital ecosystem, which is centred on an ethic-driven approach. It is indeed part of the innovation that INFINITY aims to achieve to provide guidelines and recommendations for natively embedding ethical principles in software systems (T8.2). All AI tools that will be developed (T8.4, T9.4, T10.4) and integrated into CHDO collection management systems (CMS) (WP6, WP7) will be supported by tools for ethical and legal validation. More specifically T8.7 and T9.7 are dedicated to engineer as easy-to-integrate plugins, state of the art tools for addressing user consent management, provenance tracking, bias detection, explainability, and intellectual property rights (IPR) management. These tools will be adapted to meet the specialised requirements of CHDO curation, ensuring that annotations remain fair and inclusive. All use cases will integrate these functionalities (WP6) in their CMS and support them with GUIs in association with the AI-based services that they will offer (T8.4, T9.4). Bias detection tools combined with the multi-context representation of CHDO through multidimensional KG (T8.5, T9.5) will ensure social robustness of INFINITY tools.

Open collaboration. A comprehensive Financial Support to Third Parties (FSTP) programme (T15.4 and 16.4) will be launched, encouraging external stakeholders to experiment with and adopt the project outcomes. This includes one open call with two distinct strands; one aimed at the creative industry to explore new business models using data enriched by INFINITY, and second call targeting cultural institutions to integrate INFINITY innovations into their processes and collection management systems and document their experiences. We expect to fund a total of 14 projects. The way the calls are managed is outlined in a dedicated Annex.

Public and societal engagement. In INFINITY we assume that an effective promotion and use of Cultural Heritage through cutting edge computing technologies can have a significant societal, cultural and economic impact. To test this assumption, we plan several initiatives aiming at engaging citizens as well as professionals of the Cultural and Creative public and private sectors (Section 2.2.1): sixteen Open Days (T14.1) targeting citizens to showcase demos and collect feedback. These will be connected to existing science communication initiatives e.g. European Citizen Science Platform and national campaigns. Citizens will be directly involved as main participants of the [CROWD] and [4DGRID] use cases. Three crowdsourcing and citizen science campaigns organized by [CROWD] and the validation campaigns of [4DGRID] not only will involve citizens in directly contributing to the project development but will make them aware of innovative computational tools available to them for exploring Cultural Heritage knowledge through intuitive and engaging interaction. Events dedicated to the INFINITY Stakeholder Network will engage professionals to foster adoption and awareness: four capacity building workshops (T14.2,15.2,T16.2) are planned during the project lifetime.

Interdisciplinarity; integration of cultural heritage, humanities, and computational methods is needed to address data life cycle challenges. According to the Social Construction of Technology (SCOT) theory[22], the design outcome of technology is not solely based on technical requirements but is deeply influenced by the surrounding sociocultural context, norms, values, and power dynamics. Hence, INFINITY adopts a participatory approach to identify specific

[22] Bijker, W. (2015). Social Construction of Technology. In J. D. Wright (Ed.), International Encyclopedia of the Social & Behavioral Sciences, 2nd edition (pp. 135-140). Elsevier Science.

scenarios and extract the requirements for routing adoption. The sectors that could benefit from the project outcomes will inform the knowledge models used to classify CHDO and collections and will guide the design of the technical solutions. A formal study of workflows and activities related to creation and management of CHDO, most adopted ontologies for their description, ethical requirements, IPR, licenses and reuse practices, and business models based on repurposing of CHDO will provide a comprehensive context wherein to validate our requirements and test our solutions. For example, the identification and cataloguing of digital assets is a crucial process for preservation and valorisation of cultural heritage. However, the effort required to identify, annotate, collect, and curate these objects is significant and constitutes a major investment for researchers and institutions. INFINITY relies on (and contributes to) areas of the cultural heritage sector, including: research in social science and humanities, crowdsourcing and citizen science, tourism, education, management, and creative business exploitation. While these areas may have different established methodologies and research interests, they may also benefit from being informed by their mutual experience and findings. However, this is very hard to achieve with existing technologies and practices. INFINITY addresses this matter by developing five use cases to validate its computational solutions as applied in real-world scenarios defined by the main actors involved in cultural heritage. In the INFINITY vision, a *next generation of knowledge graphs* (KG) relying on a neuro-symbolic approach i.e. *multi-dimensional KGs* (mKG), provide a platform wherein to handle the CHDO life cycle. Therefore, a specific technical expertise in Semantic Web technologies and Artificial Intelligence is required to handle issues related to data contextualization, interlinking, interoperability, provenance, trustworthiness and sharing policies. To empower the heritage handled and produced by the project at scale, **AI methods for assisting in CHDO annotations (e.g. named entity recognition, entity linking and knowledge extraction from text, images and audiovisual content, geolocation extraction from historical maps and images, manuscript transcription) will be applied** according to early-adopters' requirements and real scenarios analysed at the beginning of the project, and evolving during its development according to new stakeholders and scenarios addressed. From software engineering, we borrow the concept of building common interfaces to facilitate software integration into multiple software systems as well as to hide the heterogeneity of the handled data and to capture and render the multi-perspective nuances of CHDO. Cultural Heritage ontologies and mKG realise such an interface and allow the diverse CH actors to access, explore, and (re)use the plethora of heterogeneous data that are available in the field. In fact, CH knowledge graphs provide a standard semantic layer to interaction and managing systems and a standard representation to integrate the results of (manual and automatic) knowledge extraction from heterogeneous unstructured and structured sources (e.g. audiovisual, images, texts). INFINITY relies on (and contributes to) the following computer science areas: Semantic Web, Ontology and Knowledge Graph engineering, Artificial Intelligence, and Human-Machine Interaction.

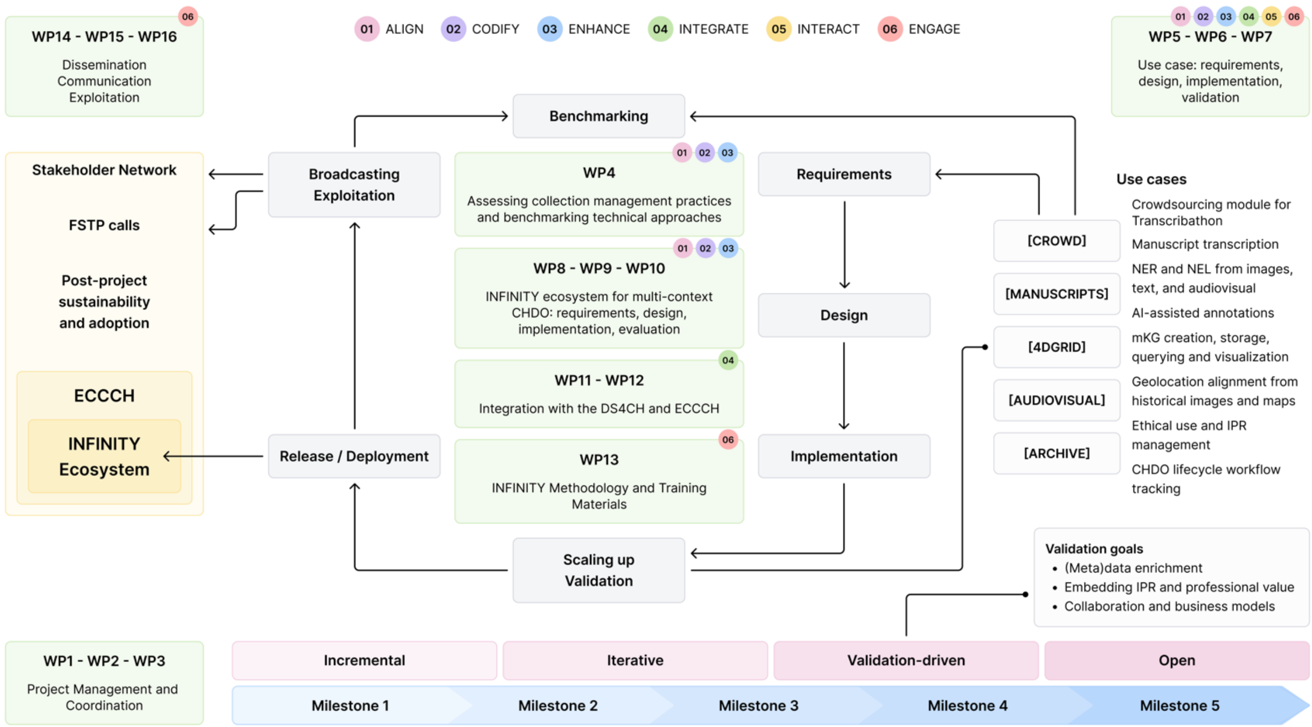
The INFINITY Consortium has representatives from cultural institutions and organizations, the EIT Culture & Creativity, technology providers of collection management systems, SSH and computer science researchers and developers that have contributed to the definition of the use cases.

Iterative design. INFINITY is structured over 42 months with 16 interlinked work packages, detailed in Section 3, to harness the consortium's diverse skills and disciplines. The project methodology (Figure 1.2.2) includes continuous cycles of problem analysis, design, implementation, evaluation, and validation. Each cycle culminates in the release of an updated version of the technology, progressively enhancing the INFINITY digital ecosystem:

- **Analysis of the problem (requirements) and the input sources, from use case requirements / feedback (WP5, WP8)**: The requirements analysis will adopt a user-centric approach, ensuring the needs of the end-users and stakeholders are central to the project. They will be defined along with a technical roadmap (T5.1) that will guide the implementation of five use cases (WP5-6-7). This will ensure a quick start as from the beginning of the project the focus will be very clear. The use cases will also serve as a testbed for building a benchmark (WP4) at the early stage of the project, for validating existing practices and tools for collection management against three core *validation goals* in line with the objectives and scope of the call: **1) (meta)data enrichment, 2) embedding IPR and professional value, and 3) collaboration and business models**. The result of this early validation will contribute to bootstrap the requirement analysis process (WP5, WP8). The use cases' definition and design (WP5) will serve as a basis for the design and development of the software components of the INFINITY digital ecosystem (WP8-9) that will be integrated in the use case CHDO collection management system (WP6) and provide a reference to the technology providers (WP9-10) as well as a validation context (WP7). Requirements for the ecosystem's software components (WP8) will be defined in synergy with the requirements of the use cases (WP5), and with the requirements for addressing integration with the ECCCH and DS4CH (WP11, WP12), promoting interdependence and alignment. A Technical Board (WP8, WP9, WP10) will oversee this process and ensure this alignment is constantly respected. To identify these requirements, workshops and interviews with stakeholders, including members of the Stakeholder Network (WP14, WP15, WP16), will be conducted. Feedback from these interactions will inform the technological needs, guided by the use cases. This validation-driven process ensures that the components developed are relevant to the specific use cases and reusable outside the project.

- **Design of the solution also by analysing the state of the art and possibly reusing / extending existing methods (WP5, WP8)**: The design phase will involve an in-depth analysis of the state of the art, identifying existing methodologies and technologies that can be reused or extended. The Technical Board will deliver a Rulebook (WP8, WP9, WP10), which will provide standardised guidelines for developers, ensuring consistency and interoperability within the INFINITY ecosystem and the broader ECCCH. This phase will also involve iterative design reviews and consultations with stakeholders to refine the architecture and ensure the solution addresses both immediate and broader technological needs. Design considerations will prioritize modularity and scalability to enhance reusability across various application contexts.
- **Implementation of the designed solution (WP6, WP9)**: The implementation phase will follow an agile development methodology, organizing the work into iterative sprints to deliver incremental progress. Development teams will produce intermediate versions of the ecosystem components and of the use cases, which, while not fully stable releases, will be shared with early adopters for testing and feedback. These iterations will culminate in three major releases: *Alpha* at M20 (cf. Milestone 2), *Beta* at M32 (cf. Milestone 3), and the final *1.0* release at M40 (cf. Milestone 4). Each release of the software components will be included in a corresponding release of the INFINITY digital ecosystem (respectively maturity level 1, 2 and 3), encompassing software, associated documentation, training materials, and guidelines.
- **Evaluation / tuning of the developed methods against state-of-the-art approaches (WP10):** Developed methods will undergo rigorous evaluation and tuning against state-of-the-art benchmarks. Performance metrics and user feedback will guide adjustments and improvements. Regular evaluations will ensure that the solutions meet both technical and practical standards. These assessments will involve both internal validation and feedback loops with early adopters and use case implementers, allowing for continual refinement throughout the project lifecycle.
- **Validation of the developed methods in the use case setting, that will provide feedback (WP7):** Validation activities will be tightly integrated with the use case implementations, ensuring that the developed methods perform effectively in real-world settings. The five use cases are categorized and validated based on their alignment with the three *core validation goals*, which ensures comprehensive coverage of validation areas and avoids duplication of effort. The use cases are described in detail in the next Section (1.2.3), while a detailed plan for their execution is elaborated in WP5-6-7. Feedback from these validations will be used to fine-tune the components and guide future iterations. This phase will emphasize collaboration with early adopters, whose insights will be instrumental in improving usability and functionality.
- **Release of a new version of the technology within the INFINITY digital ecosystem (WP7, WP10):** The release phase will see the deployment of the ecosystem's components in three stages: Alpha, Beta, and the final 1.0 version. Each software release will be incorporated in a release of the INFINITY digital ecosystem, encompassing software accompanied by comprehensive documentation, training materials, and guidelines, to facilitate adoption. The Technical Board will validate each release, to ensure it meets maturity criteria and aligns with the Rulebook, including alignment and integration with the ECCCH. Intermediate versions, made available between releases, will foster early adoption and iterative feedback, driving continuous improvement.
- **Broadcast through communication, dissemination and exploitation activities (WP14, WP15, WP16):** To maximize impact, communication, dissemination, and exploitation activities will broadcast the ecosystem's advancements. Capacity-building workshops and tailored training materials will empower stakeholders and early adopters to experiment with and reuse the software. Additionally, an open FSTP (Financial Support to Third Parties) call will fund approximately 14 projects, with half focusing on software early adopters and half focusing on experimenting with new business models enabled by the availability of INFINITY's innovations. These projects will provide resources to support experimentation and integration, further strengthening the ecosystem's adoption and relevance.

Figure 1.2.2 Iterative design methodology of INFINITY



1.2.3 Design and validation through five use cases

Five use cases (see also Section 3, WP5-6-7 for their associated implementation workplan) play a central role in INFINITY to validate its results against three core validation goals in line with the call's main objectives: **(meta) data enrichment, embedding IPR and professional value, collaboration and business models**. All use cases are based on already well-established concepts and proven demands and support the validation of tools and data of different types with large audiences. The use cases develop and test innovative tools for human-driven acquisition, categorisation and annotation of digital objects covering all types of CHDO (texts, images, 3D models, sounds and videos) combined with new AI-based approaches, resulting in high-quality content and metadata. Their execution and validation contribute to the development of the ECCCH by 1) sharing the data produced on the DS4CH and showing the exploitable nature of the INFINITY software components.

[CROWD] Supporting Citizen Science and Crowdsourcing. (Lead: EF, Participants: F&F, TMO, PSNC). **Target users**: citizen (scientists), cultural heritage institutions, researchers, educators. **Early adopters:** Europeana, UWr, CHI **Problems:** Supporting **citizen science** and **crowdsourcing** in generating **ethical** CH datasets, enrich existing collections and integrating with existing knowledge graphs, at large scale. Supporting automated transcription and **contextualization** of handwritten material with AI tools. Improving data quality, enhancing historical knowledge, preserving heritage knowledge at risk of loss, with public contributions. **Datasets:** Europeana 1945, Local Time Machines, post-WWII collections and archives. **Type of CHDO**: Multimodal with specific attention on handwritten text-documents. **Description:** the use case runs the "Peace in Europe- Europeana 1945" campaign, which aims to gather and transcribe post-World War II (WWII) memorabilia (such as letters, diaries, official documents, leaflets, photos) from citizens in Germany, the Netherlands, and Poland. Crowdsourcing, AI, and citizen science will be used to increase the representativeness of specific types of cultural heritage assets by digitizing personal artifacts and transcribing handwritten texts that are at the risk of being ignored or fully lost. Metadata created in the form of stories and linking to our mKG shared over the ECCCH and DS4CH will increase the multidimensionality of existing datasets. The open-source tool Transcribathon will be enhanced with a crowdsourcing module and thoroughly documented. Community engagement through collection days and transcribathons will foster participation, while video guides and templates will ensure broader adoption across other EU initiatives. **Validation goals: (meta) data enrichment, embedding IPR and professional value.**

[4DGRID] Geotemporal Alignment of Historical Image Data. (**Lead: TMO**, Participants: EPFL, UNI Jena). **Target Users:** General public, SSH researchers (spatial history, human geography, architecture). **Early Adopters:** Local Time Machines (LTM) as city specific networks**. Problems:** to geolocate and temporally align maps, city photographs, and vedutes (a genre of landscape painting) to world views. Prototypes to create and view world scale 4D space and time models exist at TRL4-5. **Types of CHDO:** Images as maps, city photographs, and vedutes, textual information as image captions, toponyms objects as architectural and sculptural 3D models**. Datasets:**

Digitised city photographs, ancient maps, genealogical records, and spatial datasets from LTM networks. **Description**: This use case develops a toolchain to geotemporally aligned historical images, maps, and vedutes, visualizing them in 4D worldviewers. Local Time Machines (LTM) will supply datasets through community engagement. Selected data will be enriched, stored in Zenodo, and integrated into a spatio-temporal knowledge graph (mKG). Citizens and researchers will access the graph via 4D viewers. A contest will identify LTMs to participate, contributing data for enrichment and visualization of city-specific historical information experiences. **Validation goals: (meta) data enrichment.**

[MANUSCRIPTS] Enhancing Manuscript Access and Enrichment through Knowledge Graphs. (Lead: UWr, Participants: PSNC). **Target Users:** Researchers, general public, cultural heritage institutions. **Early Adopters:** UWr, PSNC. **Problems:** To increase accessibility and usability of manuscripts for researchers and the public through metadata enrichment. To fill gaps in manuscript metadata descriptions and create a coherent **knowledge graph**. To optimize transcription workflows using AI tools and integrate it into manuscript metadata. To integrate enriched manuscript data into the knowledge graph and reinject it into the Digital Library to improve discoverability and engagement. **Types of CHDO:** Digitized manuscripts. **Datasets:** Manuscript collection of UWr (approx. over 6,000 digitized) **Description:** UWr houses a significant collection of manuscripts, but the current Digital Library (DL) platform (dLibra) lacks the depth of metadata and contextual data required for advanced research and discovery. Key details - such as dimensions, page counts, and the materials used (parchment or paper) - are often missing or incomplete. Critical elements like provenance, origin, illuminations, musical notations, and binding types are scattered across different tools, making it difficult to form a comprehensive understanding of each item. Also, manuscripts with *multiple languages* lack clear annotations specifying where each language appears. Bibliographic references and cross-links to related manuscripts are also limited, restricting deeper exploration. Currently, this data is dispersed across multiple tools within the UWr ecosystem. Researchers require consolidated, enriched, and structured access to this information. This use case aims to bridge these gaps by enriching the objects in DL with *harmonized metadata* (obtained from multiple tools) and linking manuscript data with mKG. Tools from the UWr ecosystem will enable enrichment of metadata and data (e.g. transcriptions). Additionally, the solution will facilitate the re-import of the enriched metadata into the Digital Library (DL), enhancing the presentation and accessibility of digital objects. Enrichments will be started automatically or manually using an adapted editor interface in dLibra (for users that create CHDO descriptions). Finally, they will also be visible in the user view. Wherever possible, these changes will be implemented as plugins to UWr ecosystem tools. **Validation goals: (meta) data enrichment.**

[AUDIOVISUAL] Context-Aware Enrichment of Audiovisual Heritage. (Lead: NISV, Participants: EIT CC SW). **Target Users:** Researchers, general public, cultural heritage institutions (CHI), journalists, Creative industry. **Early adopters:** NISV, National (Dutch) and European Research Infrastructure for the Arts and Humanities (CLARIAH, DARIAH), National Dutch Digital Heritage Network, Europeana. **Problems:** Enriching audiovisual collections across contexts with **multidimensional and multimodal knowledge graphs**. Improving metadata quality, accessibility, and content discoverability for time-based media. Generating/extracting named entities from video, descriptions, and Automated Speech Recognition (ASR) outputs while ensuring trust and ethical use. Making AI-driven processes transparent through comprehensive data sheets[23] describing operating characteristics, test results, recommended uses, and other information. **Types of CHDO:** digitised and born digital film, television, radio and podcasts. **Datasets:** Audiovisual datasets with varied access conditions (copyright, privacy) from NISV and European audiovisual archives available via the EUscreen network of which NISV is member. **Description:** This use case explores the use of INFINITY tools to create enriched, inclusive and interconnected audiovisual archives, validated by researchers and the public. By integrating context-aware descriptions and enabling access to and interlinking distributed collections (via platforms like CLARIAH Media Suite and the Data Space for Cultural Heritage), it enhances engagement with more contemporary heritage collections, most of which are still in copyright. It also examines the ethical, copyright, and privacy challenges surrounding AI-driven entity generation, ensuring transparency and accountability in workflows. **Validation goals: (meta) data enrichment, embedding IPR and professional value, collaboration and business models.**

[ARCHIVE] Unlocking Archival Knowledge. (Lead: MIC, Participants: UNIBO, EIT SW CC). **Target users:** General public, cultural heritage institutions, SSH researchers, researchers, digital humanists, educators, journalists, creative industry professionals, citizens. **Early adopters:** ICAR, State Archives, public and private archival institutions. **Problems:** To make archival portals more intuitive and engaging for non-experts and creative professionals. Enhancing usability, accessibility and **cross-sector reuse** of archival data. To better communicate

[23] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2021. Datasheets for datasets. Commun. ACM 64, 12 (December 2021), 86–92. https://doi.org/10.1145/3458723

conditions for appropriate content reuse, detailing who to contact, permission processes and potential tariffs. Making archival data more accessible for creative repurposing while preserving contextual integrity and respecting conditions of reuse. **Types of CHDO:** Structured and unstructured text, digital reproduction of archival resources. **Datasets:** Data from Archives National Portal. **Description:** This use case tackles the barriers faced by non-expert users and creative professionals in accessing and reusing archival data. Online disintermediation deprives users of archivist support, making navigation difficult due to specialized terminology, incomplete metadata, and complex contextual frameworks. Additionally, archives often lack clear pathways for understanding reuse conditions, permissions, and associated costs—discouraging creative industries from leveraging archival content. The use case aims to improve portal usability through semantic enrichments and **contextual annotations in the form of mKG**, connecting archival descriptions to places, people, events, and institutions via standardized open models (e.g., thesauri, controlled vocabularies, ontologies). By enhancing clarity around **reuse rights and processes**, archives can stimulate innovation across creative sectors while ensuring that archival materials remain embedded in their **historical and documentary context**. This use case will contribute to increase the availability, visibility and accessibility of archival materials in ECCCH and Europeana. **Validation goals: (meta) data enrichment, embedding IPR and professional value, collaboration and business models.**

1.2.4 Related projects

Project	Description of synergies and expected added value for INFINITY
ECHOES (2024 - 2029) NISV, PSNC, EF, TMO	ECHOES is an ongoing project that will develop The Cultural Heritage Cloud (ECCCH) a shared platform designed to provide heritage professionals and researchers with access to data, scientific resources, training, and advanced digital tools tailored to suit their needs. ECHOES aims to build on existing open standards, fostering collaboration and adaptability to changes. It is designed to become a cornerstone for preserving, sharing, and innovating around Europe's rich cultural heritage. Its extensible data model ensures ease of adaptation and expansion, facilitated by INFINITY.
Data Space for Cultural Heritage (2022 - 2026), EF, NISV, PSNC, TMO	The common European data space for cultural heritage is a European Union flagship to accelerate the digital transformation of the cultural heritage sector and facilitate the open and trustworthy sharing of heritage data across Europe. It provides an ideal ground for applying the INFINITY methodology and tools to a massive range of cultural data. INFINITY results must be technically compatible with the data space (e.g., through interoperable data models) and aligned with its community and capacity-building initiatives.
MULTIPOD (2024-2027, HE); SPA	Storypact develops a language model and knowledge graph for supporting socio-culturally sensitive communication by mapping between terms whose interpretation differs across groups, also within the same language. This work will support and synthesize with the KG and LLM development of INFINITY for the cultural heritage domain.
POLIFONIA (2021-2024, H2020); UNIBO, NISV, DP	INFINITY integrates European musical heritage into its research, education, and exploitation activities using methods from the Polifonia project. It extends knowledge graph creation and AI-driven ontology engineering techniques to incorporate the Polifonia Ontology Network. Additionally, INFINITY utilizes and expands Polifonia's ontologies on licensing for FAIR data management in its digital infrastructure.
SPICE (2020 - 2023, H2020); UNIBO	SPICE (EU Horizon2020 Grant Agreement N. 870811) develops technologies that enable communities to select and interpret paintings, artifacts, and museum objects. These tools facilitate sharing interpretations among people, fostering learning, self-awareness, and empathy for other communities. The approach is termed Citizen Curation. INFINITY will reuse the ontologies developed in this project to model perspectives and narratives.
FAIR (2023-2026)	FAIR is a large-scale Italian research project (PNRR) involving the main research organisations in the country. Its goal is to develop foundational research for future AI. The WP8.7 (lead by the INFINITY coordinator) investigates "Human and Artificial Creativity" from cognitive and computational perspectives and develops neuro-symbolic systems and formal theories that can support achieving this goal. INFINITY can exploit perspective models and reasoning tools.
CLARIAH (since 2014, NWO Large Scale Research Infrastructure Grants NISV	CLARIAH, the Dutch national infrastructure for arts and humanities research, offers extensive access to cultural heritage data through FAIR-compliant services and virtual research environments, integrating digital tools from European collaborations like DARIAH, CLARIN, and EOSC. A key facility, the CLARIAH Media Suite at NISV, provides a secure research environment, central to the [AUDIOVISUAL] use case.

5D Culture (2023-2024, DEP); TMO, NISV; UJENA; EF	5D culture enriches the offer of European 3D digital cultural heritage assets in the data space and fosters their reuse in important domains such as education, tourism and the wider cultural and creative sectors towards socially and economically sustainable outcomes. 5DCulture contributed to the development and data collection for the 3D and 4DViewer tools utilized in INFINITY.
INDUX-R (2024-2026, HE)**;** UJENA	INDUX-R aims to transform European industries with a human-centric XR ecosystem, advancing core XR technologies and tackling societal and ethical challenges across diverse use cases.INDUX-R has already developed a set of prototypic workflows and tools for spatial enrichment which will be utilized and further enhanced in INFINITY.
DFG 3D Viewer 2021-2026, DFG) UJena	The DFG 3D Viewer establishes a national 3D model infrastructure in Germany, integrating tools for data and metadata enrichment. With over 63.000 3D mesh models, it supports developing and validating INFINITY pipelines.
3DBigDataSpace (2025-2026, DEP) TMO; PSNC; EF; UJENA	3DBigDataSpace creates and validates pipelines for spatialising, classifying and utilising 3D models at large scale with the objective to collect, enrich and contribute 50.000+ 3D meshes of European cultural heritage. The tools and datasets will be re-used to develop and validate the INFINITY pipelines.
Parcels of Venice (2019-2025) SNF EPFL	The Parcels of Venice project has enabled EPFL to build a working prototype for a geohistorical platform including more than 50 000 historical records with associated digitized sources regarding the history of the City of Venice. The resulting browser and the associated 4D grid infrastructure will be used as part of the **[4DGRID]** use case during this INFINITY project.
AI4Media (2020-2024, H2020); NISV	Ai4Media aims at establishing a network of excellence centres on Artificial Intelligence applied to media. INFINITY will use and enhance its experiences related to AI based analysis of multimedia documents.
EnrichEuropeana (2018-2023); F&F, EF; PSNC, AIT	The CEF projects EnrichEuropeana,and EnrichEuropeana+ were developing the citizen science platform Transcribathon.eu to transcribe and enrich CHDO from/to Europeana.eu. It includes modules for AI transcription and semantic enrichment and organised Transcribathon events in several countries. INFINTY will use and expand the platform for the CROWD use case.
DARIAH-EU	Digital Research Infrastructure for the Arts and Humanities is a European network that supports digital research, data management, and collaboration in the arts and humanities, fostering innovation and preservation of cultural heritage. The tools and services can be re-used to develop and validate the INFINITY pipelines.
DARIAH-PL/Dariah.lab and DARIAH-PL/Dariah.hub (PSNC, UWr)	DARIAH-PL/Dariah.lab established a national research infrastructure for art and digital humanities across Poland, featuring various labs. This infrastructure evolves under Dariah.hub Poland, enhancing the national ecosystem for interdisciplinary cultural heritage and archaeological research. The integrated tools and datasets from these projects support the development and validation of the INFINITY pipelines.
ekip (NISV, EUDIN)	The ekip collaborative platform deals with complex innovation processes involving many stakeholders from Creative and Cultural Industries (CCIs) network ekip's approach is based on open innovation principles, aiming to provide evidence-based policy suggestions while involving broad stakeholder groups. The experience of using new business models and helping the production of new products and services has parallels to INFINITY.
READ-COOP/ Transkribus (EUDIN)	Transkribus is the AI powered infrastructure to produce accurate transcriptions of historical texts under cooperative principles. Both the new business model, experience of innovation, and the experience of setting up, maintaining, and sustaining a user-based cultural heritage system provides useful input into INFINITI's aims.

1.2.5 Interdisciplinarity

Section 1.2.2 already highlighted the application of interdisciplinarity in developing cultural heritage solutions within INFINITY. This section focuses on managing interdisciplinarity as a practical methodology throughout the project's execution. Interdisciplinarity serves as both a challenge and a driver of innovation, demanding practice, patience, and collaboration to bridge gaps between disciplines. Central to INFINITY's goals, interdisciplinarity integrates expertise from cultural heritage, computer science, humanities, and creative industries. By enhancing the semantic enrichment and interlinking of cultural data, **the project fosters advanced, cross-disciplinary research**, enabling scholars to connect diverse data sources and uncover new insights into cultural phenomena. Challenges include: (i) **lack of a shared language** among diverse stakeholders, (ii) **difficulties translating creative ideas into technical requirements** and vice versa, and (iii) **siloed thinking**. Drawing on extensive experience in interdisciplinary work, partners have devised a strategy with three key components: 1) Use existing methodologies to **map ethical questions, stakeholder needs, relationships**, and power dynamics, ensuring all disciplines are represented and gaps identified. 2) Hold sessions during general assemblies **to challenge project assumptions**, promote cross-disciplinary reflection, and discuss topics like responsible communication of digital technologies and the role of immersive tech. 3) Develop a **shared glossary** to enhance clarity across disciplines, using techniques like shape language and concept mapping for abstract or technical concepts. An **Interdisciplinary Working Group** (T1.1), chaired by a senior team member on the project management board, will oversee the interdisciplinary approach.

1.2.6 SSH Integration

INFINITY adopts a human-centred approach to technological innovation, leveraging lessons from the Social Sciences and Humanities (SSH). The Social Construction of Technology (see Section 1.2) adopted by INFINITY underscores the notion that effective technology is the product of context-sensitive choices rather than merely representing a universal solution. Recognising that data analysis tools carry inherent biases that can shape societal understandings and impact specific communities (D'ignazio, 2023), INFINITY emphasizes a digital humanism perspective. This approach aims to dismantle traditional disciplinary silos, positioning inclusion and democratic rights at the core of new technology development. Utilising a design methodology championed by UEDIN, INFINITY fosters collaboration between technology and use case providers from the SSH sector (including cultural heritage data providers and academic partners) from the project's inception. This ensures that the creation and deployment of technologies are guided by human values and needs (Humbel, 2025). Use case partners will consult extensively with SSH experts to ensure that the outputs of INFINITY (i) adhere to best user experience practices, (ii) comply with ethical and legal standards, and (iii) meet broader societal needs. All technology design processes documented in an innovation binder (T14.3-16.3). The integration of SSH research methods enhances the analysis, presentation, and interpretation of varied societal viewpoints through digital heritage data (Cameron, 2021). This will promote deeper, context-aware analyses, enabling SSH scholars to visualise and dissect societal challenges through digital data more effectively. As a result of INFINITY's work, research platforms such as ECCCH, EOSC, DARIAH, CLARIAH (refer to Section 1.2.4) will incorporate new methodologies and tools, enhancing functionality and integration into SSH communities. This expansion emphasizes SSH's vital role in transforming how cultural heritage is accessed and interpreted, promoting an inclusive and ethically aware technological framework. INFINITY's integration of SSH methodologies is essential for developing supportive and innovative infrastructures for CHDO, significantly enhancing their ethical, cultural, and societal impact to meet the diverse needs of the global heritage community effectively. There are aspects of CHDO and their providers which mean that SSH methods are essential for this domain: (i) **Interdisciplinary collaboration** enriching digital curation, spearheaded by the Interdisciplinarity Working Group (T1.1), (ii) SSH methods ensure **culturally accurate interpretations** of CHDO, focusing on ethical considerations (Mindel 2021) (T8.1). (iii) Enhanced user interfaces and digital interactions are developed under Task 5.3, making technology accessible and intuitive. (iv) **Public Engagement** strategies increase accessibility and education in cultural heritage, detailed in Section 2.2. (v) **Sustainable and responsible innovation** balances technological advancement with cultural integrity preservation. (vi) **Policy development** support is provided through collaborations with projects like ECHOES and the European Heritage Hub. (vii) Ethical data management, maintaining standards of consent, privacy, and cultural sensitivity, is a focus of Section 1.2.10 and supports responsible CHDO handling.

1.2.7 Gender dimensions

The INFINITY consortium intends to represent the EU values of gender equality and attention to diversity, with the firm belief that recognising and integrating gender and wider diversity issues into research and innovation is vital for its societal relevance and quality. The gender dimension strategy centres around four key components:

- The overall design of the work plan. To act in line with the EU's ambitions expressed in the "EU Work Plan for Culture 2023-2026" and the "Women in Digital" strategy, the consortium has incorporated a gender perspective in the overall design of the work plan, actively promoting equal representation of women and non-binary people in the activities of INFINITY. With participation at the core of its activities, INFINITY embraces an inclusive and gender-conscious approach, with three main aims to: (a) ensure access to culture for all EU citizens without social categorisation (in terms of gender, ethnicity, age or disability); (b) contribute to the improvement of more inclusive cultural services offered by heritage organisations; (c) integrate different gender needs, attitudes & behaviours, ensuring proactive participation at all levels in the project.

- Specific research challenges central to the project. Gender dimensions are related directly to specific research challenges of INFINITY, given its focus on multiple dimensions and perspectives on data. INFINITY will specifically investigate biases (T8.7), including the gender dimension and even broader intersectionality of identities (ethnicity, age, ability), to ensure that KGs do not contribute to further discrimination of marginalised groups by amplifying stereotypes and demeaning attitudes. These aspects will be further investigated in T9.7 and T10.7 [add]. Gender analysis will be observed for any survey to be performed towards stakeholders as well as planning of workshop speakers and attendees.

- **Team composition.** We recognize the urgency of promoting women's empowerment in technology and science, technology, engineering and mathematics (STEM) fields. To advance gender balance, we will ensure equitable representation in work package leadership and prioritize gender equality in our recruitment strategies for project staff."
- **Communication and dissemination**. Our strategies will prioritize inclusivity and use appropriate tools to reach diverse stakeholders. We will ensure equal access to capacity building, knowledge sharing, and networking events. We support the "No Woman, No Panel" initiative by the Open Society Foundation and will host events for European Girls and Women in ICT Day. Special efforts will be made to reach these groups, such as organizing workshops via local community centres and libraries.
- **INFINITY's Measures**. From the project's start, the team will be informed of their gender-related responsibilities and follow internal policies to ensure gender equality and diversity. An interdisciplinary working group will oversee ethical, gender, and inclusivity dimensions throughout the project. This group, with equal gender representation, will produce a public report showcasing our progress on gender equality towards achieving SDG 5.

1.2.8 Open science

Open science practices are an integral part of the project's work plan. The strategy is built upon five key components:

1. Embracing existing open data, standards and tools: The basis for all CHDO metadata and knowledge in INFINITY will be existing open data and standards extended by our components to embrace multidimensionality and ethical usage. Our metadata alignment layer will support a consistent view on CHDO metadata independent of the underlying metadata model or schema (EDM, EBUCore etc.). Values of metadata properties will be aligned in meaning through reference to concepts in the mKG, which in turn are built and evolved from open knowledge such as Wikidata. Existing open tools at CHI will be extended by INFINITY plugins to incorporate our innovations.

2. Delivering outputs built on open standards: Web-based services, user interfaces and APIs will be used for giving access to the tools developed in the project, thus ensuring their easy integration and re-purposing. We will integrate mKG with Data Spaces through the reference architecture and components (e.g., Eclipse) that are provided.

3. Making research findings openly accessible: All our academic output (See Section 2.2 for KPIs) will be published under either Green Open Access or Gold Open Access models; for economic purposes, the former will be predominantly used, while UNIBO has also reserved funds allowing at least two key publications to be distributed as Gold Open Access. All publications will also be archived in Zenodo, and the project's website, while pre-prints of key publications will also be pre-published in arXiv upon submission.

4. Ensuring the reproducibility & re-use of our research outputs: Software implementations of our methods, used for performing the experimental evaluation reported in our publications, will be open-sourced by default in GitHub. The related datasets will also be publicly released, following the FAIR principles. Supporting training materials (T14.2), such as short training videos published on YouTube, will also be released.

5. Contributing to EU open science platforms and communities: The **Open Research Europe** platform will be used as one of the instruments for publishing our open-sourced outputs, in parallel to traditional peer-review publication venues and global software repositories such as GitHub. We will also contribute to the **European AI-on-demand** platform and the **European Language Grid** by submitting relevant open-source software and/or open licence datasets. The consortium members have already been adhering to open science principles (e.g., ensuring the open access, reproducibility and re-use of their research outputs) for many years. For example, the involved UNIBO team offers a repository for publications, IRIS-IR1, and another repository, AMS Acta, that can be used for collecting and disseminating unpublished literature. Both repositories are linked synchronized with the OpenAIRE platform and comply with international standards. JENA team has open-sourced code for 4D desktop and mobile world viewers and a scripted Zenodo storage and enrichment pipeline, is coordinator of the DFG 3D Viewer national infrastructure, and holds 65.000 3D models and 250.000 image datasets, EPFL team has open-sourced code for Time Atlas browser and associated 50 000 historical records, UEDIN has released cultural heritage datasets via the EC funded CrossCult project and a variety of datasets and open publications via Creative Informatics.

1.2.9 Data Management

INFINITY's research data and outputs, excluding publications, will adhere to the FAIR principles (Findable, Accessible, Interoperable, Reusable) to ensure effective data management throughout the project lifecycle. Our commitment to these principles will be detailed in our Data Management Plan (DMP, Deliverable D1.1), scheduled for release to the consortium by Month 2. It contains **data collection and security protocols** to guarantee that all data is collected securely and stored safely, preventing unauthorized access or use. We will implement these protocols across all project activities, particularly those involving sensitive data collection, such as user studies or dissemination efforts. A **data steward** will be appointed to the Technical Board (WP1) to provide guidance on and oversee data management practices.

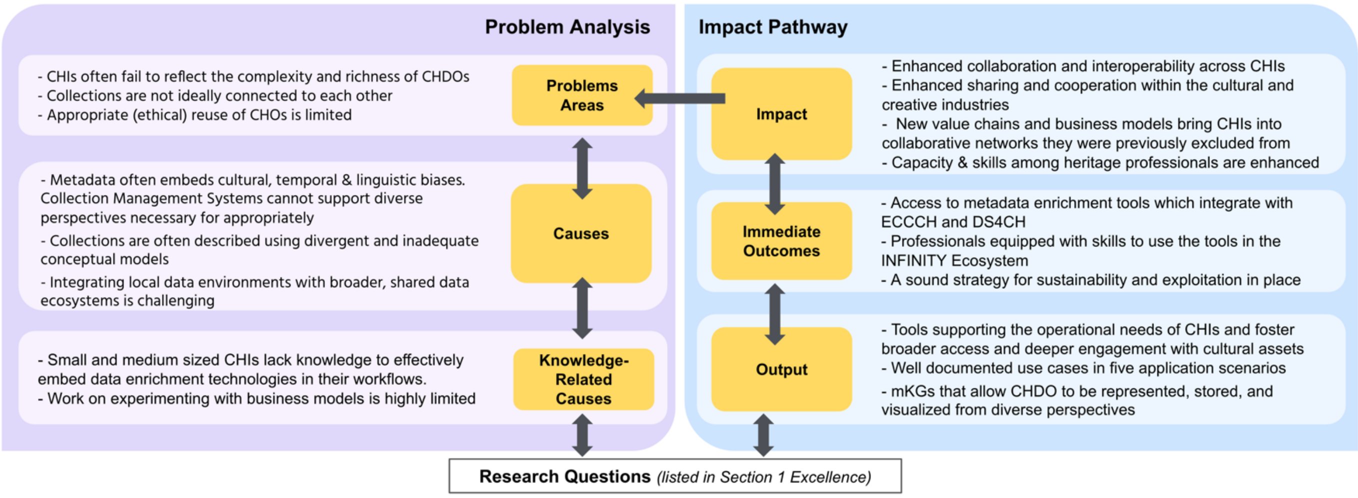
The approach regarding **regulatory compliance** will be meticulously aligned with relevant national and EU regulations, including the General Data Protection Regulation (GDPR – Regulation (EU) 2016/679) and the EU Charter of Fundamental Rights (Article on Protection of Personal Data). These standards will apply universally to all consortium partners, including those outside the EU, such as EBU, to ensure consistent data protection measures are maintained. The DMP will specify the types of data transferred between consortium partners, including non-EU members, with a strong emphasis on data minimization and the use of pseudonymization and anonymization techniques wherever feasible.

More specifically, the DMP addresses aspects related to: (i) The DMP will describe the **various types of data** collected and processed by INFINITY, such as datasets for AI training, social media metadata, and outputs from user validation studies. The plan will cover both newly collected data and the use of existing datasets, ensuring all personal data is handled with the same rigorous standards to mitigate privacy risks. (ii) Which repositories (such as the ECCCH, Zenodo, and the Data Space for Digital Heritage and European Open Science Cloud (EOSC) will be evaluated for storing INFINITY's outputs, which will be assigned Persistent Identifiers to enhance findability and long-term accessibility (iii) All data released by INFINITY will be made accessible to consortium partners during the project and will be published under licenses that promote reusability, such as the Creative Commons CC BY-NC for datasets and CC BY for software components. And, where relevant, align with principles for indigenous data governance[24]. We will use standard formats to ensure data interoperability. (iv) Responsibilities for data curation and preservation will be assigned within the consortium, with all related costs accounted for in partner budgets. The management of intellectual property rights will be governed by the Consortium Agreement, ensuring that all data, particularly those containing personal information, are processed only after obtaining appropriate consents.#§CON-MET-CM§# #§COM-PLE-CP§# #§REL-EVA-RE§#

2. Impact #@IMP-ACT-IA@#

This section highlights INFINITY's role in addressing and magnifying impacts specified in the work programme, utilising the Theory of Change (McLellan 2020) to illustrate the connection between problem analysis and societal impact through an impact pathway analysis.

Figure 2.1 INFINITY Theory of Change



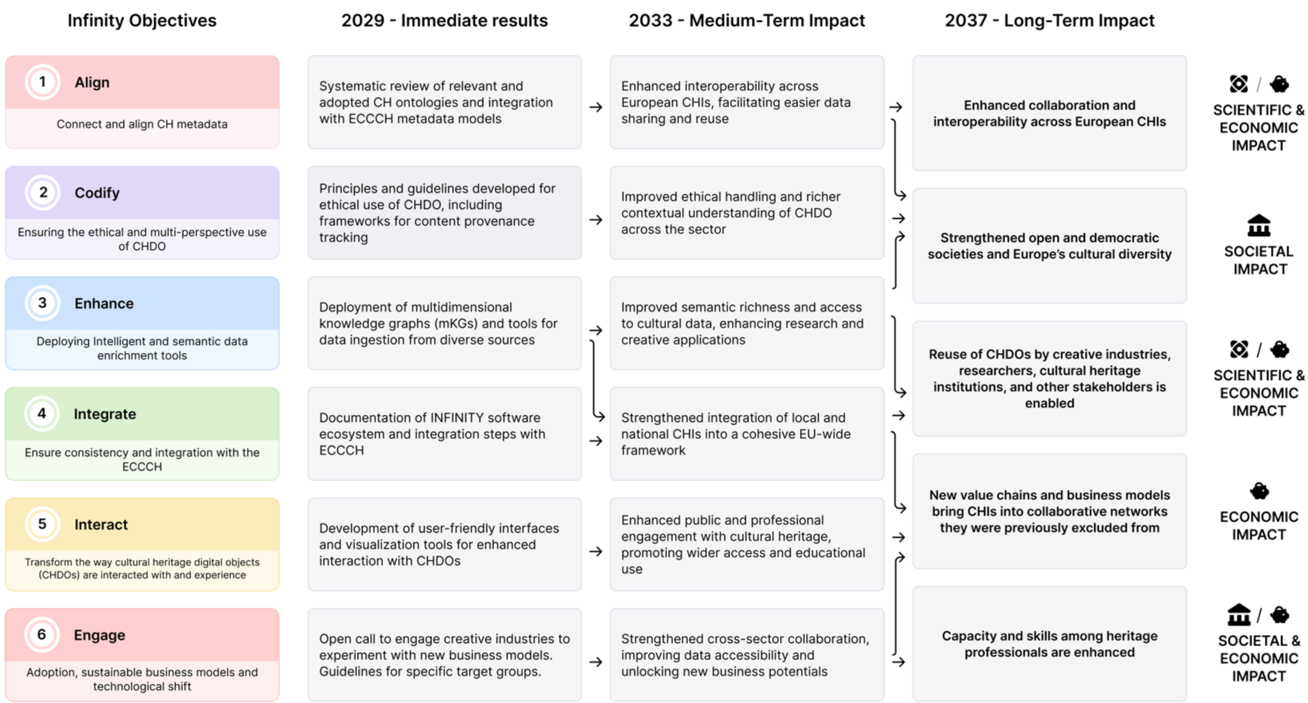
INFINITY seeks to address persisting barriers that restrict access to CHDO and limit the value of cultural heritage in innovation contexts (Terras, 2022). The project aims to impact this area by developing interdisciplinary methods, guidelines, and tools for multi-contextual data enrichment, ethical reuse, and enhanced interconnectivity. As summarised in Figure 2.1, there are various causes that require interventions in order for the ECCCH to succeed. The impact pathway (right side of Fig. 2.1) shows how RQs lead to project impacts, discussed further in the subsequent section. In this context, INFINITY is designed to achieve a long-term vision that reinforces the ECCCH objectives and impacts the way CHDO are being enriched, accessed and reused. Figure 2.2 depicts how project objectives will drive societal, economic, and scientific impacts over the next decade. It summarises how the project's immediate results (we consider the M42 milestone finalising this stage) lead to anticipated mid-term outcomes (initiated by dissemination and exploitation activities in the last year of INFINITY, continuing up to four years after project end) and long-term Impact (4+ years after INFINITY). The medium-term impacts are aligned with the expected outcomes (EOs) of the call - this is unpacked in Section 2.1.1.

[24] https://www.gida-global.org/care

2.1.1 Contributions towards the expected outcomes

The project's pathway to impact is unpacked in the table below. For each expected outcome of the call, we identify the immediate results and outcomes that will, through the amplifying dissemination and exploitation activities, lead to anticipated medium-term (up to 4 years after INFINITY) and long-term (more than 4 years after INFINITY) impacts. Finally, we describe the expected scale and significance of these impacts four years after the end of the project (medium-term impact).

Figure 2.2 INFINITY's ten-year vision with pathways towards scientific, economic and social impact



Expected outcome 1: "The European Collaborative Cloud for Cultural Heritage (ECCCH) is widely used by European cultural heritage professionals and researchers for metadata enrichment of digital cultural heritage objects."
Immediate results and outcomes: (i) Systematic review of relevant and adopted CH ontologies and archival description standards (ii) Integration of outcomes with ECCCH and DS4CH metadata models (schema mapping and extension tools) (iii) Multidimensional knowledge graphs deployment, query, update tools (v) Plug-in and web service architecture for smooth integration of ecosystem components into existing platforms (e.g., ECCCH, dLibra, DInGO, Transcribathon, CLARIAH Media Suite) central to the use cases. (vi) The INFINITY rulebook for developers aligns with ECCCH guidelines (vii) Integration with ECCCH and DS4CH metadata models.
Medium term impact: Enhanced data interoperability and integration across platforms and collections [achieved through SO1 ALIGN and SO4 INTEGRATE] **1. Access to specialised tools through ECCCH**: Communities using the ECCCH will have access to specialised tools tailored to their specific needs, facilitating tasks that are crucial for their research and professional activities. The ECHOES project will see active engagement from numerous professionals utilising these tools. **2. Integration with Collection Management Systems (CMS)**: CHI are encouraged to integrate ECCCH services with their existing CMS, thereby streamlining operations and enhancing data coherence across platforms. Aim: ≥100 CHI experiment with at least one tool, 40% of them integrate one or more in the medium-term. **3. Use Case-Specific Impacts**: Each use case associated with the ECCCH initiative will generate specific medium term (= four years post project) impacts. These cases are not only demonstrations of the potential uses of multidimensional CHDO but also serve as models that can be replicated across the ECCCH user base and beyond. The impacts of these use cases extend to both ECCCH users and CHI not directly connected to the project, illustrating the broad applicability of these innovations: **[CROWD]:** Coordinated by EF and F&F this use case enriches CHI collections and makes field testing and research data widely available, enhancing the involvement of researchers and CH professionals across Europe. A new business model is introduced based on active use and financial support of the platform by CHI. Aim: number of CHI actively participating grows ~20% annually. **[4DGRID]:** Repository operators and spatial researchers benefit from improved data quality and accessibility, enabling the creation of detailed historical narratives and virtual tours. This comprehensive approach allows for innovative educational applications, such as virtual city tours and enhanced learning materials for students. Aim: Facilitate ≥1.000 user sessions of the 4D frontend tools weekly.

[MANUSCRIPTS]: It employs the dLibra CMS system within the DInGO structure to foster greater cooperation among ≥150 GLAM institutions across Europe, facilitated by the UWr. Facilitate ≥25 joint research projects using shared data within three years post project. **[AUDIOVISUAL]:** Building trust among content creators and users, fostering respect for diverse cultural narratives and enhancing the educational impact of archival materials. Ethical guidelines that are adopted by at least 50% CHI in the EUscreen network (via EUscreen network coordinator NISV). **[ARCHIVES]:** Portal users benefit from improved navigational tools and content presentation, easing access to archival materials. Archival institutions gain centrality in cultural discourse, contributing significantly to the educational and societal value of historical documents. Aim: Improve navigational tools and user interfaces for ≥ 100 archival collections.

Long term impact: Enhanced collaboration and interoperability across European cultural heritage institutions is achieved, leading to a richer, more accessible digital heritage landscape.

Scale and significance: In a recent survey conducted in the context of the ECCCH, nearly a thousand heritage professionals participated. The findings revealed that 46% of respondents identified the absence of a digital collaboration platform, specifically one with tools tailored to the needs of CH professionals, as the primary obstacle in the sector's digital transition (European 2022). INFINITY's efforts are directly linked to these identified needs and monitored in two ways. First, that the Europeana Network includes 4,000 CHI, our goal to **improve collaboration** through INFINITY's outcomes across 400 European CH institutions (10%) is considered realistic. Second, INFINITY is expected to improve the effectiveness and usage of research infrastructures such as EOSC, DARIAH, CLARIN, and E-RIHS. The project's success will be evaluated using engagement metrics and the adoption rates of ECCCH tools, which are expected to enrich the digital heritage and research landscape. This will be quantified by increases in collaborative research projects, scientific publications derived from ECCCH data, and the expanded use of collections.

Expected outcome 2: "Scientific and professional value as well as intellectual property and other associated rights are effectively embedded in the digital objects of the ECCCH throughout the digital content production chain, thus enabling and boosting cooperation, sharing and re-use."

Immediate results and outcomes: (i) Multilingual, multimodal entity recognition and linking toolset as plugin for Collection Management Systems, (ii) Framework for understanding appropriate, responsible and transparent reuse of CHDO, and practical guidelines, (iii) Tool for user consent management for AI-generated datasets (iv), Open call (FSTP) to engage creative industry in identifying and experimenting with innovative business models enabled by the data produced through INFINITY use cases and shared on the ECCCH and DS4CH, (v) formal collaborations with European Digital Innovation Hubs (EDIH), Enterprise Europe Network (EEN), European Cultural and Creative Sectors and Industries Policy Platform (ekip) and the EIT Culture and Creativity established.

Medium term impact: Increased legal and ethical compliance in data handling and use, as a result, CHI feel more confident and encouraged to share their CHDO to the ECCCH due to trust in the implemented IP framework.
[achieved through O2 CODIFY]

Long term impact: Wide **reuse of CHDO by creative industries, researchers, cultural heritage institutions and other stakeholders is enabled** by clear, legally compliant rights and values. This fosters trust, encourages broader sharing and cooperation, and ensures the appropriate and sustainable (re)use of cultural digital heritage objects, benefiting CHI and the wider cultural and creative sectors.

Scale and significance: By enhancing legal frameworks and clarity, INFINITY will aim to increase the reuse of CHDO, and contribute to the European data strategy[25], particularly in managing digital rights and fostering a creative digital economy. The initial success metric is the extent to which rights metadata is enriched. Given the complexity of measuring this across various CHI and heritage platforms, we will use the Europeana Collections as a representative sample to gauge this metric. Given 78% of the Europeana collections (~40 million objects) are available for reuse, i.e. labelled Public Domain Mark, all CC licenses or InC-EDU. Based on earlier work of EF on metadata enrichment, 40% of the metadata for these collections are enriched by the collection owners (collaborating in the Europeana Aggregator Forum) within five years of project completion. Another key metric is **engagement**, focusing on integrating CHDO into mainstream creative industries through robust legal frameworks and innovative business practices. Although Pan-EU baselines for CHDO reuse are unavailable, the InDICES project highlights a direct correlation between high-quality and readily accessible rights metadata and creative reuse (Tartari, 2022). Europeana Impact Framework will be expanded (T16.3) to explore how various impacts such as economic output and job creation in creative sectors can be effectively measured.

[25] https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/european-data-strategy_en

Expected outcome 3: "European cultural heritage professionals and researchers are aware and make use of new collaboration- and business models based on values and rights embedded in the digital objects throughout the multi-actor value chain."

Immediate results and outcomes: (i) Multilingual AI-driven cross-collection, interlinking tools, (ii) Manuscript transcription tools (OCR, TEI format integration), (iii) AI-driven geospatial data alignment tool (iv) Extended Transcribathon toolset and crowdsourcing tools (v) Open call (FSTP) to engage creative industries and cultural institutions with their technology providers in early adopting INFINITY ecosystem of components and integrate them in their platforms.

Medium term impact: New collaborative networks and business models emerge, facilitated by shared data resources. [achieved through 03 ENHANCE and O6 ENGAGE]

Long term impact: INFINITY's digital tools and methods **are fostering new value chains and business models that bring cultural heritage institutions (CHI) into collaborative networks they were previously excluded from**, particularly due to their collections not being openly accessible. These advancements align with ethical standards and legal frameworks, supporting sustainable practices and enhancing the availability of cultural heritage and knowledge.

Scale and significance. 100 large (more than 100 FTE) and 300 small CHI are actively embedded in collaborative networks within the Cultural and Creative Sectors they were not part of yet. For instance, through joint projects, pilots, and knowledge exchange activities. This integration will support the EU's strategy to enhance digital skills and inclusion, fostering equitable access to digital heritage resources. Success will be tracked through participation rates and the breadth of new digital services offered. This aligns with the objectives of EIT Culture and Creativity, to stimulate innovation in cultural practices and participation. Long-term outcomes will be measured by the sustained engagement of CHI in these networks and their impact on local and European economies, promoting a culturally rich and participatory democratic society. **Baseline:** the EIT-CC Strategic Agenda indicates 12k jobs in existing business in the creative businesses are sustained through innovations supported by the EIT-CC by 2023. (EIT 2024) INFINITY outcomes will positively influence this.

Expected outcome 4: "European cultural heritage professionals and researchers are provided with clear information as well as targeted training modules on the innovative tools and methods developed."

Immediate results and outcomes: (i) Training material designed for the specific target groups (Section 2.2) Workshops and hackathons for training stakeholders and disseminating the methodology material, (ii) Web interfaces to access INFINITY data such as mKG and annotated collections, or to check the legal status of each CHDO, (iii) GUIs to interact with AI tools, whether for transcription, ASR, entity recognition, video annotation, or metadata enrichment, offering manual and automatic options and supporting Explainable AI.

Medium term impact: Professionals who gained training and experience through INFINITY feel confident in using a wide range of advanced tools for heritage management made available via the ECCCH and are able to experiment with different solutions to achieve desired goals. Their awareness of how to use data enrichment techniques in ethical and critical ways is increased. [achieved through O5 INTERACT]

Long term impact: Capacity and skills among heritage professionals are enhanced, enabling them to effectively utilise advanced digital tools and fully engage with the European Collaborative Cloud for Cultural Heritage (ECCCH). Through INFINITY, CHI are introduced to this ecosystem and gain the knowledge needed to leverage its tools and opportunities. This not only enriches their working practices but also opens new pathways for collaboration, innovation, and broader impact within the cultural heritage sector.

Scale and significance. The project targets enhancing the digital skills of 400+ heritage professionals, incorporating advanced digital tools and methodologies into their practices. This effort is in line with the EU's Digital Education Action Plan, aiming to enhance digital literacy across Europe. Metrics for evaluation will include training completion rates and the application of digital tools in heritage preservation tasks. INFINITY will form the basis to establish a continuous learning culture in the sector, the aim is to engage 600+ heritage professionals annually in advanced training modules offered through the ECCCH, Europeana and the EIT CC. INFINITY supports the lifelong learning objectives of the EU[26], aiming to integrate continuous professional development into the cultural heritage sector. Success will be measured by the adoption rate of ECCCH tools. Baseline: the ECHOES project has set a target to train over 3,000 entities in the use of ECCCH tools by 2029.

[26] https://education.ec.europa.eu/focus-topics/digital-education/action-plan/action-1#Proposal

2.1.2 Contributions to wider impacts

Wider impacts from the destination: *"The full potential of cultural heritage, arts and cultural and creative sectors as a driver of sustainable innovation and a European sense of belonging is realised through a continuous engagement with society, citizens and economic sectors as well as through better protection, restoration and promotion of cultural heritage."*
INFINITY will contribute to these expected wider impacts in the following ways: - **Realisation of the FARO Convention[27] goals towards democracy**. By capturing complex multidimensional contexts of CHDO, INFINITY's methods enact the FARO Convention's principles that advocate for the importance of meanings and usages that individuals and communities assign to cultural heritage. Data enrichment strategies and use cases introduced will convincingly showcase the value of including citizen perspectives and representation of different cultural viewpoints and how this can advance discussion about societal topics that are key to healthy democracies. This will encourage CHI to lower existing public engagement barriers and create participation opportunities that encourage citizens to take an active role in accessing and interpreting cultural heritage. The project will highlight this importance of participation in heritage in guidelines towards CHI and other stakeholders created by WP13. - **Improved European competitiveness in the global data economy.** INFINITY will boost diverse innovation ecosystems by providing scientists, technologists and industry adopters with advanced metadata enrichment strategies, multidimensional data and mKG. The ability to capture and represent different interpretations and cultural meanings of data will stimulate societal, technological and scientific innovations (e.g. journalists researching societal trends, publishers or startups in tourism launching new services based on multidimensional data) that will give European companies a competitive advantage in the global market. The project's connection with the EIT Culture and Creative (via partner EIT-CC-SW) will facilitate this connection with innovation ecosystems, f.i. by organising capacity building workshops. - **Reinforced ability to act on urgent societal challenges.** Scientific and technical innovations introduced by INFINITY will enable researchers and professionals from a variety of sectors to more critically reflect on trends and narratives in the historical and contemporary data - especially in relation to biases, gender representation, colonial narratives and visibility (or lack thereof) of marginalised communities, as well as the rights of community to determine how data about them should be described and used. The availability of multidimensional data and critical engagement facilitated by CHI that play a key role in shaping knowledge and opinions with it will create a path towards a more fair and equal representation of social issues and shared topics in cultural heritage, education and media. Results produced by the INFINITY use cases (e.g. data visualisations, platforms for citizen science collaborations) will exemplify this critical engagement with societal challenges.

2.1.3 Requirements and potential barriers

This high-level PESTLE analysis provides a framework for considering risks that may be triggered by external factors and need to be mitigated during the project lifecycle. It will be extended and regularly reviewed (T1.5) to minimise its impact.

POLITICAL - Financing of ECCCH, DS4EU, and other infrastructures relevant to INFINITY is dependent on EU policy. While INFINITY cannot directly influence these policies, the project will deliver actionable data and insights to policymakers, demonstrating alignment with and progress toward EU policy priorities. Additionally, the complex landscape of EU regulations concerning intellectual property, ethics, and end-user engagement necessitates meticulous navigation. INFINITY will facilitate compliance by providing guidance to its partners and coordinating with other related projects to ensure effective adherence to EU regulations. **ECONOMIC -** CHI often face significant challenges in securing adequate funding, which can restrict their ability to grow and innovate, such as integrating new tools or experimenting with emerging technologies. This situation can adversely affect their long-term sustainability. INFINITY will address these financial obstacles through guidelines developed in Work Package 13, tailoring recommendations to enhance accessibility and utility for both large and smaller CHI, thereby fostering broader institutional resilience and growth. The cultural heritage and creative sectors are highly diverse and fragmented, which may challenge the widespread adoption of INFINITY's outputs. INFINITY will conduct thorough market analysis and stakeholder engagement activities to identify and understand the specific needs and characteristics of different market segments. **SOCIAL -** Engaging local communities in cultural heritage projects is crucial yet challenging, particularly when segments of the population feel disconnected from public institutions. INFINITY aims to build trust at a grassroots level by successfully executing its use cases and will disseminate these experiences widely to foster deeper community engagement and participation in cultural heritage initiatives. **TECHNOLOGICAL -** The rapid evolution of technology could outpace the project's outputs, necessitating ongoing adjustments to integrate the latest innovations. **LEGAL -** CCIs and CHI must contend with complex and constantly evolving copyright laws that affect how cultural content is used and shared. Many may lack direct access to legal expertise. To address this, INFINITY will ensure that all partners,

[27] https://www.coe.int/en/web/culture-and-heritage/faro-convention

especially smaller organisations, have access to legal counsel. This will be achieved by facilitating connections between larger organisations, which often have specific legal experts, and smaller institutions requiring legal guidance. **ENVIRONMENTAL** - The energy-intensive nature of processing data with AI systems poses environmental challenges. INFINITY will leverage the expertise of partners like UNIBO and SPA, who have experience in developing energy-efficient algorithms, and PSNC, which operates an energy-efficient data centre, to address these challenges.

2.2 Measures to maximise impact - Dissemination, exploitation and communication #@COM-DIS-VIS-CDV@#

All consortium partners bring extensive expertise in producing high-impact innovations, and have devised the initial **Plan for the Exploitation, Dissemination and Communication of Results (PEDR)** by combining good practices and proven strategies from previous EU projects and similar initiatives. PEDR is envisioned as an agile instrument responsive to the market needs, technological or societal trends and European policies. The plan will be constantly reviewed and when necessary, additional DEC instruments will be added during the course of the project to reach the envisioned impact. Activities in this plan are divided into three distinct stages: **Stage 1 - Awareness building**. At the start of the project, the communication strategy will focus on raising awareness among the stakeholders community of the rationale and ethos of the project. To connect with the ECCCH, INFINITY will join the ECHOES Stakeholder Network and (as part of WP11 and WP12) the ECHOES Implementation task force. The project will establish a Stakeholder Network with representatives from all the stakeholders groups. **Stage 2 - Participation**. Support target groups understand the concepts and outcomes of INFINITY. As the first project results (f.i. the use cases) become available (Milestone 2), outreach activities will become more frequent. **Stage 3 - Action**. INFINITY outcomes are broadly accessible, supporting future research activities through the ECCCH and further exploitation. The 8 target groups impacted by INFINITY's outcomes are listed below, highlighting the key value propositions for each.

Table 2.1 INFINITY Target Groups

Target group	Description & value proposition
Cultural heritage institutions (CHI)	Museums, archives, and libraries enhancing CHDO in the ECCCH through the use of INFINITY tools for advanced metadata enrichment and ethical data sharing. We will target organisations of different sizes, including small- and medium sized CHI. **Value proposition**: Leverage advanced digital tools and AI technologies to enhance the accessibility and preservation of cultural artifacts. Use of KGs, multimodal data analysis, and AI to enrich metadata and user experience.
Creative industries (CI)	Innovators in media, design, and technology transforming CHDO into commercial and educational products using INFINITY tools for innovation and sustainability. **Value proposition**: Capitalise on digital innovations and enriched cultural content to create new products and services. Integration with technology developers can lead to novel creative expressions and digital experiences.
Technology developers and service providers (TEC)	Professionals building and maintaining CHDO tools (notably Collection Management Systems) and infrastructure, benefiting from INFINITY's frameworks for interoperability and ethical AI integration. **Value proposition**: Opportunity to develop and implement cutting-edge technologies in AI, cloud computing, and data management within the cultural heritage sector. Collaboration with CHI and CI can open new markets and applications for existing technologies.
Researchers in humanities and social sciences (RES-SSH)	Academics advancing multidisciplinary research with access to semantically rich, ethically curated CHDO. **Value proposition**: Access to richer, multidimensional data sets for research, powered by AI and machine learning tools, enhances the depth and scope of research outputs.
Researchers in computer science (RES-CS)	Experts in AI, semantic web, and machine learning leveraging INFINITY's innovations in metadata and knowledge graph technologies. **Value proposition**: contribute understanding of how technology can be applied responsibly and effectively in culturally significant contexts. Access to training data.
Data infrastructures and Standards developing organisations (DATA)	Entities like the Data Space Support Centre (DSSC), Gaia-X,W3C, PARTHENOS and SSHOC help ensure INFINITY tools align with global standards for interoperability and adoption. **Value proposition**: INFINITY provides insights helping develop standards for data interoperability and ethical management across digital platforms.
European citizens (ECIT)	General public and participants in crowdsourcing projects like Transcribathon, contributing to and using enriched cultural heritage data. **Value proposition:** improved access to cultural heritage through digital platforms and interfaces, also allowing for engagement through citizen science collaborations.
Cultural heritage	Decision-makers who are ensuring ethical governance and sustainable practices for CHDO

policy makers and managers (POL)	using INFINITY frameworks. **Value proposition**: Inform policy development with insights on new business models and evolving societal impact of CHI as data in enriched and more widely available

2.2.1 Dissemination and communication strategy

INFINITY partners identified key dissemination and communication instruments, detailed in Table 2.2.

Dissemination strategy

Dissemination instruments will help target groups understand and benefit from INFINITY's insights and tools. An initial analysis of the target groups has led to the creation of 10 dissemination instruments that have been chosen for their proven effectiveness, utility, and complementarity.

	DISSEMINATION										COMMUNICATION					
	INFINITY Stakeholder Network	Guidelines for professionals	Benchmarking workshops	Partner networks	Use case demonstrators	INFINITY showcase events	Capacity building workshops	INFINITY open call webinar	Professional events	Academic outreach channels	Project website	Press releases	Social media channels	Mailing lists & newsletters	Open days	Broader comm. outreach
Cultural heritage institutions (CHIs)	o	o	o	o	o	o	o	o	o		o	o	o	o	o	o
Creative industries (CI)	o	o		o	o	o		o	o		o	o	o	o	o	o
Technology developers & service providers (TECH)	o	o		o	o						o	o	o	o		o
Researchers in humanities and social sciences (RES-SSH)	o	o		o	o	o				o	o	o		o		o
Researchers in computer science (RES-CS)	o	o		o	o					o	o	o		o		o
Data infrastructures and Standards developing organisations (DATA)	o	o	o	o	o		o		o		o	o		o		o
European citizens (ECIT)					o						o		o		o	o
Cultural heritage policy makers (POL)	o				o				o		o	o	o	o		o

Table 2.2 Dissemination and communication instrument mapping for different target groups

- The **INFINITY Stakeholder Network** will be set up (starting in month 3) to guide the project throughout its duration. All representatives of the stakeholder groups from across the EU are invited to join. It is an informal network of professionals, consisting of at least 80 professionals by the end of the project. Members will receive regular progress updates, and will be invited to contribute to the evaluation process. To ensure their active participation, representatives of the Stakeholder Network will be invited to join dissemination activities as participants, co-organisers, speakers, experts, and critical validators;
- WP13 will deliver **Guidelines and other materials targeted at four professional stakeholder groups**: (i) cultural heritage sector - focusing on applying multi-contextual data enrichment (ii) cultural heritage infrastructures - explaining how existing (inter)national infrastructures could endorse and incorporate new business models for data management; (iii) creative industries - exemplifying new business and innovation opportunities, and (iv) SSH academics - providing insights into how INFINITY can be useful for research and teaching. These materials will be digital, story-driven, stored online and made publicly available under Creative Commons licenses. These guidelines will play a central role in the project's capacity building and exploitation efforts;
- **Benchmarking Workshops** (T4.4): Relevant stakeholders from CHI, CIs and TECH (acronyms defined in the table above) will be actively involved to evaluate the performance and implementation approaches of different collection management systems' components (SoA of functionalities to be developed and extended in INFINITY), providing insights to inform the work in WPs 8-10;
- INFINITY will maintain close ties with **Partner Networks and Related Projects**, especially those working on CL2-2023-HERITAGE-ECCCH-01 and related calls (WP11, WP12) as well as the projects listed in Section 1.2.4;
- Use Case Demonstrators (WP5) will be central to building INFINITY's community of practitioners. The implementation and evaluation activities will attract CHI and CI representatives. The project will also investigate the applicability of its demonstrators to other industrial and societal challenges (WP5, WP6, WP7) thus attracting a broader range of stakeholders from other disciplines and sectors;
- The project will plan ad-hoc **Showcase Events (T14.1, T15.1, T16.1)** to open up the project's processes and discuss its results with CHI, CI, and RES-SSH. This includes organising onsite or online demonstrations during existing conference and industry events, co-organising activities with related activities or projects, and participating in innovation showcases organised by local or sectoral ecosystems connected to the consortium partners.
- The project will organise eight **Capacity-Building Workshops** (four online and four on-site) for CHI and CI (T14.1, T15.1, T16.1). INFINITY will host 2 workshops and 2 hackathons to drive innovation and collaboration between CHI, creative industries and developers. Workshops will focus on integrating digital objects and creating sustainable business models for cultural heritage data. Hackathons will develop digital solutions, such as adaptive exhibition, immersive AR applications, fostering partnerships and scaling innovation. Attention will be paid to ensure that these workshops have a wide geographic reach - online workshop formats will remove travel barriers and efforts will be made to organise physical workshops outside of capital cities and Western Europe.
- One **Open Call Webinar** will be organised to provide an overview of the FSTP call's scope and eligibility criteria and to serve as a forum for addressing questions from potential applicants.
- INFINITY will present its outcomes at **Professional Events** all over Europe, including: (i) events hosted in the context of the ECCCH, European data space for cultural heritage, and other large-scale initiative such as the European Heritage Hub, the EIT Culture and Creativity, and AI4LAM; (ii) domain-specific conferences, such as MuseumNext, ICOM,

CIDOC, NEMO's European Museum Conference, Time Machine Conference or the Digital Heritage Conference; (iii) national events.

- Research and technology partners will present the project's results via various **Academic outreach channels**. This includes: (i) (Inter)national academic conferences, f.i.: Conference on Language Resources and Evaluation, International Conference on Theory and Practice of Digital Libraries (TPDL), ADHO and EADH Digital Humanities Conferences, ICOMOS CIPA Symposium, Euromed, 3DArch, IEEE Digital Heritage and ACM Multimedia, Web3D, EuroVis; (ii) academic journals, f.i.: Journal of Cultural Policy and Management, IEEE Trans. on Pattern Analysis and Machine Intelligence, Journal of the Association for Information Science and Technology (JASIST), International Journal of Heritage Studies (IJHS), IEEE Trans. on Multimedia, ACM Trans. on Intelligent Systems and Technology, IEEE Trans. on Knowledge and Data Engineering, ACM Journal on Computing and Cultural Heritage (JOCCH), ISPRS journals, PLOS One.

Dissemination activities will be **dynamically aligned with exploitation strategies**, ensuring that they remain relevant to current conditions. As the project progresses and engages with target groups, new dissemination efforts may be introduced to bridge identified knowledge gaps and meet emerging needs.

Communication strategy

Communication activities are executed throughout the lifecycle of the project, aimed at enhancing visibility and awareness among targeted groups. The strategy will involve a coordinated effort by all partners and will leverage the expertise of communications professionals working on the ECHOES project to align with the overarching communication strategies of the European Collaborative Cloud for Cultural Heritage:

- Project Website, integrated in the ECCCH website (echoes-eccch.eu) with basic information about the project, including use case descriptions, a repository for open documents and a blog with project updates;

- Press Releases and direct media contacts will provide journalists with information about objectives / outcomes, to be published in local and national mainstream (targeting the broader society) and specialised media outlets;

- Social Media accounts to be set up in M1. Emphasising culturally relevant, dynamic content and recognising the B2B nature of the ecosystem, the focus will primarily be given to professional social media channels such as LinkedIn and BlueSky;

- Newsletters to be issued periodically (5 times/yr.) and circulated in the relevant communities, with adherence to the requisite GDPR regulatory requirements.

- Existing **Mailing Lists** (including Europeana Tech, Europeana Research, AI4LAM, etc.) will be used to crowdsource knowledge, engage in community discussions and invite participation in the project's activities;

- Events Organisation. It is planned to organise 6 workshops (4 online and 2 in person) and 2 hackathons to have contacts with stakeholders and industry representatives. (See Section 2.2.1.)

- Open Days will be organised in each country (T14.1) represented in the consortium and demos will be presented to citizens, collecting feedback INFINITY-enabled experiences, in connection with existing science communication events, f.i. organised by EU STEM Coalition, the non-profit network supports better STEM education in Europe and annual science communication related events in EU member states (f.i. the Netherlands *"Weekend of Science"* organised every October).

- Broader communication outreach. Existing communication channels of the consortium members and their business networks (see Section 3.2), as well as established EU communication channels such as CORDIS, the Horizon Magazine, Open Research Europe, the AI Media Observatory and the Horizon Results Platform will facilitate a broad diffusion on the goals and outcomes of INFINITY.

In all its dissemination and communication activities, the project will aim to use **accurate and appropriate visuals** and language. Specifically, the project will adhere to the guidelines from the Better Images of AI initiative[28] to ensure that visuals used do not create a distorted or unrealistic understanding of AI but rather help to demystify it. Table 2.3 lists KPIs for both Dissemination and Communication. To gauge the effectiveness of ongoing strategies and adjust them as needed, structured impact measurement and feedback mechanisms will be put in place (T14.1, T15.2, T14.1).

DISSEMINATION KPIs	COMMUNICATION KPIs
# professionals part of the stakeholder network at project end: ≥ 90	# unique website visitors: ≥ 1.000 p.m. (M18), ≥ 5.000 (M42)
# Benchmarking Workshops: 4	# website page views: ≥ 5,000 p.m. (M18), 25,000 (M42)
# participants in Benchmarking Workshops: ≥ 800	# project video views: ≥ 500 (M18); ≥ 1.500 (M42)
# Memorandum of Understanding with Partner Networks ≥ 25	# LinkedIN followers: ≥ 500 (M18), ≥ 1.500 (M42)
# Use Case Demonstrators: 5	# BlueSky followers: ≥ 200 (M18), ≥ 800 (M42)
# Guidelines for Professionals: ≥ 800 downloads by M42	# mainstream media mentions: ≥ 5 p.a.
# Capacity Building Workshops: 4. Attended by 80 pax each	# newsletters ≥ 18 published by M42
# refereed scientific publications: ≥ 25	# mailing list subscribers: ≥ 200 (M18); 750 (M42).
# average journal impact factor: ≥ 1.5 SCI R	# Broader comm. outreach: ≥ 80 times main stream media attention
# scientific presentations: ≥ 10 p.a.	# Open Days: 16 (2 in each of the 8 countries)
# co-hosted events: ≥ 2 p.a.	# open day attendees: ≥ 50 participants per event
# industry presentations: ≥ 10 p.a.	# press releases: ≥ 3 (in ≥ 4 languages)
# INFINITY Open Call webinar: 110 professionals registered	

Table 2.3 Dissemination and communication KPIs

[28] https://blog.betterimagesofai.org/better-images-of-ai-guide/

2.2.2 Exploitation of project outcomes

Table 2.4 presents an initial inventory of Key Exploitable Results (KER). These services are crucial not only to empower CHI in effectively deploying INFINITY tools but also to offer partners opportunities to expand their impact (Column "Market" - See table 2.2 for an explanation of the acronyms used) and generate additional value through premium services. This strategy emphasises aligning deployment and integration of tools and applications with the broader objectives of ensuring ethical, sustainable, and impactful contributions to the cultural heritage landscape. Exploitation of these KERs happens directly **in the ECCCH** ecosystem (Section 2.2.2.1) and **outside of it** (Section 2.2.2.2) by the consortium members.

Table 2.4 Key Exploitable Results of INFINITY. The technology type is one or more of the following: MOD: a model or methodology specification; LIB: a reusable software component; API: an operating software exposed as Web API or Service; USE: a tool for our target consumers (cultural heritage and creative industry professionals and researchers, academic and education professionals, citizens).

KER ID	Type	Description, related task and type, lead partners	Market(s)
1	MOD	Ontology network and mapping schemas. Knowledge graphs of cross-collections alignments.	CHI, TEC, DATA, CI, RES-SSH, RES-CS
2	LIB API	Multilingual, multimodal (text, images) cross-collection entity and concept recognition/linking and metadata alignment.	CHI, TEC, DATA,CI
3	LIB API	Multilingual AI-aided manuscript transcription.	CHI, TEC, DATA, RES-SSH
4	LIB API	AI tools for geolocalisation of images, data synchronization with authority data providers, toponym extraction.	CHI, TEC, DATA, RES-SSH, RES-CS
5	LIB API	AI-assisted annotation tools supporting multi-context representation of CHDO.	CHI, TEC, DATA, RES-SSSH, RES-CS
6	LIB	(Large) Language Models tuned to learn multilingual text and concept embeddings, and fine-tuning pipeline.	CHI, TEC, DATA, RES-CS
7	LIB API	Ethical validation and bias detection tools.	CHI, TEC
8	LIB API	Content provenance tracking tools, user consent management for AI-generated datasets.	CHI, TEC, CI
9	LIB API	IPR tools to encode and manage ownership, usage rights and reusability.	CHI, TEC, DATA, CI
10	MOD LIB	mKG data model and software for creation, deployment, querying, evolution management.	TEC, DATA, RES-CS
11	USE	Web interfaces to edit, browse, query, and visualise multidimensional KGs. KGs. (T6.1-6.5) - (i) - F&F, SPA, UNIBO, NISV, UNI JENA, PSNC	DATA, TEC, RES-CS, CI, ECIT
12	USE	GUIs to interact with AI tools offering manual and automatic options, providing explanations, and validation support	TEC, DATA, CHI, CI, ECIT
13	LIB API	Tools for embedding and tracking the lifecycle of CHDO as mKG: skills, methods, preservation status, history of changes.	CHI, TEC, DATA
14	USE	Transcription interface: interactive tool for transcribing handwritten materials with AI-assisted suggestions.	CHI, DATA, TEC, RES-SSH, ECIT
15	USE	4D multilingual visualization tools (geospatial and temporal layering of data).	CHI, DATA, TEC, RES-SSH, ECIT
16	USE	Permission and rights management interface.	CHI, DATA, TEC, ECIT
17	LIB API	Extended Transcribathon toolset and crowdsourcing tools.	CHI, TEC, DATA, CI CHI, TEC, DATA, ECIT
18	USE	GUI for collaborative multi-perspective annotation enrichment and verification. (T6.1-T6.5) - (i), (ii) - F&F, UNIBO, NISV, UNI JENA,	CHI, DATA, TEC, ECIT

		PSNC	
19	LIB API	Plug-in and web service architecture for integrating INFINITY components into existing platforms (e.g., dLibra, CLARIAH Media Suite). (T6.1-T6.5, T9.2) - (i) - UNIBO, PSNC, SPA	CHI, DATA, TEC, RES-SSH
20	MOD	Guidelines for the ethical annotation and use of CHDO . (T8.1, T9.1) - (iv) - UEDIN	CHI, TEC, CI, RES-SSH, RES-CS, POL
21	USE	GUI for encoding, querying and visualising lifecycle history of CHDO.	CHI, TEC, DATA, ECIT
22	USE	GUI for bias detection and ethical validation.	CHI, TEC, DATA, ECIT
23	MOD	Benchmark for validating the capability and maturity level of CHDO collection management processes and tools. Guidelines and training material for reusing INFINITY ecosystem software components.	CHI, TEC, CI

2.2.2.1 Exploitation in the context of ECCCH

The Exploitation, Dissemination, and Communication strategy (T14.3, T15.3, T16.3) will be developed collaboratively with the ECHOES consortium and other projects and initiatives within the ECCCH network. KERs are designed to be modular and scalable. incremental adoption and integration with existing systems. For each KER, costs related to maintenance, operation, training will be assessed to guide financial planning. INFINITY's involvement in the ECHOES Integration Task Force will inform the exploitation conditions. UEDIN will bring their experience in the READ-COOP, the European Cooperative Society that successfully sustains the Transkribus platform. The precise modalities and business requirements for integrating the outcomes—whether tightly or more loosely—will shape the sustainability planning IP management (see Section 2.2.5), and contractual arrangements.

2.2.2.2 Further exploitation activities

Exploitation opportunities for INFINITY extend well beyond validated use cases and their respective applications. This project not only provides opportunities for the technology and research partners involved but also opens doors for many other organisations with complementary competences outside the consortium, including the ECCCH community. INFINITY's tools and services are open source but come with a licensing structure that allows for the development of commercial applications. By integrating with existing workflows in CHI, INFINITY fosters collaboration, partnerships, and new market ventures, especially benefiting small and medium-sized enterprises. The growing market of Collection Management Software (CMS) is particularly relevant in this context since INFINITY's tools (or their extensions) could significantly extend the capabilities of these systems. Recent research valued the global CMS market at USD 312.10 million in 2023, a figure expected to reach USD 349.65 million in 2024. The estimated compound annual growth rate (CAGR) is 13.98%, reaching USD 780.19 million by 2030[29]. Major players in this market include Ex Libris, Axiell, Adlib (used by NISV), TMS Collections, and MuseumPlus, many of them European companies. Alongside commercial systems, open source CMS platforms such as dLibra (used by UWr), CollectiveAccess, and Omeka-S, are not included in these market numbers, but also signify considerable economic value. Given this context, INFINITY's outcomes and enriching integrations are poised to make a significant impact.

Four service providers will bring results to the market: SPA offers an innovative AI-based text editor for storytelling, built on KERs 1, 6, 10b, and 19. This tool integrates multilingual, multi-contextual metadata into existing content management systems (CMS), helping institutions expand their audience engagement while preserving cultural sensitivity in presenting heritage data. / F&F DS expands its Transcribathon platform using KER 17, enabling CHI to organise citizen science campaigns that enrich their digital collections. The platform combines open access with tailored customisation services, ensuring sustainability and active public participation. / DP provides consulting and services for implementing ethical frameworks in metadata management. Offer solutions customised for cultural institutions aiming to comply with global data standards and ethical guidelines KER 13,16, 20, 21] / PSNC exploits KERs 10a, 11, and 13 by providing cloud-based services for large-scale deployment of INFINITY components. PSNC, using KERs 18, 19, enables CHI to integrate high-performance knowledge graph tools, modernising their data workflows and infrastructure. When exploiting INFINITY results, these **four partners will consider two models** that have proven to be successful in this market: (i) selling new service offerings that have been developed utilising the open-source implementations (ii) charging for consultancy services.

Other partners have also anticipated their exploitation plans: UNIBO focuses on incorporating INFINITY tools into its educational programs, such as master's degrees in Artificial Intelligence and Digital Humanities, while advancing research in cultural heritage technology. Leveraging KERs 2, 5, and 10a, UNIBO collaborates with public

[29] https://www.giiresearch.com/report/ires1587279-collection-management-software-market-by-component.html

administrations and digital companies, facilitating the integration of knowledge graph applications and multilingual metadata tools into local and national heritage projects. / UNI JENA provides tools for spatial and temporal visualisation of cultural data, targeting CHI and research organisations. Focus on enhancing storytelling through geospatial alignment (KER 4). / EF utilises KERs 14, 17 and 21 to launch EU-wide participatory campaigns that enrich cultural collections. Through crowdsourcing and community-driven initiatives, EF demonstrates how INFINITY tools can foster engagement and make cultural heritage more accessible to the public. KER 20 is used to ensure the interoperability between the ECCCH and the DS4CH and the integration of the project results as part of those two initiatives. / NISV integrates INFINITY tools (KERs 11, 12, 13, 16, and 18) into the CLARIAH Media Suite, providing advanced metadata enrichment and visualisation capabilities to the Dutch research community. Additionally, NISV applies KER 21 guidelines to design future AI-driven solutions, enhancing the accessibility of audiovisual collections. / TMO integrates and opens INFINITY tools to the network of local time machines and enables the processing of local data (KER 15). Additionally it will promote the use of KER 13, 14 and 16 through “Time Machine Academies” educational sessions directed to the hundreds of institutions of its network. / UEDIN promotes KER 20 and 21 through the development of openly licensed ethical toolkits for linked open data. By training CHI and technology providers, UEDIN ensures the responsible use of INFINITY tools, fostering trust and compliance in managing cultural data, while providing guidance on scaffolding activities towards innovation. / UWr leverages KER 17 through the dLibra CMS, enhancing collaboration among CHI institutions. By facilitating joint projects, UWr enables CHI to share and enrich digital collections, expanding their accessibility. / EIT-CC-SW explores partnerships with creative industries, focusing on immersive storytelling products powered by KER 21. These collaborations demonstrate how enriched CHDO can unlock new business models and creative opportunities. / EPFL will integrate KERs 1, 4, 15 into the geohistorical platform it develops, directly targeting CHI and research organisations. / CNR capitalises on KERs 7, 8, and 9 by developing tools for bias detection, explainability, user consent management, and ethical validation. Through its spinoff, BUP Srl, CNR provides consulting and technology solutions to CHI and public administrations, ensuring responsible and inclusive AI adoption in cultural heritage workflows.

2.2.4 Contribution to Standardisation efforts.

Linked Data & Knowledge Graphs. UNIBO, SPA and EF engage with W3C Linked Data Platform (LDP) and the Knowledge Graph Construction Community Group to enhance the interoperability and scalability of cultural heritage data. The W3C Dataset Exchange WGs DCAT vocabulary guides INFINITY's data cataloguing practices, supporting FAIR principles. Collaborations with the Linked Data Benchmark Council ensure compliance with graph benchmarks, fostering robust and scalable graph-based solutions. Active participation by EF and PSNC in the DSSC and Gaia-X will ensure INFINITY outcomes will also inform the European Data Spaces. **AI Ethics.** INFINITY contributes to ethical guidelines such as the AI HLEG Ethics Guidelines for Trustworthy AI, OECD AI Principles, and standards like ISO/IEC 24029-1:2021 (robustness of neural networks), ISO/IEC TR 24027:2021 (bias in AI systems), and IEEE 7001-2021 (transparency of autonomous systems).

2.2.5 Intellectual property (IP)

An Intellectual Property Rights (IPR) Register (Task 14.2) will be set up to document third-party software usage, consortium member agreements on access rights, and licensing terms for post-project technology use. This register is key for managing and protecting intellectual property, consistent with Horizon Europe standards and ECCCH requirements. INFINITY will default to a free-to-use policy for its software to promote early adoption and facilitate integration into existing industry practices. The project will adopt open-source strategies that align with the commercial activities of consortium members, including SPA, PSNC, and F&F, who will primarily handle the direct exploitation of developed technologies. The consortium will ensure IP measures are established in the Consortium Agreement and adjusted as needed. Rights for IP transfer or licensing will be considered, especially for members like non-profit research centres that might integrate these technologies into broader applications. Protection strategies will prevent the unauthorised disclosure of project results, termed "Foreground", until protection decisions are finalised. Background IP will remain with the original owner, but access rights will be adjusted as necessary for the project, as specified in the Consortium Agreement. Rights for joint exploitation will be detailed in separate Business Agreements to support strategic project planning. #§COM-DIS-VIS-CDV§

2.3 Summary

SPECIFIC NEEDS	EXPECTED RESULTS	D&E&C MEASURES
CHI often fail to reflect the complexity and richness of CHDO, presenting overly simplistic or monolithic representations that neglect the multi-dimensional interpretations of CH. Collections are often not connected to each other, resulting in fragmented access and limited opportunities for collaborative exploration. Furthermore, appropriate reuse of CHDO is limited, hindered by insufficient metadata, non-interoperable systems, and a lack of clear guidelines. These challenges highlight the **need for robust solutions that prioritise cross-collection connectivity, enhance metadata quality, and foster interoperability**, enabling the ethical reuse of CHDO and supporting a holistic understanding of their contexts. The integration with DS4CH and ECCCH further supports advanced data enrichment, embeds professional value, and facilitates innovative business models within the sector.	- Creation of **tools that enable deep, contextual metadata enrichment** for multimodal and digital heritage assets, improving their accessibility and interpretability. - Development and deployment of **Multidimensional Knowledge Graphs** that facilitate a nuanced understanding of CHDO from multiple perspectives, incorporating diverse cultural, historical & social contexts. - Implementation of AI technologies to **enhance the ethical (re)use and management** of digital cultural heritage, ensuring compliance with both legal standards and community ethical norms. - Establishment of **interoperable data models** that support seamless integration of INFINITY's tools and methods in existing (heritage) infrastructures and collection management systems. - Provision of **extensive training materials**, guidelines, and documentation to support cultural heritage institutions in adopting and maximising the benefits of INFINITY's innovations. - **Strengthening of a network** of CHI professionals, academics, technologists, and policymakers through workshops, seminars, and FSTP to foster widespread adoption and knowledge exchange.	(i) **Awareness Building**: Engaging with the CH community through workshops, seminars, and digital media to raise awareness about the project's goals and benefits. (ii) **Participation**: Encouraging active participation of target groups through targeted training sessions and demonstrations, ensuring they understand and can implement the project's tools and methods. (iii) **Action**: Ensuring the integration of project outputs into everyday practices through policy recommendations, guidelines, and standard-setting activities. These measures are supported by a dynamic Plan for the Exploitation, Dissemination, and Communication of Results, which adapts to emerging trends and feedback from stakeholders.

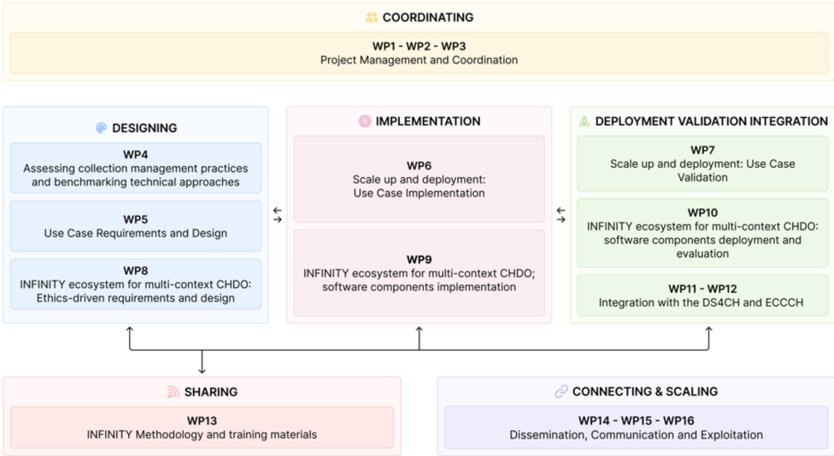
TARGET GROUPS	OUTCOMES	IMPACTS
Cultural Heritage Institutions (museums, archives, and libraries) will use the project's tools for digital preservation and public engagement. **Creative Industries** will leverage enriched cultural data for product development and market expansion. **Technology Developers** focusing on digital solutions for cultural heritage will integrate INFINITY's innovations into their offerings. **Data infrastructures** will develop standards for data interoperability and ethical management across platforms. **Researchers** in digital humanities who will use the project's data sets for research and teaching, in computer science they will reuse tools & data. **Citizens** particularly through participatory platforms like Transcribathon benefit from increased access to and engagement with CHDO. **Policy makers** will gain actionable insights.	-Access to metadata enrichment tools which integrate with ECCCH and DS4CH - Professionals equipped with skills to use the tools in the INFINITY Ecosystem. - A sound strategy for sustainability and exploitation in place. - Support structures and scaffolding for innovation with CHDO content	ECCCH becomes a central hub for enriching, sharing, and ethically using cultural heritage content across Europe. - **Scientific**: Advancements in digital heritage technologies will foster new research methodologies and insights. - **Economic**: The creative and digital sectors will see growth through new business models and increased efficiency. - **Societal**: Enhanced public access to cultural heritage promotes cultural diversity and societal cohesion, contributing to a stronger European identity and democratic values. These impacts align with the goals of Horizon Europe, driving sustainable innovation and reinforcing a sense of European belonging through cultural heritage.

#§IMP-ACT-IA§#

3. Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#

3.1 Work plan and resources

Figure 3.1 PERT chart



The general approach and workflow of INFINITY are shown in Figure 1.2.2. The PERT chart (Figure 3.1) shows the connection between the WPs. WP4 concerns the evaluation of the state of play of existing practices and technologies for CHDO management and metadata annotation. WP5-6-7 is where the five use cases are executed, WP8-9-10 are the technology providers, WP11-12 provide socio-technical supervision to the technology providers to ensure a seamless and sound integration with ECCCH and DS4CH. WP13 collects, organizes and produces training material for the relevant target stakeholder categories (cultural heritage professionals, heritage infrastructures, creative industry, academic and education professionals). WP14-15-16 deal with dissemination, exploitation, and communication activities. WP1-2-3 address the overall project coordination and management and provide for a robust FAIR-driven Data Management Plan and guidelines on self-assessment of ethical principles to the overall project. WP4, WP5, and WP8 address the requirements and design phase of the project and provide the foundation for the implementation phase, that is responsibility of WP6 and WP9. WP7, WP10 and WP11-12 realize the deployment, integration (with ECCCH and DS4CH) and validation of the INFINITY digital ecosystem building on the outputs of the implementation WPs (WP6, WP9). All these WPs contribute incrementally to develop the INFINITY methodology substantiated through a set of organized materials for supporting training and expertise gaining of cultural heritage professionals, maintainers and practitioners of CH research infrastructures, creative industry and academic research and education. WP4 to WP13 represent the research and development workpackages, they interact constantly with WP14-15-16 (dissemination, communication and exploitation) and WP1-2-3 that concern ensuring engagement, adoption and smooth realization of the project goals.

Figure 3.2 Gantt chart

WPs	Year 1												Year 2												Year 3												Year 4					
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42
WP1-WP2-WP3 Project Management and Coordination			D1.1			D1.2						D1.3										D2.1														D3.1						
WP4 Assessing collection management practices and benchmarking technical approaches						D4.1 D4.2						D4.3																														
WP5 Use Case Requirements and Design												D5.1 D5.2																														
WP6 Scale up and deployment: use case implementation																														D6.1												
WP7 Scale up and deployment: use case validation																																										D7.1
WP8 INFINITY ecosystem for multi-context CHDO: Ethics-driven requirements and design						D8.1						D8.2 D8.3																														
WP9 INFINITY ecosystem for multi-context CHDO: software components implementation																				D9.1				D9.2																		
WP10 INFINITY ecosystem for multi-context CHDO: software components deployment and evaluation																																D10.1				D10.2				D10.3		
WP11 Integration with the DS4CH and ECCCH																								D11.1																		
WP12 Integration with the DS4CH and ECCCH																																										D12.1
WP13 INFINITY Methodology and Training Materials																																										D13.1
WP14-WP15-WP16 Dissemination, Communication, Exploitation			D14.1									D14.2 D14.3												D15.1						D15.2 D15.3						D16.1						D16.2 D16.3
MILESTONES								M1												M2												M3								M4		M5

Table 3.1a: List of work packages

WP No	Work Package Title	Lead Parti. No	Lead Part. Short Name	Person-Months	Start Month	End month
1	Project Management and Coordination 1	1	UNIBO	18,4	1	12
2	Project Management and Coordination 2	1	UNIBO	18,7	13	30
3	Project Management and Coordination 3	1	UNIBO	18,9	31	42
4	Assessing collection management practices and benchmarking technical approaches	3	SPA	28	1	12
5	Use Case Requirements and Design	4	UNI JENA	50,9	1	12
6	Scale Up and Deployment: Use Case Implementation	8	PSNC	94,5	6	30
7	Scale up and Deployment: Use Case Validation	8	PSNC	47,9	31	42
8	INFINITY ecosystem for multi-context CHDO: Ethics-driven requirements and design	13	UEDIN	67	1	12
9	INFINITY ecosystem for multi-context CHDO: software components implementation	1	UNIBO	89,5	9	30
10	INFINITY ecosystem for multi-context CHDO: software components deployment and evaluation	1	UNIBO	71	24	42
11	Integration with the DS4CH and ECCCH	5	EF	11,5	1	24
12	Integration with the DS4CH and ECCCH	5	EF	10,5	25	42
13	INFINITY methodology and training materials	6	NISV	34,3	19	42
14	Dissemination, Communication, Exploitation	9	DP	28	1	12
15	Dissemination, Communication, Exploitation	9	DP	37	13	30
16	Dissemination, Communication, Exploitation	9	DP	33,6	31	42

Table 3.1b: Work package description

Work package number	**1**
Work package title	Project Management and Coordination 1

Objectives. The overall objective of the coordination and management WP is the smooth realization of the project. The following specific objectives are set: 1) to coordinate and supervise the project research and innovation (R&I) activities to be carried out according to the Work Plan, also by monitoring and ensuring quality and timing of project deliverables; 2) to carry out the overall administrative and financial management and reporting of the project; 3) to manage contacts with the EU Commission and establish effective internal and external communication; 4) to solve possible conflicts within the consortium; 5) to assure the data management and protection of the outputs generated by the project (DMP); 6) To set up an Advisory Board including an Ethics Advisor and a representative from ECHOES and maintain regular interaction with them 7) To oversee that coordination mechanisms are put in place in WP11-12 to ensure synchronized planning and synergy with the integration TF lead by ECHOES.

T1.1 Coordination, project meetings and reporting (Lead: UNIBO; Participants: All; M1-M12). The coordination and management structure of the project is made up of the following bodies. **The Coordinator**, (UNIBO), will be the ultimately responsible for the overall project coordination and will act as intermediary between the Consortium and the European Commission. The Coordinator will organize, implement and follow up a kick-off meeting and periodic meetings (at least two per year). The Coordinator, assisted by the Project Manager, will convene and manage the project meetings with the support of the project partner. The **General Assembly**, comprised of one member from each partner institution, will be the **steering and decision- making body. Work Package Leaders** will supervise respective WP activities, as the executive body. The **Project Manager** will be appointed within the European

Program and Project Office of UNIBO, he/she will assist the General Assembly and the Coordinator in all administrative and financial duties. Their specific tasks and procedures will be described in the Consortium Agreement adopted by the consortium before the project start. Moreover, regular coordination mechanisms in order to ensure synchronised planning as well as synergy and/or complementarity of deliverables and outcome will be periodically carried out. The **Technical Bord** (TB. WP8) will be part of the coordination and management structure in order to support the advancement of the project. The data steward is part of the TB**.** Efficient Communication with the European Commission will be guaranteed by the Coordinator, who will represent the only intermediary between the Consortium and the European Commission. The activities involve developing and maintaining a detailed project plan, tracking progress, and addressing any arising technical or organisational issues. Additionally, the task incorporates quality management to ensure that all deliverables meet the set standards and benchmarks. An **Interdisciplinarity Working Group** will also be installed (chaired by a senior project member) to guarantee that the project's interdisciplinary approach is adhered to. An **Advisory Board** (AB) will be set up including external experts from Cultural Heritage, Creative Industry, Social Science and Humanities, and Computer Science communities. The AB will also include an ***Ethics Advisor*** and a senior representative from the ECHOES project. Regular communications with the AB are planned to manage quality assurance for scientific deliverables and ensure ethical compliance.

T1.2 Overall legal and contractual management (Lead: UNIBO; Participants: All; M1-M12). UNIBO as Coordinator, assisted by the Project manager, will supervise and guarantee the coherence between the project activities and the procedures described in the Grant Agreement and the Consortium Agreement. The Coordinator will lead discussion on amendments and revisions of these Agreements if necessary.

T1.3 Financial and administrative management (Lead: UNIBO; Participants: All; M1-M12). This task ensures the financial integrity of the project. It entails establishing and maintaining accurate financial records, coordinating the submission of cost statements by all partners, and managing European Commission payments and distribution of shares according to consortium agreements. A Project Handbook will consolidate general operational information, including contact details, partner roles and responsibilities, processes, templates, and procedures for deliverables preparation. The Project Manager will guarantee assistance to the consortium on administrative and financial issues.

T1.4 Data Management (Lead: UNIBO; Participants: All; M1-M12). The aim of this task is to deal with the definition and sharing of policies and the identification of actions for data management and associated software produced in INFINITY. A Data Management Plan will be created to support the management of the outputs of the project during its development and after is end. The FAIR data principles will be the leading element in the development and implementation of the DMP. It will ensure that personal data generated or acquired during the project execution will be ethically managed according to the General Data Protection Regulation (GDPR), the national laws implementing ePrivacy Directive 2002/58/EC (and its forthcoming revision), and the EU Charter of Fundamental Rights. The DMP is a dynamic document to which all partners will actively contribute throughout all the project duration. It will provide the identification elements and the descriptions of the data sets and will include details regarding how the research data will be handled during the project and how they will be preserved after it is completed. It will specify which methodologies and standards will be used in the data creation and management and how and when the data will be shared and made open for re-use. It will be continuously updated according to possible needs, evolving expectations and framework conditions. A first DMP will be delivered at month 6 and updated when necessary.

T1.5 Quality, risk control and ethical compliance (Lead: UNIBO; Participants: All; M1-M12). This task aims to monitor and control project quality and risks. Activities include tracking the progress of technical tasks, identifying potential issues that could delay milestones, and ensuring interoperability among project components. A quality assurance system with peer review mechanisms for all deliverables will be established. Additionally, the task involves organizing regular meetings with the **Ethics Advisor.** This task will also set the ethics requirements that the project must comply with, including the procedures to identify/recruit research participants, the informed consent procedures and associated templates and data protection measures: submitted as deliverable at M12.

Work package number	**2**
Work package title	Project Management and Coordination 2

Objectives. The objectives of WP2 are the same of WP1 and imply a continuation of the activities of WP1 to achieve a smooth realization of the project. After the first 12 months of research activities, the first results will be produced and shared in accordance with the Grant Agreement and the Consortium Agreement provisions, following the Communication and Dissemination Strategy and Plan.

T2.1 Coordination, project meetings and reporting (Lead: UNIBO; Participants: All; M13-M30). See WP1.
T2.2 Overall legal and contractual management (Lead: UNIBO; Participants: All; M13-M30). See WP1.
T2.3 Financial and administrative management (Lead: UNIBO; Participants: All; M13-M30). See WP1.

T2.4 Data management (Lead: UNIBO; Participants: All; M13-M30). An updated DMP will be delivered in M22.
T2.5 Quality, risk control and ethical compliance (Lead: UNIBO; Participants: All; M13-M30). See WP1.

Work package number	**3**
Work package title	Project Management and Coordination 3
Objectives. The objectives of WP3 are the same of WP1 and WP2 and imply a continuation of the activities of WP1 and WP2 to achieve a smooth realization of the project**.**	
T3.1 Coordination, project meetings and reporting (Lead: UNIBO; Participants: All; M31-M42). See WP1. **T3.2 Overall legal and contractual management (Lead: UNIBO; Participants: All; M31-M42).** See WP1. **T3.3 Financial and administrative management (Lead: UNIBO; Participants: All; M31-M42).** See WP1. **T3.4 Data management (Lead: UNIBO; Participants: All; M31-M42).** A final updated DMP will be delivered at month 36. **T3.5 Quality, risk control and ethical compliance (Lead: UNIBO; Participants: All; M31-M42).** See WP1	

Work package number	**4**
Work package title	**Assessing collection management practices and benchmarking technical approaches**
Objectives. This WP assesses existing practices and tools for CHDO collection management and technical components: 1) enhancing understanding of the various (technical, strategic, sustainability, etc.) choices underpinning decision making in CHI, 2) creating a **benchmark** regarding metadata enrichment, IPR management and appropriate reuse of CHDO, with a focus on multi-context, copyright-aware, and ethics-preserving approaches; 3) **evaluating** the ~20 commonly used CMS in use in European CHI to assess their enrichment and reuse capabilities. We also contribute to requirements collections to complement WPs 5 and 8, providing input to the "technology provider" WPs (WP9-10). Objectives contributed to: **O2** and **O3**.	
T4.1 Analysis of current practices and systems for Collection Management and Metadata Enrichment (Lead: PSNC; Participants: UNIBO, SPA, UNI JENA, EF, NISV, F&F, MIC, UEDIN; M1-M6). This task evaluates the current state of practices and systems for collection management (CMS) and their capabilities for enriching CHDO with additional data. The assessment, performed through consultation with CH professionals, researchers and institutions, will focus on ethical, and multi-contextual aspects of commonly used cultural heritage systems as well as identifying the key workflows in digitisation, conservation, restoration, and dissemination of CHDO. Key outputs include defining the technical requirements needed to enhance CHDO metadata, guided by the use cases (WP5). The output of this task will feed WPs 5 and 8, providing the identified gaps and weaknesses of current practices and processes, contributing to the requirements and design of the INFINITY ecosystem. **T4.2 Review of IPR, Copyright Regulations, and Business Models (Lead: EIT CC SW, Participants: UNIBO, CNR, SPA, EF, NISV, MIC, UEDIN; M1-M6).** This task involves a comprehensive review of the current landscape surrounding intellectual property rights (IPR), copyright regulations, publishing, licensing, and business models related to the appropriate reuse of CHDO and possible exploitation possibilities. The objective is to thoroughly understand existing frameworks and standards used by CHIs for managing data. It will also identify any gaps or inconsistencies that need to be addressed to promote sustainable and appropriate reuse of CHDO in ways that respect copyright value, consider ethical and moral aspects and support viable business strategies. The output will feed WP5 and WP8 to provide a reference for the requirements and design of the INFINITY ecosystem. **T4.3 Benchmark design (Lead: SPA; Participants: UNIBO, UNI JENA, EF, NISV, TMO, PSNC, F&F, MIC, UEDIN, EIT CC SW; M4-M12).** In this task, we design benchmarking experiments to evaluate the state of the art in the various technical approaches which will be supported by the INFINITY ecosystem (cf. Section 1.2): CHDO annotation (with use of language models and KGs, e.g. NER/NEL), mKG construction and evolution, ethical and FAIR metadata generation (provenance, usage rights), metadata model and lexical-conceptual alignment (using embeddings). The result will be a benchmark to assess the capability level of current CMS against several dimensions related to multi-contextual (meta)data enrichment, embedding of IPR and professional value, and supporting collaboration and business models. **T4.4** Benchmarking **workshops (Lead: SPA, Participants: UNI JENA, EF, NISV, TMO, PSNC, F&F, MIC, UEDIN, EIT CC SW; M4-M12).** This task involves organizing and conducting benchmarking challenges and workshops to comparatively assess the performance and implementation approaches of the different component types within existing CMS. Workshops may be held internally, online (as challenges like Kaggle Competitions) or as public	

events alongside relevant conferences, depending on partner availability and opportunities. Participants may be CMS users and/or the developers of the participating components.

T4.5 Validation and Requirements Refinement (Lead: SPA, Participants: UNIBO, CNR, UNI JENA, EF, NISV, TMO, PSNC, F&F, MIC, UEDIN, EIT CC SW; M7-M12). Benchmarking results are aggregated and analysed to establish a baseline for the current state of CMS components. This task evaluates different approaches, compares performance, and extracts technical requirements for the INFINITY ecosystem (WPs 9, 10).

Work package number	**5**
Work package title	**Use case requirements and design**

Objectives. WP5 prepares the basis for INFINITY's technology validation. As such it formalises the requirements of the use cases and their design. Its goal is twofold: 1) to ensure that the software ecosystem is effectively designed to drive innovation into Collection Management Systems; 2) to contribute innovating in relevant, though specific, cultural heritage use cases. Its key outputs contribute to (1) the functional requirements for the software ecosystem (WP8-10) and the use case platforms (WPs 6-7), (2) the non-technical requirements of the use cases, e.g. with regards to training (WP13) and (3) the integration with ECCCH and DS4CH (WP11, WP12). It includes three tasks: T5.1 provides a technical roadmap to all partners, ensuring supervision for the **co-creation** process and the definition of a shared **evaluation protocol** for each use case (providing the basis for WP7). This development is supervised by the Technical Board (T8.1), which ensures continuous integration and synergy with the software ecosystem development (WP9-10); T5.2 defines five use cases (described in Section 1.2.3) for collecting, processing and publishing Cultural Heritage Digital Objects (CHDO) in the ECCCH and provide a testbed for **(meta)data enrichment, embedding IPR and professional value, collaboration and business models.** T5.3 formalises the requirements of the five use cases and explores and leverages synergies for technology development. Expected outcomes of the WP are (1) a **roadmap and validation protocol for all use cases (T5.1),** (2) the **scoping of the use cases** (T5.2), and (3) **formal requirements and design e.g. as user story, system specifications (T5.3)**. Objectives contributed to: **O1** to **O5**

T5.1 Technical roadmap (Lead: UNI JENA; Participants: UNIBO, SPA, EF, NISV, TMO, PSNC, F&F, MIC, UEDI, UWr, EIT-CC-SW, EPFL; M1-12). This task provides a reference document defining a common framework (technical roadmap) to all use cases (T5.2) for: 1) coordinating collection of resources for use in the pilots, 2) gathering requirements from internal and external adopters/stakeholders, 3) setting objectives and challenges and mapping them from specific domains to the technology provider work packages (WP8-10), 4) align approach with the ECCCH through the ECHOES Integration Task Force (see also WP11 and WP12), 5) defining a validation protocol for each use case with shared principles and quality standards. This comprises the formalisation of evaluation instruments (e.g. protocol data, questionnaires, focus groups), samplings and evaluation settings (WP7). The goal is to maximise synergy and methodological soundness. The validation protocol aims at proving, through measurable indicators, the effectiveness of the INFINITY software ecosystem against the three core goals of the call: **(meta)data enrichment, embedding IPR and professional value, collaboration and business models**.

T5.2 Use case definition and scoping (Lead: TMO; Participants: UNIBO, SPA, UNI JENA, EF, NISV, PSNC, DP, F&F, MIC, UWr, EIT-CC-SW, EPFL; M1-12). This task develops in-depth documentation of each use case, outlining their scope and purpose, specifying the types of digital objects and the AI/crowdsourcing tools required, identifying target users, stakeholders and required actions and their timing. The task uses a co-creation, user-driven approach to formalize the overall conditions, requirements and action plans of all use cases.

T5.3 Formal requirements and design (Lead: PSNC; Participants: UNI JENA, EF, NISV, PSNC, F&F, MIC, UWr, EIT-CC-SW, EPFL; M1-12). The task provides a formalisation of each use case's requirements according to the result of T5.2 and by complying with the technical roadmap from T5.1, identifying shared technical components across use cases, facilitating the reuse of tools from the ecosystem (WP8-9-10), the compilation of modular toolchains and thereby reducing development costs. The task also prepares requirements for the integration with the ECCCH and DS4CH (WP11-12). This task serves as a basis for the design and development of the software ecosystem (WP8-9-10) and the integration of the INFINITY tools into use-case specific applications and platforms.

Work package number	**6**
Work package title	**Scale up and deployment: use case implementation**

Objectives. This WP deals with: (1) the integration of the INFINITY software components (WP8-10) in the end-user applications for collaborative CHDO annotation [CROWD], 4D world viewing [4DGRID], manuscript transcription [MANUSCRIPTS], media processing [AUDIOVISUAL] and archival record enrichment [ARCHIVES]; (2) the setup and execution of the use cases (contests, campaigns, etc.) and therefore an initial data collection and testing of the INFINITY software tools. The focus is on the technical development of the use cases defined in WP5 and the integration of INFINITY software components (from WP8-10). The work begins with leveraging the requirements for the use cases (WP5) and uses outputs from WP11-12 for ECCCH integration. The development process is coordinated by the Technical Board (T8.1) and follows the technical roadmap outlined in T5.1. It is driven by the three core validation goals. This ensures a coherent and efficient transition from the design to the practical applications. The implementation involves iterative design, prototyping, and integration steps. Objectives contributed to: **O1 - O5**. Expected outcome of the WP will be (1) **extended GUIs and end-user applications integrated with the INFINITY software** and (2) a set of **realised use case demonstrations** (as events, contests, etc.) to contribute data and contribute to the testing and iterative development of the INFINITY software component in WP9. The software components will be integrated as plugins (T9.2) for existing software (e.g. content management systems), wherever possible. **CMS tools used for use cases will have** interfaces to edit, browse, query, and visualise multidimensional KGs as well as GUIs to interact with AI tools offering manual and automatic options, **providing explanations, and validation support** (T9.7). The outcomes of this work package feed into WP7, where the use cases will undergo validation and evaluation and contribute to the development of the training material (WP13). The structure includes five tasks, one per use case.

T6.1 [CROWD] (Lead: F&F; Participants: EF, TMO, PSNC, DP, UEDIN, AIT; M9-M30): Peace in Europe - Europeana 1945, post WW2 digitisation and transcription campaigns in three countries (Poland, Netherlands, Germany). Each participating country will undertake a campaign to engage the public in the collection, digitization, transcription, and annotation of private documents and memorabilia from the time of peace following WWII. The campaign will start with an online project kick-off for all three countries followed by high-profile national launch events featuring public crowdsourcing collection days and the launch of an online Transcribathon. Over the campaign, countries will host two to three public collection days, paired with physical Transcribathons for collaborative transcription efforts. A professional PR agency in each country will drive awareness through strategic campaigns, aligned with national launch events and activities. Collaborations with national cultural heritage institutions, Ministries of Culture, and other stakeholders will ensure alignment with local priorities. Local TMOs will be encouraged to help mobilise participants, provide digitisation support, and engage schools and communities. The existing Transcribathon toolset will be improved and expanded with a **crowdsourcing module** and new features to enhance its usability for citizen science and crowdsourcing campaigns. Key elements include: **Scalable Campaign Tools:** User-friendly systems for contributions, scalable database and image storage solutions (including IIIF-server support), and **user/rights management features** (T9.7) with API connectivity for integration with other systems. **Transcription and Annotation Tools:** Enhanced capabilities for **handwritten text recognition** (HTR) and **AI-assisted annotation** (T9.4) to improve efficiency and accuracy. **Citizen Science Engagement:** Gamified features, personalized feedback, and community tools to encourage sustained public participation. AI-driven tools will also be integrated to provide semantic enrichment using Natural Language Processing (NLP) and Named Entity Recognition (NER) (T9.4). These tools will be integrated from the INFINITY digital ecosystem and will enable the automated creation of structured metadata and contextual linking of data, which will be integrated into European repositories such as the common European data space for Cultural Heritage (DS4CH) and ECCCH.

T6.2 [4DGRID] (Lead: TMO; Participants: UNI JENA, EPFL; M6-30) 4D geo information system: Within the task a contest will be launched to identify 3 suitable regional and local Time Machine as data holders and demo case providers for city scale demo cases. This task prepares and conducts a contest to identify and select suitable demo case providers (contest documents, evaluation, capacity) for the provision of data and metadata and/or the utilisation of the 4D tools in scenarios as education or information. (2) According to the specific demo case data and scenarios, the functionalities for the **processing of spatial data** (in T9.4) and the **extension of the 4D world viewer interfaces** to collect user data (e.g. images, re-photography, audio inputs, texts) will be implemented. Specific functional extensions comprise (a) the AI-supported **geolocalisation of images**, (b) **Speech-to-text** and (c) data **synchronisation with authority data providers**, (d) the **extraction of toponym information** and (e) the **storage and ingestion of metadata in the mKG DB (T9.2-9.5) and CERN/Zenodo** for files. (3) The improvement of the 4D mobile and desktop viewer tools include the **data visualization from mKG**, and the provision of **4D views and information according to the scenarios** (e.g. city guiding, information access, large scale city postcards) and **multi-language capability** to realize city-specific demo cases to be validated in WP7.

T6.3 [MANUSCRIPTS] (Lead: PSNC; Participants: UWr; M9-30). This task aims to deliver changes needed to address manuscript presentation and description in UWr tools. It will focus on implementing necessary changes in dLibra CMS and other supporting tools to enable seamless integration with **mKG** using components from WP9. This includes but is not limited to the usage of INFINITY software components and applying automatic data enrichment with enhancement of object descriptions. Enrichments will help to harmonize data scattered across different tools in

UWr (using results from T9.3, T9.4) as well as provide connections with mKG (T9.2, T9.5). Appropriate changes to the GUI of dLibra CMS, to interact with AI tools, will be implemented. Moreover, this task will implement changes needed to optimize the **transcription creation workflow** in UWr systems. Those include the creation of manuscript structure-aware annotations, multilingual AI-aided manuscript transcription, and implementing the advanced manuscript presentation view in dLibra CMS. Dedicated manuscript datasets will be prepared and published in dLibra. Mechanisms related to delivering data to mKG will be implemented in respective tools from the UWr systems. dLibra GUI will be also extended to enable encoding, querying and visualising the lifecycle history of CHDO (T8.6, T9.6).

T6.4 [AUDIOVISUAL] (Lead: NISV; Participants: DP, UEDIN, EIT-CC-SW; M9-30). This task will leverage the Dutch National CLARIAH Research Infrastructure for the Arts and Humanities as a foundation to validate INFINITY tools for effectively connecting national research environments with European data spaces. Specifically, we will utilize the CLARIAH Media Suite—an online research environment developed over the past decade in collaboration with the Dutch research community and hosted at NISV—as a testbed for implementing INFINITY components using APIs developed in T9.2 and pipelines for AI and data representations: (i) ontologies (T9.3) to support multi-context representation, and especially IPR/copyright management, provenance, and the tracking of the creation/evolution workflow of CHDO to support business model development (in conjunction with T8.3), (ii) AI tools (T9.4) for entity extraction, linking, and semantic annotation, geospatial data alignment and automated transcription, and (iii) embedding of entity linking pipelines and AI models into the multidimensional knowledge graphs framework in conjunction with (T9.5). Via ECCCH and DS4CH (T11.1 and T11.2) this use case will enable researchers to engage with corpora composed of both local and European datasets. The task will follow a phased approach, beginning with minimal viable scenarios and gradually increasing complexity as the project progresses. From the outset, we will actively involve the Media Suite Research Community and the European SSH Research Community via DARIAH, ensuring user feedback informs development and testing at every stage. This iterative strategy aims to ensure practical usability, foster engagement, and strengthen the interoperability of research environments across national and European contexts. The resulting enhanced collection and the interaction with them, as far as IPR management is concerned, will be tested by Digital Paths with the goal of expanding audiovisual sources to support the marketing campaigns of their clients, as consultant for cultural and creative organisations.

T6.5 [ARCHIVE] (Lead: MIC; Participants: UNIBO, DP, UEDIN, EIT-CC-SW; M9-30). This task aims to improve the accessibility and usability of archive portals, ensuring access to resources to any type of user. Producers of archival data and metadata (State Archives and Superintendencies) will be involved through a call for proposals aimed at identifying the suppliers of the demonstration cases. At the same time, ontology **defined in WP9** will be used for the semantic annotation of the conditions of use of digital resources and archival metadata. The aim is to produce significant data for the execution of the demonstration case. The data will report the conditions of use and appropriate reuse of the published resources, based on current legislation, also providing useful examples to explain the aforementioned conditions. The examples will also have the purpose of inspiring potential users for a correct use of archival sources. The actions include (a) the deployment of ontological modules and mKG, through plugins based on the output of **WP9** to describe and explain the conditions of use of data, metadata and digital resources in archival domain, (b) the evaluation of data modelled on the basis of the same ontologies, (c) the publication of mKG applying automatic data enrichment on a SPARQL endpoint, (d) the implementation in the public consultation portal of GUI dedicated to supporting users in understanding the conditions of use according to guidelines and associated ontology and software functional requirements/design **defined in WP8**, (e) the implementation in the public consultation portal of functionalities aimed at monitoring the use of the resources made available. The expected results of this activity are (1) a collection of ontologies and a mKG of archival resources, (2) the development in the public data consultation portal of a specific section to encourage and evaluate reuse, (3) the integration of the INFINITY software components, for explainable AI-assisted annotation, in the public data consultation portal using components **from WP8 & WP9** (4) the documentation of the results as best practices, and (5) the alignment and sharing of the resulting data on the European common dataspace for cultural heritage.

Work package number	**7**
Work package title	**Scale up and deployment: use case validation**

Objectives. Final technical deployment of the use case platforms, as evolution of the outcomes of WP6. It aims to evaluate and validate INFINITY technologies through the use case-associated campaigns. This provides a complimentary user-centric evaluation of the INFINITY software components (WP10) and supports, through a feedback-look approach, their refinement. This process is driven by the validation protocol defined T5.1. This WP will closely interact with WP10 in a feedback-loop way to ensure systematic validation of the software components deployed in the INFINITY digital ecosystem through the use case platforms and with WP11-12 for the continuous integration with ECCCH and DS4CH. The validation will involve large groups of stakeholder representatives and is organized around three tasks reflecting the three core validation goals defined in WP5: **(meta)data enrichment, embedding IPR and professional value, collaboration and business models.** All validations will be conducted by assessing usability, user satisfaction, and the functionalities of the developed graphical user interfaces (GUIs). A mix of methods and established metrics (e.g. effectiveness, efficiency, error rate) will be used and formally defined in the technical roadmap, such as heuristic evaluation e.g. Nielsen's 10 heuristics, task-based testing, surveys and questionnaires, and focused groups. The output of this WP will contribute to the development of the INFINITY methodology (WP13) by reporting the lessons learned from which good practices can be distilled. Objectives contributed to: **O1** to **O6**.

T7.1 Validating use cases results on (Meta)data enrichment (Lead: PSNC, Participants: UNIBO, SPA, UNI JENA, EF, NISV, TMO, F&F, MIC, UEDIN, UWr, EPFL, AIT; M31-42). This task performs the validation of all five use cases against the validation protocol defined in T5.1 (technical roadmap) for the goal "**(meta)data enrichment**". The validation will include conducting usability tests with target users (representatives of stakeholders - CH and creative professionals, researchers, scholars, citizens and the general public - in accordance with the definition of each use case). The validation will focus on the solutions performance, effectiveness, efficiency, and user satisfaction across all defined use cases. Specific attention will be given to evaluate the interaction with mKG (annotation, querying, browsing, visualization), with AI-assisted metadata enrichment (including user consent), collaborative curation tools, 4D viewer, manuscript transcription. Ethical aspects of CHDO metadata enrichment will be also addressed in this validation task e.g. detecting potential bias in CHDO description, and developing best practice, structured approaches to ensuring a plurality of voices and viewpoints are included in documentation, via structured check-lists, benchmarking, and close reading of AI-created content to understand integration with human-created documentation.

T7.2 Validating use cases results on embedding IPR and professional value (Lead: UEDIN, Participants: UNIBO, CNR, SPA, EF, NISV, PSNC, DP, F&F, MIC, EIT-CC-SW; M31-42). This task performs the validation of [CROWD], [AUDIOVISUAL] and [ARCHIVE] use cases against the validation protocol defined in WP5 for the goal "**embedding IPR and professional value".** Representatives of stakeholders and final users of the developed platforms will be involved in validation sessions focusing on processes for interacting with CHDO as for annotating and receiving IPR information and conditions of reuse, being supported for checking compliance with their planned reuse/exploitation of CHDO against licensing and IPR conditions, receiving feedback on potential data protection infringement in CHDO, and annotating as well as retrieving information about the lifecycle history of CHDO (expertise, actions, provenance).

T7.3 Validating use cases results on collaboration and business models (Lead: EIT CC SW, Participants: SPA, EF, NISV, PSNC, DP, MIC, UEDIN; M31-42). This task performs the validation of [AUDIOVISUAL] and [ARCHIVE] use cases against the validation protocol defined in WP5 for the goal "**collaboration and business models".** The task evaluates the use cases' ability to foster interdisciplinary collaboration, develop sustainable business models, and address the needs of stakeholders across cultural heritage institutions, creative industries, and the public. Validation activities will include stakeholder feedback collection, and benchmarking against industry best practices. Insights gained will inform refinements to ensure that the use cases provide replicable and scalable frameworks for collaboration and innovative business strategies, contributing to the broader goals of INFINITY. Digital Paths will contribute to validate this aspect within the [AUDIOVISUAL] use case.

Work package number	**8**
Work package title	**INFINITY ecosystem for multi-context CHDO: Ethics-driven requirements and design**

Objectives. Understanding and formalising the requirements for the INFINITY ecosystem with the goal of enhancing CHDO metadata quality through multi-contextual descriptions, and their appropriate reuse by ethics-driven processes and IPR management. Key outputs are guidelines and software requirements/design for ethical CHDO annotation and appropriate reuse, starting from a systematic review of the state of the art on data ethics. This WP specifies the activities that characterize digital content production chains, and provides guidelines and associated ontology and software functional requirements/design for (1) embedding in CHDO scientific and professional value, IP and copyright-related information throughout the workflow, (2) embedding in CHDO their potentially different cultural, historical, biographical perspectives, (3) integrating AI in metadata enrichment services, including cross-collection alignment, and (4) ethical and sustainable reuse of multi-context CHDO. The envisioned technical solution is based around the novel idea of multidimensional knowledge graphs (mKG). mKG will be technically specified to support their representation, storage, browsing, visualization, querying and evolution management. As such, this WP provides a horizontal perspective on the functional requirements that the INFINITY software ecosystem (WP9-10) shall address and their design, as a complement to the vertical perspective given by WP5. The requirements and design activities rely on WP4 outputs and are conducted in close collaboration with WP5 to reflect and address the technical solutions required for the integration of components in the use case platforms. A **Technical Board** (TB) supervising the ecosystem's component development will be set up to ensure the ecosystem aligns and integrates with ECCCH and DS4CH (WP11-12) and feed into the methodology materials (WP13). Objectives contributed to: **O1-O2-O3.**

T8.1 Technical Coordination and overall digital ecosystem architecture (Lead: SPA; Participants: UNIBO, CNR, UNI JENA, EF, NISV, PSNC, F&F, UEDIN, EPFL, AIT; M1-M12). This task will set up and run a Technical Board (TB) which, according to the roadmap defined in T5.1, will supervise the technical development of the INFINITY digital ecosystem (software components) making sure that synergies are identified in a timely manner and issues around compatibility and technical interoperability are addressed harmonically during the development phase. The TB will define the common methodological approach and identify the supporting collaborative tools to be used, and release guidelines through a *Rule Book*. It will also design and release the **overall architecture** of the ecosystem's software components, defining the APIs required to support their release as web services and the development of CMS platform plugins. It will ensure that all the developed components are released according to the FAIR principles, documented (for WP13) and respect the alignment and integration constraints identified by WP11-12 for interoperability with the ECCCH and DS4CH. The TB, led by Dr. Lyndon Nixon (SPA), includes one representative per technical partner, a data steward, and at least one representative participating in the integration task force coordinated by ECHOES. It takes decisions by ⅔ majority.

T8.2 Guidelines on data ethics and ethical requirements and design for the software ecosystem (Lead: UEDIN; Participants: UNIBO, EF, NISV, TMO, MIC, EIT-CC-SW; M1-M6). Systematic review of data ethics and requirements for embedding ethical processes in the ecosystem components. This task will feed T8.1 as for ethical principles to be defined in the Rulebook for developers. It will also interact with all other tasks to oversee that the requirements and design of the software components are grounded in and compliant with ethical principles for CHDO annotation, reuse and integration of AI tools in software pipelines. These will include representation and bias, consent and rights, data integrity and authenticity, technological design (explainability, use of open standards, sustainability), cultural sensitivity and access and equity (e.g. multilingual support).

T8.3 Ontologies, vocabularies, and mapping schemes (Lead: UNIBO, Participants: UNIBO, CNR, NISV, TMO, MIC, UEDIN; M1-9). Systematic review of CH ontologies (CIDOC CRM[8], EDM[7], ArCo[9], ECHOES[11]), archival standards (EAC-CPF, OAD, ARKIVO, Spectrum, Records in Context), and ontologies/vocabularies addressing IPR/copyright, provenance, workflows and contextualization. Alignment and extensions (if required) of the reviewed ontologies: design of mapping schemes, and pipelines for metadata harmonization across platforms and collections. Developing ontology module extensions if required, and plan release on ECCCH and DS4CH. Existing tools for ontology alignment will be reviewed and identified for reuse and possible extension. This task will ensure that the INFINITY ecosystem is supported by ontologies that comprehensively address multi-context representation of CHDO, IPR/copyright management, provenance, and the tracking of the creation/evolution workflow of CHDO. The output will constitute the foundation for T9.3.

T8.4 AI tools for enriched (meta)data annotations: requirements and design (Lead: UNIBO, Participants: All; M1-12). Requirements and design (and identification of reusable existing) AI-driven tools for knowledge extraction, semantic annotation (text, images, audiovisual), geospatial data alignment (for historical maps/photos) [4DGRID], multilingual transcription ([CROWD], [MANUSCRIPTS]). Definition of pipelines for AI model fine-tuning to produce multilingual concept and text embedding layers to support metadata mapping (T8.3) and knowledge graphs (T8.5). This task will take on board the benchmarking results and recommendations of WP4. The output will constitute the foundation for T9.4.

T8.5 Multidimensional KG Architecture design (Lead: SPA, Participants: UNIBO, EF, PSNC, F&F; M1-12). Formal definition of multidimensional KGs (mKG). Designing components for mKG creation, storage, querying, and updates. Pipelines for extracting mKG from internal data models such as dLibra. Designing components for KG completion and evolution using embeddings (T8.4). Defining mKG integration with entity linking and annotation pipelines. Developing plans for large-scale mKG deployment and maintenance (linked to DS4CH/ECCCH). The output will constitute the foundation for T9.5.

T8.6 Capturing CHDO lifecycle and associated workflows: requirements and design (Lead: PSNC, Participants: UNIBO, CNR, SPA, UNI JENA, NISV, F&F; M1-12). Based on input from T4.1 this task specifies the requirements, design and identification of reusable software components and models (T8.3) to support embedding in CHDO their lifecycle history: tracking and embedding of methods, skills, and processes that influenced the creation and evolution of CHDO, including version control and AI vs human contributions. The output will constitute the foundation for T9.6.

T8.7 Ethical and legal validation tools: requirements and design (Lead: UEDIN, Participants: All; M1-12). This task will identify reusable software components for supporting legal and ethical validation associated with the use of AI (T8.4) for annotation enrichment and knowledge extraction integrated in the use cases (WP5). A systematic study of the state of the art will be conducted and the most advanced open-source software available will be evaluated against established benchmarks as well as with respect to the quality of their documentation and easiness to deploy. The main functionalities that will be addressed are: user consent management, provenance tracking, bias detection, explainability, and intellectual property rights (IPR). The output will constitute the foundation for T9.7.

Work package number	**9**
Work package title	**INFINITY software ecosystem for multi-context Cultural Heritage Digital Objects: software components implementation**

Objectives. Technical development of the INFINITY software ecosystem as a set of components responding to the requirements and design defined in WP8. Support for the integration of the software components for enhancing the CMS used by the use cases (WP6). WP6 and WP9 work in synergy, the Technical Board will continue monitoring and validating the development activities ensuring constant integration between use cases, the ecosystem components, the ECCCH and DS4CH infrastructures, and to avoid effort duplication. The development is based on an iterative, incremental approach including prototyping and integration with WP6 (cf. Section1.2). The first release of the ecosystem (*Alpha)* will be delivered at M20. The outcomes of this WP feed WP10 (deployment and evaluation), and WP13 (training material and methodology targeting developers (T13.2)). Objectives contributed to**: O1-O2-O3.**

T9.1 Technical coordination (Lead: SPA; Participants: UNIBO, CNR, UNI JENA, EF, NISV, PSNC, F&F, UEDIN, EPFL, AIT; M13-M30). The Technical Board steered in T8.1 monitors the development process, ensures adherence to the Rulebook (T8.1), the DMP and the Technical Roadmap (T5.1), and addresses interoperability issues as they arise. TB coordinates with WP6 to ensure that components align with use case requirements and integrate seamlessly into the CMS. Regular development sprints will be planned, reviewed, and validated to align with release schedules. A release calendar will be established to synchronise development cycles across the consortium, and TB will facilitate issue tracking and collaborative problem-solving to ensure smooth component integration. One releases of the ecosystem of software components will be planned during this period resulting in one deliverable combining software and documentation for all components developed in the seven tasks of this WP. The task will also ensure that ontologies identified and extended in WP8 are consistently applied, and further extensions may be developed to address emerging requirements during development. Additionally, the TB will evaluate when a component or the content it produces should be released on the ECCCH or the DS4CH and will ensure adherence to the ECHOES infrastructure and data models design.

T9.2 API and Plugin Implementation for Ecosystem Integration (Lead: PSNC; Participants: UNIBO, SPA, UNI JENA, NISV, F&F, EPFL, AIT; M6-M30). This task ensures the implementation of APIs and plugins required to integrate ecosystem components into the use case CMS. It works in close collaboration with WP6 which will provide input to align APIs with the CMS specifications. This phase will also address backward compatibility, facilitating the incremental addition of components without disrupting existing workflows. The implementation will reference the ecosystem architecture defined in T8.1, ensuring alignment with the overall design and structural coherence. The task will adhere to ontologies identified in WP8, ensuring compliance and alignment across platforms and collections.

T9.3 Cross-Collection Alignment and Metadata Harmonization (Lead: UNIBO, Participants: EF, NISV, TMO, PSNC, MIC, UEDIN, UWr; M9-24). This task continues the work initiated in T8.3 by developing tools for cross-collection alignment and ontology-based metadata harmonization. The focus will be on enhancing metadata coherence across platforms, ensuring that the extended ontologies from WP8 are consistently applied and refined as needed. Harmonization pipelines and mapping tools will be implemented to facilitate interoperability and seamless integration of data from different collections. The task will also explore additional ontology extensions that arise from cross-platform alignment challenges. The output feeds WP6, and T10.4 where they will be formally evaluated and deployed.

T9.4 AI Tool Development and Pipeline Implementation (Lead: UNIBO; Participants: UNIBO, SPA, UNI JENA, NISV, PSNC, F&F, EPFL, AIT; M9-M30). Building on T8.4, this task focuses on the development of AI tools to be integrated in all use case CMS for entity extraction, linking, and assisted semantic annotation, geospatial data alignment (for integration in [4DGRID)], automated transcription ([CROWD], [MANUSCRIPTS]). Definition of pipelines for AI model fine-tuning: multilingual embeddings, multidimensional vector spaces. Developers will fine-tune models and integrate them into annotation pipelines. The task will also involve testing tools across multilingual datasets and refining pipelines to improve performance and accuracy. Collaboration with the other tasks and WP6 will ensure that AI tools are aligned with ontology frameworks and metadata standards and can be deployed in combination with models for robustness, ethical and validation (T9.7).

T9.5 Multidimensional Knowledge Graph (mKG) Development (Lead: SPA; Participants: UNIBO, PSNC; M9-

M24). This task involves the development of the mKG infrastructure as designed in T8.5. Developers will implement components for mKG creation, querying, and versioning, ensuring scalability and robust performance. The task will also include embedding entity linking pipelines and AI models into the mKG framework to support large-scale deployments. A knowledge graph from publicly available knowledge, e.g. Wikidata, will be used as seed and aligned to the cultural heritage domain by entity and relation extraction over publicly available CHDO metadata, e.g. Europeana. All data produced by the use cases (WP6-7) will incrementally feed the mKG. A language model will be trained (fine tuning an open-source foundational model) with datasets of historical records as well as contemporary documents (e.g. news articles) to learn the multiple contexts of interpretation of the same words or terms (across languages: we will include multilingual NLP in the model training). This model will provide the embedding layer from to extract explicit multicontextual interpretations of the graph entities. Ontology extensions identified during the development process will be integrated into the mKG to ensure consistent representation and interoperability. Evaluation and deployment will take place in T10.5.

T9.6 Workflow Tracking and Lifecycle Management Tools (Lead: PSNC; Participants: UNIBO, CNR, SPA, UNI JENA, NISV, F&F; M9-M24). This task develops tools for capturing and tracking the lifecycle of CHDO, as specified in T8.6. The tools will enable documentation of CHDO lifecycle processes, embedding skills, and tracking methods that influenced it evolution. Existing models identified in T8.3 will be extended and integrated into these tools, ensuring comprehensive lifecycle management. Compliance with ontology extensions and harmonization processes from WP8 will be a key focus, ensuring alignment across workflows. Evaluation and deployment in T10.6.

T9.7 Ethical and Legal Compliance Validation (Lead: CNR; Participants: UNIBO, SPA, UNI JENA, NISV, PSNC, F&F, UEDIN, EPFL, AIT; M9-M24). In alignment with T8.7, this task focuses on embedding ethical and legal compliance tools into workflows for cultural heritage digital objects (CHDOs), ensuring that the development and curation processes meet high standards of transparency, fairness, and accountability (robustness). The task builds on the results of T8.7 and implements the identified components in adherence with the ecosystem architecture (T8.1) and the defined API and plugin design from T9.2. State-of-the-art tools will be engineered to address user consent management, provenance tracking (e.g. PROV-O[Fout! Bladwijzer niet gedefinieerd.] and ccREL[30] combined with interoperable repository such as DSpace and the International Image Interoperability Framework (IIIF) and with workflow tracking provided by T9.6), bias detection (e.g. Fairlearn[31] (Bird et al, 2020) and IBM's AI Fairness 360[32] (Bellamy et al, 2019)), BERT (Devlin, 2018), RoBERTa (Liu et al, 2019), explainability (e.g. TextAttack[33] (Morris et al, 2020), LIME[34]), and intellectual property rights management (e.g. RightsStatements.org[35]). Validation checkpoints will be incorporated to ensure compliance with ethical and legal standards throughout the CHDO lifecycle. The identified tools will be tailored to the specific needs of the CHDO curation workflows, according to the input from T8.7. Compliance with ontology extensions and harmonization from WP8 will be ensured. These components will be integrated in and validated through the use cases (WP7) and deployed in T10.7.

Work package number	**10**
Work package title	**INFINITY software ecosystem for multi-context Cultural Heritage Digital Objects: software components deployment and evaluation**

Objectives. Finalization of the technical deployment of the INFINITY software ecosystem, as a continuation of the work conducted in WP9. All components undergo a rigorous evaluation against state of the art baselines. The objective is to validate the performance, interoperability, scalability, and compliance of the developed tools and systems, ensuring their readiness for deployment. WP10 leverages the use cases validation processes from WP7 to ensure practical applicability and user-based evaluation. During the duration of this WP, two iteration cycles will be completed, at M32 with the second release of the ecosystem (*Beta*) and at M40 the final release (1.0). A report will describe the deployment and formal evaluation of the software components. Lessons learned to WP13. Objectives contributed to: **O1** to **O4.**

[30] https://opensource.creativecommons.org/ccrel/
[31] https://fairlearn.org/
[32] https://aif360.res.ibm.com/
[33] https://github.com/QData/TextAttack
[34] https://github.com/marcotcr/lime
[35] https://rightsstatements.org/en/

T10.1 Technical coordination (Lead: SPA; Participants: UNIBO, CNR, UNI JENA, EF, NISV, PSNC, F&F, UEDIN, EPFL, AIT; M31-M42). This task extends the efforts of T9.1 by focusing on the technical evaluation and deployment of components post-implementation. The Technical Board (TB) will oversee the evaluation and deployment process. The TB will conduct structured assessments to monitor interoperability, scalability, and adherence to ontologies established in WP8 and to the ECCCH infrastructure. Following successful evaluation, components will be deployed across the INFINITY ecosystem and the use case platforms. The TB will also coordinate with WP7 for overseeing the user-oriented validation. Deployment sprints will address issues, track performance, and refine components based on feedback from use case tests (WP7). The TB will assess readiness for release on ECCCH and DS4CH, ensuring seamless ecosystem integration.

T10.2 API and Plugin Evaluation and Deployment (Lead: PSNC; Participants: UNIBO, SPA, UNI JENA, NISV, F&F, EPFL, AIT; M31-M36). Building on T9.2, this task evaluates and deploys the implemented APIs and plugins, ensuring robustness, efficiency, and backward compatibility. The task assesses API performance under different use case scenarios provided by WP7, ensuring alignment with CMS platform specifications. This includes stress testing, security evaluation, and functionality checks. Post-evaluation, APIs and plugins will be deployed, ensuring minimal disruption to workflows. The deployment will adhere to the ecosystem architecture (T8.1) and ontology frameworks from WP8, addressing any emerging compatibility issues. Updates will flow into T13.2.

T10.3 Cross-Collection Alignment and Metadata Harmonization Validation and Deployment (Lead: UNIBO; Participants: EF, NISV, TMO, PSNC, MIC, UEDIN, UWr; M31-M36). Following T9.3, this task evaluates and deploys tools for cross-collection alignment and ontology-based metadata harmonization. The validation process focuses on metadata coherence, ontology consistency, and data integration across platforms. Harmonization pipelines will be deployed to enhance interoperability. Benchmarking against state-of-the-art solutions and feedback from WP7 will guide refinement and deployment. Ontology extensions identified during the process will be integrated into the final deployed solution.

T10.4 AI Tool Performance Testing, Optimization, and Deployment (Lead: UNIBO; Participants: SPA, UNI JENA, NISV, PSNC, F&F, EPFL, AIT; M31-M38). This task evaluates, optimizes, and deploys the AI tools and pipelines developed in T9.4. AI tools for entity extraction, semantic annotation, geospatial alignment, and manuscript transcription will be tested across multilingual datasets and refined based on benchmarks against domain-specific baselines. The AI tools will be deployed into operational use case CMS, ensuring scalability and performance (WP7). Continuous feedback loops from WP7 will drive further optimization during deployment.

T10.5 Multidimensional Knowledge Graph (mKG) Performance Evaluation and Deployment (Lead: SPA; Participants: UNIBO, PSNC; M31-M38). Expanding on T9.5, this task evaluates and deploys the mKG infrastructure, ensuring scalability, performance, and interoperability. The evaluation will validate entity linking pipelines, querying efficiency, and embedding quality. The mKG infrastructure will be deployed to integrate seamlessly with CHDO metadata and use case CMS. The final deployed mKG will support large-scale, multilingual, and multicontextual applications within the INFINITY ecosystem and use cases.

T10.6 Workflow Tracking and Lifecycle Management Validation and Deployment (Lead: PSNC; Participants: UNIBO, SPA, UNI JENA, NISV, F&F; M31-M38). This task evaluates and deploys tools developed for workflow tracking and lifecycle management in T9.6. The evaluation ensures CHDO lifecycle-supporting processes, embedding methods, and tracking functionalities align with ontology frameworks from WP8. The tools will be deployed to manage CHDO life cycles across use case CMS, ensuring interoperability and compliance.

T10.7 Ethical and Legal Compliance Review and Deployment (Lead: UEDIN; Participants: UNIBO, CNR, SPA, UNI JENA, NISV, PSNC, F&F, EPFL, AIT; M31-M38). Continuing from T9.7, this task ensures all ecosystem components meet ethical and legal standards prior to deployment. Compliance tools will be deployed across the use case platforms. Compliance with privacy, bias detection, and user consent protocols will be validated and tools evaluated against existing benchmarks complementing the user-based validation performed in WP7. This task will also ensure the identification of needs in the area to provide guidelines for post-project development to keep alignment with evolving regulations.

Work package number	**11**
Work package title	**Integration with the DS4CH and ECCCH**

Objectives. This work package implements coordination mechanisms with the ECCCH and the common European data space for cultural heritage (DS4CH) to ensure relevance of the project results in the complex digital data ecosystem they must be integrated in. The work package uses the European Interoperability Framework layers of interoperability (governance, legal, organisation, semantic, technical) as general guidance for action. The specifications of use cases (WP5) and INFINITY ecosystem components (WP8) is a key source of input for the work package interfacing actions.

In turn, the work package identities and communicates precise requirements and call to actions to properly align with ECCCH and DS4CH during the design of use cases and core ecosystem components (WP5, WP8), development (WP6, WP9), deployment and validation (WP7, WP10), capacity building and dissemination (WP13-16). This task works in close collaboration with the TB (WP8-9-10). To ensure efficiency and agility, we aim to define the integration opportunities and needs with a minimal core team of partners already involved in these digital initiatives, and bring the results to the other partners wherever they are doing the work that would be impacted by the integration requirements (and benefits). Objective contributed to: **O4.**

T11.1 Alignment and integration with ECCCH (Lead: PSNC; Participants: NISV, EF, TMO; M1-M24). This task contributes directly to the overarching objectives of the ECCCH initiative via (1) establishment and maintenance of regular communication channels (e.g. via the ECHOES Integration Task Force), (2) engagement in joint discussions on technical matters to assure interoperability (e.g. data model alignment, conformance with software development guidelines), (3) technical collaboration and alignment with other projects and initiatives contributing to ECCCH to achieve synergy (e.g. reuse of technology, collaborative developments, development sprints), (4) reuse of ECCCH resources (e.g. datasets, tools, services, low level libraries) in the developments of INFINITY, (5) support for joint capacity building activities with ECHOES or other contributing projects and initiatives (e.g. technical workshops, hackathons). This task also includes coordination with EOSC - following an approach that will be determined when ECCCH and EOSC will have determined their own alignment mechanisms.

T11.2 Alignment and integration with DS4CH (Lead: EF; Participants: NISV, PSNC, TMO; M1-M24). This task ensures seamless integration of the project results into the common European data space for cultural heritage. As T11.1 does with ECCCH, T11.2 establishes communication channels and engages in joint discussion on technical matters and other activities. It communicates the requirements for different layers of interoperability with the DS4CH (technical, data, capacity building). Focus points include (1) compatibility with relevant DS4CH framework such as the Europeana Data Model and the Europeana Licensing Framework, (2) technical interfacing for data exchange with Europeana APIs, publication of capacity building material in the Europeana Academy (including quality checks), and dissemination towards the relevant communities of the Europeana Network Association.

Work package number	**12**
Work package title	**Integration with the DS4CH and ECCCH**

Objectives. This work package continues and finalises the work on interfacing the project results with the ECCCH and the common European data space for cultural heritage (DS4CH). Objective contributed to: **O4.**

T12.1 Alignment and integration with ECCCH (Lead: PSNC; Participants: NISV, EF, TMO; M25-M42). This task continues the feedback loop with the ECHOES Integration Task Force to ensure that the project results will be fully integrated into the ECCCH. This also includes coordination with EOSC.

T12.2 Alignment and integration with DS4CH (Lead: EF; Participants: NISV, PSNC, TMO; M25-M42). This task continues to provide guidance on the interoperability of the project results with the data space during the use case and core components implementation phase (WP9-10).

Work package number	**13**
Work package title	**INFINITY Methodology and Training Materials**

Objectives. This WP will oversee the consolidation, standardisation and packaging of INFINITY results into a variety of formats to address the specific knowledge needs of four target audiences: (i) CHI, (ii) national cultural heritage and research infrastructures, (iii) creative industries and (iv) academic research and education. These will be shared widely (T14.1) and will play a key role in capacity building (T14.2) activities. Input will come from WP4, WP5-6-7 and WP8-9-10. Objective contributed to: **O6.**

T13.1 Guidelines for the cultural heritage sector including developers (Lead: NISV; Participants: UNIBO, TMO, MIC, UEDIN, UWr, EIT-CC-SW; M25-42). This task will create a set of practical resources, such as guidelines for good practices, case studies and step-by-step guides, that will help CHI to effectively apply multi-contextual data enrichment and appropriate reuse strategies and navigate ethical questions.

T13.2 Guidelines for cultural heritage infrastructures (Lead: PSNC; Participants: EF, NISV, TMO; M25-42). Addressing actors that act as distributors, facilitators and brokers of CHDO (including service providers of content management systems - developers), this task will create recommendations and guidelines on how existing (intern)national infrastructures could endorse and incorporate new business models for data management and sharing proposed by INFINITY.

T13.3 Reuse scenarios for creative industries and tourism (Lead: EIT-CC-SW; Participants: NISV, DP, UEDIN, EIT-CC-SW; M25-42). This task will invite representatives from creative industries to engage in two expert consultation sessions to co-design scenarios for appropriate data reuse, showcasing the reuse and business potential of multi-contextual CHODs in various contexts. The scenarios will be captured in playbooks, including concrete examples from various creative sectors, including publishing, design and fashion.
T13.4 Materials for academic education and research (Lead: UEDIN; Participants: UNIBO, SPA, NISV, UWr; M25-42). This task will create materials that could be used in academic education for students in courses related to computer sciences and heritage studies. This includes (i) sample datasets, (ii) exercises that could be used by students to experiment with multi-contextual metadata descriptions and (iii) a white paper focussing on new research avenues.

Work package number	14
Work package title	**Dissemination, Communication, Exploitation**

Objectives. To plan, implement and monitor the Exploitation, Dissemination and Communication of results (PEDR) activities throughout the project and to manage the FSTP Open Call. To ensure visibility of the project based on formats to address multiple target groups: (i) ensuring that the public and external organisations know and understand the objectives and achievements of INFINITY, obtain feedback and actively participate in events and webinars to track results, (ii) supporting the visibility and long-term impact of the project, guide innovations and (iii) the future exploitation of the INFINITY ecosystem, both in the context of the ECCCH and beyond. Objective contributed: **O6**

T14.1 Dissemination and Communication Activities (Lead: DP; Participants: All; M1-12). This task will ensure broad visibility of the project among the targeted communities. It encompasses three key activities. First, the establishment of a **Stakeholder Network** that will be consulted at key milestones in the project. Second, the effective execution and monitoring of the various **Dissemination activities** (see Section 2.2.1), such as the showcase events, workshops (T12.2), webinars, disseminating guidelines for professionals and academic output and participation in events. And second, the effective execution and monitoring of **Communication Activities** (see Section 2.2.2) such as creating an INFINITY brand (project image), creating and maintaining the project website, issuing press releases, engaging via social media, distributing newsletters, and organising open days (we planned sixteen in total over the duration of the project), for broader outreach. Performance metrics like views, site visits, newsletter signups, social media engagement, and event attendance will be meticulously tracked to assess the impact and fine-tune the PEDR for maximum effectiveness.
T14.2 Capacity building workshops (Lead: EIT SW CC; Participants: all; M1-12) In the initial phase, WP14 will organize one **capacity-building workshop** tailored for CHI and creative industries. This workshop will focus on introducing INFINITY tools and methodologies, such as AI-driven metadata enrichment and multilingual metadata management. A "train-the-trainer" component will be included, enabling participants to share knowledge within their organisations. Online sessions will ensure accessibility, with detailed metrics to monitor attendance and engagement. This workshop (and the following ones in WP15-WP16) will act as a foundation for broader adoption and feedback.
T14.3 Exploitation Strategy and Business Planning (Lead: EIT CC SW; Participants: all; M1-12). This task will outline how the transition between funded project and post-project innovation and commercial exploitation is managed. It will include an overview of planned exploitation activities, analyse potential and generate recommendations for future activities. Key elements of this strategy are the creation of a detailed Intellectual Property (IP) Rights Register and a formalised agreement and the ongoing refinement of business models based on initial cost, revenue, and return on investment projections. The strategy aligns with the governance of the ECCCH, established in the ECHOES project, where INFINITY participates as a member of the Stakeholder Network. A bi-annual market-watch policy will monitor significant developments to timely adjust exploitation planning. Standardization efforts will be coordinated with the Data Space Support Centre and W3C. The task includes the initial development of the innovation binder.

Work package number	15
Work package title	**Dissemination and Communication, Exploitation**

Objectives. To plan, implement and monitor the Exploitation, Dissemination and Communication of results (PEDR) activities throughout the project. In this phase, efforts focus on visibility, engagement, and capacity-building. Showcase events, webinars, and targeted outreach highlight findings. Workshops empower professionals, and academic outputs strengthen dissemination. Monitoring ensures effectiveness and prepares for impactful results. Objective contributed to: **O6.**

T15.1 Dissemination and Communication Activities (Lead: DP; Participants: all; M13-M30). See WP14.
T15.2 Capacity building workshops (Lead: EIT SW CC; Participants: all; M13-30). We will organise two interactive workshops to train stakeholders—including CHI, creative industries, and policymakers—on effectively

using INFINITY tools. These workshops will provide hands-on experience with features like AI-driven metadata enrichment, multilingual knowledge graphs, and ethical data management. Tailored sessions will address the unique needs of each audience, encouraging practical application and feedback to refine the tools. By fostering engagement and building capacity, these workshops will ensure stakeholders are well-equipped to adopt and maximize INFINITY's technologies.

T15.3 Exploitation Strategy and Business Planning (Lead: EIT SW CC; Participants: all; M13-30). See WP14. We will focus on developing a sustainable exploitation strategy and business plan for INFINITY outcomes. Activities will include market analysis, designing business models for open-source and commercial applications, and engaging with CHI and industry leaders to align the tools with market needs. Strategic partnerships will be explored to support adoption and commercialization, ensuring long-term impact. The resulting exploitation strategy will provide a clear roadmap for sustaining INFINITY's innovations beyond the project's duration. The task continues the development of the innovation binder.

T15.4 Open call management (Lead: NISV; Participants: All; M13-30). This will include: i) defining objectives, scope, and eligibility criteria for the cascading grants, ii) promotion of calls for proposals, iii) management of the application submission and including peer- reviewed evaluation (see Annex), iv) selection and awarding v) monitoring of the awarded projects.

Work package number	**16**
Work package title	**Dissemination and Communication, Exploitation**

Objectives. To plan, implement and monitor the Exploitation, Dissemination and Communication of results (PEDR) activities throughout the project. In the final phase of the project, the focus shifts to securing long-term impact, ensuring the sustainability of results, and maximizing outreach. This period emphasizes consolidating project achievements through a comprehensive exploitation plan, aligning outputs with international standards, and preparing for future applications. High-impact events, such as the final conference and exhibitions, will showcase findings to stakeholders and the public. Objective contributed to: **O6.**

T16.1 Dissemination and Communication Activities (Lead: DP; Participants: All; M31-42). See WP14. In this task a report for Policy Makers addressing guidelines, issues and lessons learned about the integration and alignment with ECCCH and DS4CH will be produced.

T16.2 Capacity building workshops (Lead: EIT SW CC; Participants: All; M31-42). See WP14. This task will deliver a workshop planned towards the end of the project. The aim of the event will be twofold: 1) to showcase the results of the projects funded through the FSTP (T15.4 and T16.4), 2) to train CHI, creative industries, and stakeholders on INFINITY tools and methodologies. Topics will include AI-driven metadata enrichment, multilingual knowledge graphs, ethical data management, and sustainable business models. Sessions will focus on practical applications and how to exploit the results, such as integrating tools into existing workflows and developing innovative uses like AR/VR heritage experiences. Participant feedback will guide tool refinement, ensuring long-term adoption and impact.

T16.3 Exploitation Strategy and Business Planning (Lead: EIT SW CC; Participants: All; M31-42). See WP14-15. We will finalize the exploitation strategy and business plan for INFINITY tools to ensure their long-term sustainability and impact. Building on earlier efforts, it will validate market potential, refine business models for both open-source and commercial use, and establish strategic partnerships with CHI and industry stakeholders. A comprehensive roadmap will be developed, outlining licensing strategies, adoption pathways, and alignment with international standards, ensuring INFINITY's innovations remain impactful beyond the project's duration. The task includes the finalization of the innovation binder.

T16.4 Open call management (Lead: NISV; Participants: All; M31-42). See WP14.

Table 3.1c: List of Deliverables

WP	ID	Deliverable title	Deliverable description	Lead	Type	Diss	Date
1	D1.1	Project Handbook	General operational information, including contact details, partner roles and responsibilities, processes, templates, and procedures for deliverables preparation and review	UNIBO	R	SEN	M3
1,2,3	D1.2 D2.1, D3.1	Data Management Plan	Guidelines on how to handle research data during the project and how to preserve it after it is completed	UNIBO	DMP	SEN	M6, M22, M36

1	D1.3	Ethics requirements	Procedures to identify/recruit research participants, the informed consent procedures and associated templates and data protection measures	UNIBO	R	SEN	M12
4	D4.1	Survey on collection management practices and systems	Assessment of current CMS and methodologies capabilities and gaps in CHDO enrichment and management	PSNC	R	PUB	M6
4	D4.2	Survey on IPR regulations and business models involving CHDO	Existing standards and framework used by CHI for IPR management of CHDO	EF	R	PUB	M6
4	D4.3	Benchmark design and evaluation	Measures, criteria and tools for benchmarking CMS and results of benchmarking experiments	SPA	R	PUB	M12
5	D5.1	Technical Roadmap	Methodology and tools for use case execution and validation protocol	UNI JENA	R	PUB	M12
5	D5.2	Use case definition, formal requirements and design	Use case scoping and detailed definition, requirements and design of use case CMS software extensions	UNI JENA	R	SEN	M12
6	D6.1	Use case implementation	Software release documentation of use case CMS	PSNC	R	PUB	M30
7	D7.1	Use case deployment and validation	Results of user-driven validation and report on use case CMS deployment	PSNC	R	PUB	M42
8	D8.1	Guidelines on data ethics	Review on data ethics and guidelines for ethical use CHDO	UEDIN	R	PUB	M6
8,9,10	D8.2, D9.2, D10.2	Developers Rulebook	Guidelines, development methodology and supporting platform, release constraints	SPA	R	PUB	M12, M24, M36
8	D8.3	Ethics-driven requirements and design for the INFINITY ecosystem	Requirements and design of software components and for embedding ethical procedure and IPR management	UEDIN	R	PUB	M12
9	D9.1	Release (*Alpha*) of the INFINITY ecosystem	Release of software components and documentation, use case tools, and associated material	SPA	Other	PUB	M20
10	D10.1, 10.3	Release (*Beta*) and (*1.0*) of the INFINITY ecosystem	See D9.1	SPA	Other, R	PUB	M32, M40
11	D11.1	Alignment and integration with ECCCH and DS4CH	Coordination mechanisms, actions for interfacing with ECCCH and DS4CH	EF	R	PUB	M24
12	D12.1	Final Report on Alignment and integration with ECCCH and DS4CH	See D11.1	EF	R	PUB	M42
13	D13.1	INFINITY Methodology	Guidelines, reuse scenarios, and training materials for the CH sector, CH infrastructures, creative industries and tourism, and academic education and research	NISV	R	PUB	M42
14	D14.1	INFINITY image and public presence	Brand, website, and social network profiles	DP	Other, R	PUB	M3
14, 15,	D14.2,	Plan for the Exploitation,	Exploitation, dissemination and	DP	R	SEN	M12,

16	D15.1, D16.1	Dissemination and Communication of results (PEDR)	communication strategies and actions.				M24, M36
14,15,16	D14.3, D15.2, D16.2	Capacity building workshops	Report on workshops, hackathons, including statistics of participation	EIT SW CC	Other, R	PUB	M12, M30, M42
15,16	D15.3, D16.3	Outcomes of the INFINITY Open Call	Objectives, scope, eligibility and dissemination. Management of submissions and selected projects	NISV	R	PUB	M30, M42
16	D16.4	Report for Policy Makers	Guidelines, issues and lessons learned about the integration and alignment with ECCCH and DS4CH	EF	R	SEN	M42

Table 3.1d: List of milestones

Milestone number	Milestone name	Related WPs	Due date	Means of verification
1	Project iterative cycle kicked off	WP1, WP4, WP5, WP8, WP11, WP14	M8	Call for Stakeholder Network is out. Early draft of technical roadmap for use cases, Rulebook, and early set of requirements collected. Project image, website and social network profiles ready. Ecosystem publishing infrastructure e.g. GitHub, configured and ready to use. Benchmarking experiments concluded and surveys completed. Surveys and ethics guidelines available. Participation to Integration TF (ECHOES) established and activities for integration with ECCCH and DS4CH started.
2	INFINITY digital ecosystem maturity level 1	WP6, WP9, WP11, WP13, WP15	M20	Requirements and design phase concluded. First release of INFINITY digital ecosystem, including first release of use case platforms and software components (*Alpha*) and their documentation completed. Project products are ready for beta testing (developers) and early version of training material available. Integration with ECCCH and DS4CH on track. Stakeholder Network created (≥30 items) and the first capacity building workshop concluded. Use cases are in their second iteration. First version of PEDR and market analysis available. FSTP open call published.
3	INFINITY digital ecosystem maturity level 2	WP6, WP9, WP12, WP13	M32	Implementation phase concluded. Second release of INFINITY digital ecosystem, including software components (*Beta*) and documentation, completed. Use cases are in their third iteration and include new requirements from the Stakeholder Network. Project products are ready for reuse by developers and stakeholders. Alignment with ECCCH and DS4CH is systematic. A first stable version of training material available. Three Open Days organized. Second version of PEDR released, and two additional capacity workshops organized and concluded. Beneficiary of FSTP call selected, and first results available.
4	INFINITY digital ecosystem maturity level 3	WP7, WP10, WP12, WP13	40	Full integration with ECCCH and DS4CH. Deployment and validation phase concluded. Training material ready for publication. Final release of the INFINITY digital ecosystem available (stable version of software components 1.0). Final PEDR released. Results of the project funded through FTSP calls presented in an open dissemination event. All guidelines ready for publication.
5	Project completion	All WPs	42	All communication and dissemination activities completed, exploitation plan in place, project management concludes with all work done in budget and time, final report to EU.

Table 3.1e: Critical risks for implementation #@RSK-MGT-RM@#

Description of risk (indicate level of (i) likelihood, and (ii) severity: Low/Medium/High)	WP(s) involved	Proposed risk-mitigation measures

Consortium partner is unable to take up their responsibilities (**low, medium**)	ALL	The consortium is highly complementary, allowing us to source alternative competencies if a key skill is unavailable from a partner. The experienced project coordinator, previously leading the H2020 project POLIFONIA, will promptly identify and address any task or work package issues, proposing solutions to the Steering Committee.
Ineffective interdisciplinary integration, leading to misaligned objectives **(low, medium)**	ALL	Cross-disciplinary meetings and a unified glossary, with oversight by an Interdisciplinary Working Group, are employed to ensure effective integration and collaboration across different fields.
Partner problems (e.g., underperforming partner; a key partner leaves the project). (**low, medium**)	WP1, WP2, WP3	The well-balanced consortium includes organizations of various sizes with complementary expertise. Should any issues arise, such as a partner underperforming or leaving, the network will be leveraged to quickly address gaps and ensure successful project delivery.
Overlap with R&I work within and beyond Horizon Europe might result in duplication of efforts. **(medium, medium)**	WP5	Related work in topic destinations "World-leading data and computing technologies" and "A human-centred and ethical development of digital and industrial technologies" is tracked in T5.1. Communication with projects that overlap technically are conducted to ensure complementary outcomes.
Lack of content/data with the necessary variability/quantity **(low, high)**	WP6, WP7	Several technology and use case partners host extensive data repositories; active data interfaces to Wikidata and DS4CH (via EF); additional resources available via NISV and TMO members, and open data repositories; GDPR challenges for provision can be lowered by applying PET/anonymization.
Integration of INFINITY components into the tools used in the use case fails **(low, high)**	WP6, WP7	While critical for the project objectives, integration problems are unlikely as we use well-established technologies for I/O (REST APIs) and partners are experienced in the integration task with others’ systems.
Use cases fail to validate the INFINITY results **(low, high)**	WP6, WP7	Use case stakeholders are involved from the very beginning in use case design and requirements solicitation. Regular rounds of feedback and adaptation ensure results closely address stakeholder needs.
AI approaches not producing sufficient or required results **(medium, low)**	WP8, WP9, WP10	While new AI methods like GNNs are experimental, past successes suggest potential for future innovation. If one method falters, experienced partners can test alternative AI and graph approaches.
Standards and specifications do not align to those used by INFINITY tools **(medium, low)**	WP8, WP9, WP10	The project initiates early definition of required extensions and proposals to specification bodies. EF, NISV, and PSNC, as active organizations in their domains, will lead in defining data specifications. It is noted that new tools often set precedents for specifications before wider adoption.
Coordination with ECCCH delayed or takes a different approach **(medium, low)**	WP11, WP12	Coordination with ECHOES will be closely managed through PSNC and EF, via participation in the ECCCH Integration Task Force.
Diss. and comm measures are ineffective, limiting the project’s impact and adoption. **(low, medium)**	WP14, WP15, WP16	Section 2.2. shows partners have concrete, high-profile channels for dissemination and communication. Execution will be adaptive to changes in those channels and how external audiences react to our activities, modifying planned activities as necessary.

#§RSK-MGT-RM§#

Table 3.1f: Summary of staff effort

	Work Packages																**Total PM per participants**
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	
1/UNIBO	6	6	6	2	2	2	1	13	28	18	0,5	0,5	3	2	3	3	**96**
2/CNR	0,2	0,2	0,2	0,5	0	0	0,5	1	5	4	0	0	0,2	0,2	0,2	0,2	**12,4**
3/SPA	0,5	0,6	0,5	6	1,5	0	1,5	5	15,5	15,5	0	0	0	1,30	2,10	1,30	**51,3**

4/UNI JENA	2,5	2,5	2,5	2	6	6	4	3	13	3	0	0	0	1	1	1	**47,5**
5/EF	0,5	1	0,5	3	4	17	3	1	3	2	5	5	2	1	2,5	1,5	**52**
6/NISV	1	1,5	2,5	3,5	3,5	9	8	4	3	3	0,5	0,5	10	2	3,5	5	**60,5**
7/TMO	0,5	0,5	0,5	0	4	7	1,5	0,5	0,5	0,5	0,5	0,5	0	1	0,5	0,5	**18,5**
8/PSNC	0,5	0,5	0,5	2,5	8	22,3	7	6	9	10,5	5	4	4	1,1	0,5	0,9	**82,3**
9/DP	1	1	1	0	1	1	1	0	0	0	0	0	6,5	10,5	14	8,5	**45,5**
10/F&F DS	0,5	0,5	0,5	0,5	5	7	3	2	3	2,5	0	0	0	1	1	1	**27,5**
11/F&F PartG	0,5	0,5	0,5	1	2	5	3	1	1	1	0	0	0	0,5	0,5	0,5	**17**
12/MIC	1,5	1	1	1	2,5	8	3,5	0,5	0	1	0	0	0	0,5	0,5	2	**23**
13/UEDIN	1,5	1	1	4	1	2	1,5	22	1,5	1,5	0	0	7	1,5	1,5	2	**49**
14/UWr	0,2	0,4	0,2	0	1,4	3,7	1,4	0	1	0,5	0	0	0,6	1,2	3	3	**16,6**
15/EIT-CC-SW	1	1	1	2	2	2	0	2	0	0	0	0	1	3	3	3	**21**
16/EPFL	0,5	0,5	0,5	0	6	1,5	6	6	6	6	0	0	0	0,2	0,2	0,2	**33,6**
17/AIT	0	0	0	0	1	1	2	0	0	2	0	0	0	0	0	0	**6**
Total PM	**18,4**	**18,7**	**18,9**	**28**	**50,9**	**94,5**	**47,9**	**67**	**89,5**	**71**	**11,5**	**10,5**	**34,3**	**28**	**37**	**33,6**	**659,7**

Table 3.1g: 'Subcontracting costs' items

Participant Number/Short Name		
	Cost (€)	**Description of tasks and justification**
Subcontracting	30000	This subcontracting sum is planned for three contracts (10.000 EUR each) that cover the costs for engaging a local PR agency in each of the three countries (Germany, Poland, the Netherlands) where the campaign Europeana 1945 (at the core of the CROWD use case) will take place. The PR agency will develop the outreach plan and build engagement with local influencers, policy makers, and the general public to ensure campaigns' success in terms of visibility, event participation and engagement. (WP6 -Task 6.1)

Table 3.1h: 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services)

1/UNIBO		
	Cost (€)	**Justification**
Travel and subsistence	€45.730,25	Part of costs for three UNIBO team members across 42 months for project meetings and dissemination meetings.
Other goods, works and services	€11.500	Costs related to open access publications (7.000); conference fees (4.500).
Remaining purchase costs (<15% of pers. Costs)	€85.269,25	
Total	€142.500,50	

2/CNR		
	Cost (€)	**Justification**
Travel and subsistence	€12.357	Costs for 2 people in 42 months for project meetings and dissemination meetings.
Remaining purchase costs (<15% of pers. Costs)	€8.843	
Total	€21.200	

7/TMO		
	Cost (€)	**Justification**
Other goods, works and services	€40.000	4*10.000 EUR – Service for demo case cooperation partners for the preparation of data, provide consultancy and advertise at the specific locations or organize an event. The work will be supervised by TMO staff. (WP6 and WP7)
Remaining purchase costs (<15% of pers. Costs)	€21.000	
Total	€61.000	

12/MIC		
	Cost (€)	**Justification**
Travel and subsistence	€6.710	Part of the costs for 2 persons in 42 months for project meetings and dissemination meetings.
Other goods, works and services	€51.400	Services for 1) publication of ontological modules developed by the Infinity project to explain the conditions of use of data, metadata and digital resources; 2) publication of enriched data on a SPARQL endpoint; 3) implementation in the web portal of a section dedicated to the conditions of use and of functionalities aimed at monitoring the use of the resources made available.
Remaining purchase costs (<15% of pers. Costs)	€13.290	
Total	€71.400	

15/EIT		
	Cost (€)	**Justification**
Other goods, works and services	€22.450	Part of the costs for materials consumables to prepare the workshops and communication materials, caterings, renting of AV and connectivity materials, rent of spaces, communications and organization external support services to organize the workshops and hackathons. (WP14)
Remaining purchase costs (<15% of pers. Costs)	€17.550	
Total	€40.000	

3.2 Capacity of participants and consortium as a whole #@CON-SOR-CS@# #@PRJ-MGT-PM@#

The INFINITY consortium (Figure 3.3) includes a group of 17 complementary institutions from 9 European countries, ranging from academic and research bodies and cultural heritage organizations to technology providers and creative companies. Each member has been selected for their proven expertise, resources, and strategic role in the cultural heritage and digital technology sectors (see Table 3.2 for details).

Table 3.2 Consortium expertise

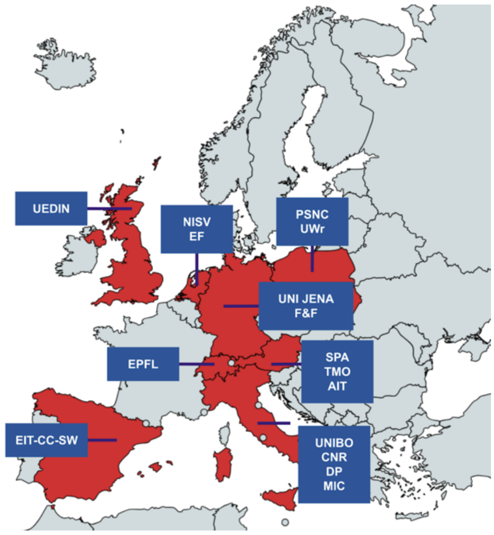
	UNIBO	CNR	SPA	UNI JENA	EF	NISV	TMO	PSNC	DP	F&F DS	F&F PartG	MIC	UEDIN	UW	EIT	EPFL	AIT
KGs, Embeddings and Multidimensionality	●		●	●	●			●									
Multimodal data analysis	●		●	●						●						●	
AI & Machine Learning for Cultural Heritage	●		●	●	●			●		●				●		●	
Digital humanities	●			●		●		●	●	●	●		●	●	●	●	
Scalable data Infrastructure and cloud technologies			●		●			●						●			●
Interoperability with reseach infrastuctures				●	●			●									
Ethical data management (usage rights, provenance)		●			●	●					●						
User-Centric Interface Design	●			●		●		●	●				●	●			
Real-World Use Cases & evaluation				●	●	●	●	●		●	●	●	●			●	
User-centred design methodologies	●					●		●	●	●						●	
Policy Alignment and Regulatory Compliance					●	●							●		●		
Business model co-design					●					●	●	●	●		●		
Creating and delivering training programmes	●				●	●		●			●	●		●	●	●	
Dissemination and communication	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Exploitation	●	●										●			●		

AI Solutions Tailored for the Cultural Heritage Sector: Computer science research groups within INFINITY are recognized for their excellence in KGs and AI advancement, particularly focusing on cultural heritage data. **UNIBO**'s Department of Languages, Literatures, and Modern Cultures expertise includes linguistic studies and discourse analysis, which intersect with semantic web technologies and AI. The team involved has a strong record in knowledge engineering research focusing on ontologies/knowledge graphs and their combination with large language (and music) models, as well as experience and expertise in software engineering and agile methodologies. The Department of Legal Studies is also involved, with expertise in IPR and associated rights and legal and ethical considerations related to the use of AI tools. They recently completed the POLIFONIA Horizon Europe project, showcasing their capability in managing large-scale, interdisciplinary projects. **UJENA** is developing a suite of 4D world visualization tools and a 4D reconstruction pipeline (to be utilized in INFINITY's [4DGRID] use case) and leads the S3 Partnership for Virtual and Sustainable Cultural Tourism as mandated by the EC to enhance alignment of innovation activities. In INFINITY, UJENA contributes to the development of technical workflows and tools for enriching and presenting image and object data. Associate partner **AIT**'s Digital Insight Lab focuses on Data Science and Digital Libraries, contributing to the [CROWD] use case with new methods for semantic text material enrichment using LMM. **CNR** is the largest public research institution in Italy. The Semantic Technology Laboratory is internationally recognised for their contribution in the area of AI, Semantic Web and software engineering. The team works in close synergy with UNIBO forming the facto a cohesive research group. Alongside these four computer science groups, two companies, F&F DS and SPA, provide specialized AI solutions; **F&F DS** in IT and web development for cultural heritage and **SPA** in developing Generative AI to enhance online communication impacts. SPA will establish advanced AI models and multidimensional KGs for contextual and socio-cultural text interpretation in INFINITY, enhancing their text editor to support ethically and culturally sensitive communication.

Embedded in the Digital Cultural Heritage Ecosystem: The INFINITY partnership has a history of contributing significant innovations within Europe's digital heritage sector, with individual partners committed to the long-term development of the field. Audiovisual archive and knowledge institute **NISV** in charge leading the [MULTIMEDIA] uses case and creation of training materials, will leverage its position in international networks (including EUscreen, UNESCO, FIAT-IFTA, ekip) to promote the uptake of the ECCCH and strengthen sectoral capacities. Building on its involvement in ECHOES, NISV will be responsible for executing the FSTP in INFINITY. The **UWr** library, with its extensive collections across all knowledge fields and special emphasis on humanities and social sciences, showcases notable music, cartographic, and graphic collections in its Digital Library through streaming and IIIF technologies. This makes it perfectly positioned to lead the [MANUSCRIPTS] use case. **F&F PartG** is a private research institute dedicated to historical research, applied history, and the public engagement of historical topics through various media formats. It has led crowdsourcing campaigns and collection days for Europeana 1914–1918 and Europeana 1989, providing the basis for their work on the [CROWD] use case. **MIC** carries out research and training in the field of description and digitization of archival assets. It promotes dissemination of international description standards, ensuring the uniformity of descriptions in archival systems and promoting their interoperability with other systems. The archival information system is the main subject of the [ARCHIVES] use cases in the INFINITY project.

Ethics-Driven Innovation and Sustainable Infrastructures: UEDIN has a track record of research excellence in all central areas of INFINITY, e.g., NLP, HCI, design, AI ethics, law and heritage The interdisciplinary Institute for Design Informatics (IDI), a collaboration between the School of Informatics and Edinburgh College of Art, specializes in data visualization, design integration, and user-centred security. **PSNC** provides advanced software and infrastructure solutions for domains like cultural heritage and digital humanities, including tools for data digitization and preservation. It offers HPC, cloud, edge, and quantum infrastructure through IaaS and PaaS platforms. It is involved in DARIAH and the ECHOES project, enhancing the integration of INFINITY and ECCCH platforms. **EPFL** is home to the EPFL Digital Humanities Laboratory (DHLAB) that develops new computational approaches to manage and enrich large digital cultural objects. In INFINITY, the EPFL DHLAB will contribute to the development of technical workflows and tools to enrich and present image and object data.

Figure 3.3 INFINITY Consortium



Networking, Outreach, and Business Exploitation: **EF**, along with 18 European partners, deploys the DS4EU, advocates the creation of robust frameworks and policies. In line with its role in the ECHOES project, EF will lead the work related to the alignment and integration of project results with ECCCH and DS4CH. EF is also key in the [CROWD] use case and supports the dissemination of the project results to its audiences and networks, including the European Heritage Hub. **TMO** is an international organization for cooperation in technology, science and cultural heritage, and a network of over 680 institutional members including world-leading scientists, innovators and civil society organizations. It supports a network of 85 Local Time Machines (geographic centres of data collections), which are instrumental for validating tools and workflows in INFINITY. **EIT-CC-SW** is one of the six co-location centres of the EIT Culture & Creative. It promotes interdisciplinary collaboration, particularly in cultural heritage, fashion, architecture, design, and audiovisual innovation linked to digital assets. It will play a key role in supporting explorations of business models for cultural heritage. **DP** is a full-service digital agency specializing in communication, design, and technology and will oversee INFINITY's communication and dissemination strategy. With a human-centred approach, the agency develops products and services that prioritise people, ensuring maximum usability and inclusivity. DP specializes in designing and developing technological solutions that comply with accessibility standards, ensuring usability for all.

Proven Collaboration and Policy Engagement: The consortium builds on already existing collaborations between all partners in related research and innovation initiatives. NISV, EF, TMO, and PSNC are working closely together on developing the DS4CH and are directly involved in ECHOES. UNIBO, NISV, and DP have collaborated in POLIFONIA. UEDIN and NISV work in ekip creating an innovative policy platform for the Cultural and Creative Industries. NISV and EF are members of the EIT-CC network. Through ekip and EIT-CC-SW, the work of INFINITY is well-grounded in EU policy making and implementation, promoting sustainable and inclusive growth in CCIs.

As Associated partner based in Switzerland, EPFL (PIC 999973971) will be fully funded by the Swiss government (SERI), according to the official financial guarantee for Horizon Europe 2024 calls. EPFL will contribute to this project with a financial contribution of 438'315 € (336'287 € personnel costs, 14'365 € other direct costs, 87'663 € indirect costs). The personnel costs will be allocated to WP1 (0.50 PM), WP2 (0.50 PM), WP3 (0.50 PM), WP5 (6.00), WP6 (1.50), WP7 (6.00), WP8 (6.00), WP9 (6.00), WP10 (6.00), WP14 (0.20), WP15 (0.20), WP16 (0.20). The person-month allocation of both associated partners (EPFL and AIT) is documented in Table 3.1f

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